Structural Accessibility in Phase-Encoded Quantum Graph Embeddings

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Abstract

A phase-encoded quantum graph circuit writes a graph G on n vertices into n qubits, applies one $R_{ZZ}(\gamma)$ gate per edge and one single-qubit mixer rotation per vertex to a product state, and is measured in the computational basis. The circuits have the alternating phase-separation and mixing architecture of QAOA, and with the uniform preparation they are depth-one MaxCut QAOA up to conventions. Sorting the 2^n Born probabilities in descending order removes the vertex labels. We determine what the sorted readout carries about G, for cubic graphs, order by order in the mixer angle β.

Under a degree-aware product preparation the Born distribution at $\beta = 0$ is layered by the number of defects, the bits that differ from the modal outcome. We prove that the first-order term of every outcome is a sum over vertices of a mixer profile evaluated at the number of defect neighbours. For the X mixer the value depends on the defect set D through two integers, the numbers of vertices inside and outside D whose whole neighbourhood lies on one side of D, and when $\tan^2(\eta/2)$ is irrational the sorted m-defect block determines the histogram of these integers over all m-subsets, for every n; the Y mixer reads the odd part of the same profile. At $m = 2$ the histogram is the adjacency–codegree pair census and nothing more, and it fixes the triangle, four-cycle and $K_4 - e$ counts. At $m = 3$, on cubic graphs of girth at least five, it is an explicit function of n and the five- and six-cycle counts; so for every n the first $K_3(n)$ ranks determine $\min(\text{girth}, 7)$, determine C_5 and C_6 at girth at least five, and agree through first order at girth at least seven. Reading the pair census from the two-defect outcomes of labelled samples costs $1/(\beta^2\mu_2)$ shots up to factors polynomial in n, where μ_2 is the weight of the two-defect layer.

The expansion sits in the Weisfeiler–Leman hierarchy order by order: the first-order two-defect block is the first round of folklore 2-WL, the order-j term of the m-defect layer is determined by $(m + j)$-WL, and K-WL-equivalent cubic graphs have sorted m-defect blocks that agree through order $K - m$. The readout at a fixed angle is not bounded by 2-WL: folklore 2-WL is injective on the connected cubic censuses through $n = 20$, and for the Cai–Fürer–Immerman pair over K_4, two 2-WL-equivalent cubic graphs on 40 vertices, an exact evaluation at the operating angle shows that the modal probabilities differ by 6.8×10^{-30}, certified by interval arithmetic, while the first 16,231 ranks agree to 1.7×10^{-18}. At second order the output purity is an exact affine function of the triangle, four-cycle and $K_4 - e$ counts, and of the $K_4 - e$ count alone under the uniform preparation, and an interval certificate shows that at the operating angle and $n \in \{14, 16, 18\}$ the first $K_2(n)$ ranks are the outcomes with at most two defects. The template, the two-defect class values and the purity closures are verified on the complete connected cubic censuses at $n = 10$, 12 and 14.

1 Introduction

1.1 The circuits and the readout

A quantum circuit can serve as a fixed feature map for graphs: the graph is written into the circuit, the circuit is run and measured, and a classical learner receives a vector built from the outcomes after an invariantization that removes the vertex numbering. This paper determines exactly what that vector carries about the graph, for one family of circuits on cubic graphs.

One qubit stands for each vertex. A product-state preparation is followed by the edge-phase operator $U_G(\gamma) = \prod_{uv\in E} R_{ZZ}^{(uv)}(\gamma)$, which is diagonal and depends on the graph only through its cut function, and then by a mixer $\prod_v R_P^{(v)}(\beta)$ with axis $P \in \{X, Y\}$. Measurement in the computational basis gives the Born distribution p_G on $\{0,1\}^n$, and sorting its 2^n entries in descending order gives the *sorted readout* $p_G^{\downarrow}$. This is the alternating phase-separation and mixing architecture of QAOA [6, 7]. With the uniform preparation $|+\rangle^{\otimes n}$ the circuit is depth-one MaxCut QAOA up to the signs and scalings of the angles; we call this member PE-QX1 and its preparation the Hadamard preparation. The anchor member PE-X1, the circuit of Miller and Lee [12], uses the degree-aware preparation $\bigotimes_v R_X(\eta d_v/d_{\max})|0\rangle$ and one X mixer; PE-Y1 changes the mixer axis to Y, and PE-X2 repeats the phase–mixer block twice with tied angles; these three are the *degree-prepared members*. The angles $(\eta, \gamma) = (23/8, 2)$, reported for PE-X1 in that work with $\beta = 1/10$ and not retuned here, are the *schedule*; the theorems are stated under angle conditions that the schedule satisfies, the first-order ones as $\beta \to 0$, and the certificates and the computations are done at the schedule with $\beta = 1/10$, the *operating angle*. The usual analysis of QAOA concerns one expectation value, the cut objective [17, 20]. Here the object is the whole measured distribution after the invariantization, and the theorems say what survives both.

The degree-aware preparation is what makes the exact analysis possible, and it is the opposite of the uniform superposition QAOA starts from: on a cubic graph it applies $R_X(\eta)$ to every qubit, and at $\eta = 23/8$ each qubit is within $c^2 = \cos^2(\eta/2) = 0.0177$ of $|1\rangle$. The edge phase is invisible to a computational-basis measurement, and the mixer creates interference between bit strings at small Hamming distance, at first order in β between strings at Hamming distance one. At $\beta = 0$ a string with ℓ zero bits, its *defects*, has probability $c^{2\ell}s^{2(n-\ell)}$ with $s = \sin(\eta/2)$, so the distribution is layered by defect count with the fixed ratio $t^{-2} = 0.0180$ between layers, $t = \tan(\eta/2)$, and the first $K_m(n) = \sum_{i\le m}\binom{n}{i}$ sorted entries are the outcomes with at most m defects. The sorted readout is a sequence of blocks, one per defect count, and the question becomes what each block determines, order by order in β.

1.2 Results

Three theorems answer the question in sequence: what the circuit encodes at first order (Theorem A), what graph structure that encoding determines (Theorem B), and where it sits relative to the Weisfeiler–Leman hierarchy (Theorem C). Two further statements follow from the same analysis: the second-order purity closures (Theorem D) and the cost of reading the pair census from samples (Proposition E). For an outcome x write D for its defect set, $k_j(D) = |N(j) \cap D|$ for the number of defect neighbours of vertex j, and $\lambda(x)$ for the *first-order value* of x, the derivative of $p_G(x;\beta)$ at $\beta = 0$ divided by $p_G(x;0)$. The *profile* of D is the pair of count vectors recording how many vertices inside and outside D have $0, 1, 2, 3$ defect neighbours; its *extremal counts* are the number $E_{\text{in}}(D)$ of vertices of D with no defect neighbour or only defect neighbours and the number $E_{\text{out}}(D)$ of vertices outside D with the same property, and its *odd readings* are the differences $q_3 - q_0$ and $q_2 - q_1$ of the count vectors, inside and outside D.

Theorem A (Template and determination; Theorems 3.2 and 3.3)**.** *Let G be a cubic graph on n vertices under a degree-prepared member, and in (2) let $t = \tan(\eta/2) > 1$, so that the sorted readout at $\beta = 0$ lists the outcomes by defect count.*

1. *For every outcome x with defect set D, $\lambda(x) = t^{-1}\sum_{j\notin D} h(k_j(D)) - t\sum_{j\in D} h(k_j(D))$, where the mixer profile h is $\cos(\gamma(2k-3))$ for the X mixer, $\sin(\gamma(2k-3))$ for the Y mixer, and $\cos(\gamma(2k-3)) + \cos(2\gamma(2k-3))$ for two tied repetitions. Under the Hadamard preparation the same holds with $t = 1$, the profile $\sin(\gamma(2k-3))$, and the two sums exchanged.*

2. *For the X mixer, if $h_{\mathrm{e}} \neq h_{\mathrm{o}}$ (for one repetition, $\cos 3\gamma \neq \cos\gamma$) and t^2 is irrational, then for every m and all sufficiently small $\beta > 0$ the sorted readout on the ranks of the m-defect block determines the histogram of $(E_{\mathrm{in}}(D), E_{\mathrm{out}}(D))$ over all m-subsets D. For the Y mixer the same holds for the histogram of the odd readings when $\sin\gamma \neq 0$ and 1, $\sin^2\gamma$, t^2, $t^2\sin^2\gamma$ are linearly independent over $\mathbb{Q}$. At the schedule t^2 is transcendental and the independence holds, so both statements hold for every n.*

At $m = 2$ the histogram is the *pair census* $(N_{a,k})$, the number of vertex pairs with adjacency a and k common neighbours, which fixes the triangle count C_3, the four-cycle count C_4 and the $K_4 - e$ count $D_\diamond$ through incidence identities. At $m = 3$ it is not a subgraph count in general, but at girth at least five it is one.

Theorem B (The first three layers; Theorems 4.1 and 4.3, Corollaries 4.2 and 4.4)**.** *Let G and H be cubic graphs on n vertices under PE-X1, PE-Y1 or PE-X2 at angles satisfying the conditions of Theorem A, and let $\beta > 0$ be sufficiently small.*

1. *Ranks $n+2, \ldots, K_2(n)$ of the sorted readout determine the pair census, hence C_3, C_4 and $D_\diamond$, and at first order nothing more: G and H have the same first-order two-defect block iff they have the same pair census.*

2. *The first $K_3(n)$ ranks determine $\min(\mathrm{girth}, 7)$. If the girth is at least five, the three-defect histogram is an explicit function of n, C_5 and C_6 (Table 1) and determines C_5 and C_6. If the girth is at least seven, the first $K_3(n)$ sorted entries of G and H agree through first order.*

The order-j Born coefficient of an m-defect outcome is a sum over extensions of the defect set by at most j vertices of a graph-independent function of the marked subgraph, which is what folklore $(m+j)$-WL computes.

Theorem C (Placement; Theorem 5.3 and Proposition 5.4)**.** *Let G and H be cubic graphs on n vertices under a degree-prepared member at fixed angles with $t > 1$.*

1. *If one round of folklore $\max(m, 2)$-WL does not distinguish G from H, their first-order m-defect blocks coincide. At angles satisfying the conditions of Theorem A, the first-order two-defect blocks coincide iff one round of folklore 2-WL does not distinguish G from H.*

2. *If G and H are K-WL equivalent, their sorted m-defect blocks agree through order β^{K-m} for every $m \leq K$.*

3. *Folklore 2-WL is injective on the complete connected cubic censuses at $n = 14, 16, 18, 20$. The Cai–Fürer–Immerman pair over K_4, two 2-WL-equivalent cubic graphs on 40 vertices, has sorted readouts under PE-X1 at the operating angle whose modal probabilities differ by 6.7957×10^{-30}, the difference certified by interval arithmetic, and whose first 16,231 ranks agree to 1.7×10^{-18}. The sorted readout at the operating angle is therefore not a function of the folklore-2-WL colouring.*

Theorem D (Second order; Theorem 6.1 and Proposition 6.3)**.** *For every cubic graph G on n vertices, the second derivative at $\beta = 0$ of the output purity $M_2(G;\beta) = \sum_x p_G(x;\beta)^2$ is $A_n + b_3 C_3 + b_4 C_4 + b_D D_\diamond$ under PE-X1 and $A_n^H + 4 \cdot 2^{-n} \sin^6(\gamma) D_\diamond$ under PE-QX1, with coefficients that do not depend on G. For PE-X1 at the operating angle and $n \in \{14, 16, 18\}$, the first $K_2(n)$ sorted probabilities are exactly the outcomes with at most two defects, by an interval-arithmetic certificate.*

Proposition E (Shots; Proposition 4.6)**.** *Let $\mu_2 = c^4 s^{2(n-2)}$ be the weight of the two-defect layer and Δ the least gap between the first-order class values. For an observer who keeps the sampled bit strings, in the first-order window $51 \log(n^2/\delta)/(\beta^2 \mu_2 \Delta^2)$ shots recover the pair census from the two-defect outcomes by rounding with probability $1 - \delta$, and any procedure that uses only the two-defect shots needs $1/(\beta^2 \mu_2)$ shots up to a factor polynomial in n.*

Theorems A to D and Proposition E are proved under their hypotheses; the certified items are the head membership in Theorem D and the rank-one difference in Theorem C; the computed items are the 2-WL injectivity and the agreement of the 16,231 ranks. The identities of Theorems A, B and D are verified on the censuses at $n = 10$, 12 and 14 (Section 7), and Appendix G lists each statement with its status and the domain of every computed check.

1.3 Techniques

The flip ratio. The pre-mixer amplitude is a product of preparation factors times the cut phase, so the ratio of the amplitudes of two strings at Hamming distance one depends only on the flipped vertex and the spins of its neighbours (Lemma 3.1). Expanding each mixer factor as $\cos(\beta/2) I - i \sin(\beta/2) X$ writes the post-mixer amplitude as a sum over flip sets; the first-order template is the one-flip term, and the order-j coefficient collects flip sets with at most j vertices.

Determination by number theory. For the X mixer the first-order value of an m-defect outcome is $(\cos 3\gamma - \cos\gamma) t^{-1} [E_{\text{out}} - t^2 E_{\text{in}}]$ plus a constant of the layer, so two outcomes collide iff an integer equals t^2 times another integer, which an irrational t^2 forbids; at the schedule t^2 is transcendental. For the Y mixer a collision is a rational relation among 1, $\sin^2\gamma$, t^2, $t^2 \sin^2\gamma$, excluded at the schedule by a Laurent-polynomial argument in $e^{i/8}$.

Extension counts. Lemma 5.2 says that the round-i colour of a k-tuple with at most $k - i$ distinct entries determines the number of outside i'-tuples of each atomic type for $i' \le i$; padding the defect set to a K-tuple turns K-WL equivalence into agreement of the order-j coefficients for $j \le K - m$.

Contraction and certification. A Born probability of the 40-vertex pair is a two-state partition function on the graph, because the mixer row $\langle x|U_M$ is a product and the edge phase is diagonal; variable elimination evaluates it exactly, and the same elimination in outward-rounded interval arithmetic at 60 digits encloses the difference of the two modal probabilities in an interval that excludes zero.

1.4 Related work

The circuits descend from QAOA [6] and the alternating-operator ansatz [7]. Wang et al. [20] show that the level-one MaxCut expectation depends on local degrees and triangles, and Szegedy [17] that on cubic graphs it carries exactly n and C_3; those are statements about one expectation value; ours are about the sorted readout, block by block. Beasley and Montanaro [1] study the output distributions of IQP circuits and of Hadamard-prepared QAOA built from a graph as isomorphism tests, with reconstruction of the graph from polynomially many samples at a small angle; they ask whether a test separates graphs, we what one fixed invariant determines; their Hamming-weight sectors are our defect layers under the degree-aware preparation. Quantum feature maps [8, 9, 15, 18]

and quantum graph circuits [11, 14, 16, 23] are the setting, the latter placed as trainable classes in the Weisfeiler–Leman hierarchy [3, 21] by which the expressivity of graph neural networks is measured [2, 4, 5, 13, 22]; our family is fixed and shallow, and the question is determination rather than expressive capacity. PE-X1 is the circuit of Miller and Lee [12]. The injectivity proved there concerns the one-repetition statevector on labelled graphs under an irrational-angle condition; its corollary for sorted distributions is not a consequence of that theorem, since sorting discards the labelling, and we treat injectivity of the sorted readout on isomorphism classes as open. What that paper reports against the hierarchy is empirical: Cai–Fürer–Immerman pairs over base graphs of treewidth at most four, where the level k of k-WL needed to distinguish a pair grows with the treewidth, were separated in noise-model simulation and on hardware up to 66 qubits; Theorem C(3) is the exact counterpart on a cubic pair, at 10^{-30}.

1.5 Limits and open questions

Every first-order statement is a statement as $\beta \to 0$, and at the operating angle $\beta\lambda(x)$ is not small (between -1.4 and 0.7 on the two-defect layer of PE-X1 at $n = 14$); what survives there is established separately, by the certificate of Theorem D and by the finite-angle map of Appendix C. The proofs use cubic regularity: the template and the order-j locality extend to d-regular graphs by replacing 3 with d, and the reduction of the X-mixer value to two extremal counts, the seven pair classes and the girth theorem are cubic counting. The first-order content is classically easy (the extremal-count histograms of m-subsets take time $O(n^{m+1})$); the theorems locate this easy part of an object that is not expected to be easy, Proposition E prices it, and Theorem C(3) shows that the rest exists, at 10^{-30}. Two questions are open: which invariants the extremal-count and odd-reading histograms determine for all n outside the girth-five class, and whether any fixed level of the hierarchy determines every Taylor coefficient of the sorted readout, level two not being one.

2 Preliminaries

Circuits. Let $G = (V, E)$ be a simple graph with $n = |V|$ vertices, and identify V with $\{0, \dots, n-1\}$ and with the qubits. With $P \in \{X, Y\}$ the mixer axis and r the number of tied phase–mixer repetitions,

$$p_G^{(P,r)}(x) = \left|\langle x|\left(U_M^{(P)}(\beta)\, U_G(\gamma)\right)^r U_{\mathrm{prep}}|0^n\rangle\right|^2, \qquad U_G(\gamma) = \prod_{uv\in E} R_{ZZ}^{(uv)}(\gamma), \quad U_M^{(P)}(\beta) = \prod_{v\in V} R_P^{(v)}(\beta), \tag{1}$$

for $x \in \{0,1\}^n$, with the Qiskit convention $R_a(\theta) = e^{-i\theta\sigma_a/2}$ and little-endian bit order. The degree-aware preparation is $U_{\mathrm{prep}}|0^n\rangle = \bigotimes_v R_X(\eta d_v/d_{\max})|0\rangle$ with d_v the degree of v and $d_{\max} = \max_v d_v$; the Hadamard preparation is $|+\rangle^{\otimes n}$. The *degree-prepared members* are PE-X1 ($P = X$, $r = 1$), PE-Y1 ($P = Y$, $r = 1$) and PE-X2 ($P = X$, $r = 2$); PE-QX1 is $P = X$, $r = 1$ with the Hadamard preparation.

The cut phase. Write $z_v = (-1)^{x_v}$ for the spin of vertex v in the outcome x. The edge-phase operator acts on $|x\rangle$ by the scalar $\prod_{uv\in E} e^{-i\gamma z_u z_v/2} = e^{-i\gamma|E|/2} e^{i\gamma c_G(x)}$, $c_G(x)$ the size of the cut encoded by x, so the graph enters the pre-mixer state only through a phase, which a computational-basis measurement does not see.

Sorting is the quantile readout. The vector p_G is indexed by qubit labels; we sort. Let $\nu_G = 2^{-n}\sum_x \delta_{2^n p_G(x)}$ be the empirical measure of the scaled probabilities.

Proposition 2.1 (Quantile readout)**.** *For simple graphs G and H on n vertices, $\|p_G^{\downarrow} - p_H^{\downarrow}\|_1 = W_1(\nu_G, \nu_H) = \min_{\sigma \in S_{2^n}} \|p_G - \sigma p_H\|_1$, and sorting is nonexpansive in every ℓ_p norm, $p \geq 1$.*

Proof. The monotone coupling attains W_1 and is one of the permutations σ; nonexpansiveness is the rearrangement inequality applied coordinatewise. □

Defect layers and blocks. Fix the degree-aware preparation on a cubic graph, so that every vertex receives the rotation $R_X(\eta)$. Put $c = \cos(\eta/2)$, $s = \sin(\eta/2)$ and $t = s/c = \tan(\eta/2)$, and assume $t > 1$, so that $c < s$ and the all-ones string is the modal outcome at $\beta = 0$; this hypothesis is in force wherever layers, blocks and ranks are mentioned. At $\eta = 23/8$, $t^2 = 55.616$. The *defects* of an outcome x are its zero bits, $D(x) = \{j : x_j = 0\}$, and $\ell = |D(x)|$ is its defect count. At $\beta = 0$,

$$p_G(x; 0) = \mu(x) = c^{2\ell} s^{2(n-\ell)}, \qquad \frac{\mu_{\ell+1}}{\mu_\ell} = t^{-2} = 0.0180, \tag{2}$$

independently of G. The *m-defect layer* is the set of outcomes with m defects. At $\beta = 0$ the first $K_m(n) = \sum_{i \leq m} \binom{n}{i}$ sorted entries are the outcomes with at most m defects; $K_2(n) = 1 + n + \binom{n}{2}$. The ranks $K_{m-1}(n) + 1, \ldots, K_m(n)$ are the *m-defect block*, and the first $K_2(n)$ ranks are the *head*. For β small enough the m-defect layer occupies the m-defect block, since the first-order values are bounded and the layer weights are separated by the fixed ratio t^{-2}. The Hadamard preparation has no layers: $\mu(x) = 2^{-n}$ for every x.

Profiles, extremal counts and the pair census. For a vertex j and a defect set D let $k_j(D) = |N(j) \cap D|$. The *profile* of D is the pair of count vectors

$$q_k^{\text{in}}(D) = \#\{j \in D : k_j(D) = k\}, \quad q_k^{\text{out}}(D) = \#\{j \notin D : k_j(D) = k\}, \tag{3}$$

for $k = 0, 1, 2, 3$. The *extremal counts* are

$$E_{\text{in}}(D) = q_0^{\text{in}} + q_3^{\text{in}}, \qquad E_{\text{out}}(D) = q_0^{\text{out}} + q_3^{\text{out}}, \tag{4}$$

the numbers of vertices inside and outside D whose whole neighbourhood lies on one side of D, and the *odd readings* are $(q_3 - q_0,\, q_2 - q_1)$ for each of the two count vectors.

For an unordered pair $\{u, v\}$ put $a = \mathbf{1}[uv \in E]$ and $k = |N(u) \cap N(v)|$, and let $N_{a,k}(G)$ count the pairs of class (a, k); the seven realizable classes on cubic graphs are $(0,0), \ldots, (0,3), (1,0), (1,1), (1,2)$. The *pair census* $(N_{a,k})$ fixes three motif counts through incidence identities,

$$C_3 = \frac{1}{3} \sum_k k N_{1,k}, \qquad C_4 = \frac{1}{2} \sum_{a,k} \binom{k}{2} N_{a,k}, \qquad D_\diamond = \sum_k \binom{k}{2} N_{1,k}, \tag{5}$$

where C_4 counts all simple four-cycles, chorded ones included, and $D_\diamond = N_{K_4 - e}$ counts not-necessarily-induced copies of $K_4 - e$, each K_4 contributing six. For a two-defect set $D = \{u, v\}$ of class (a, k) one checks $q^{\text{in}} = (2(1-a), 2a, 0, 0)$ and $q^{\text{out}} = (n - 8 + 2a + k,\, 6 - 2a - 2k,\, k,\, 0)$, so that

$$E_{\text{in}}(D) = 2(1-a) =: B, \qquad E_{\text{out}}(D) = n - 8 + 2a + k =: A. \tag{6}$$

B determines a and then A determines k. This is the bridge from the two-defect block to the pair census.

Weisfeiler–Leman refinement. Folklore k-WL, $k \geq 2$, colours k-tuples of vertices: $c_0(\bar{u})$ is the atomic type of $\bar{u}$ (which entries coincide, which are adjacent), and

$$c_{\tau+1}(\bar{u}) = \Big(c_\tau(\bar{u}),\ \{\!\{\,(c_\tau(\bar{u}[w/1]), \ldots, c_\tau(\bar{u}[w/k])) : w \in V\,\}\!\}\Big), \tag{7}$$

where $\bar{u}[w/i]$ has its i-th entry replaced by w. Two graphs are *k-WL equivalent* if the multisets of colours of their k-tuples agree after every round, which is equivalence in the counting logic C^{k+1} [3], and they are *not distinguished by one round* if the multisets agree after the first; 1-WL denotes colour refinement. For $D \subseteq V$ and $T \subseteq V \setminus D$, the *pattern* $\pi(D, T)$ is the isomorphism type of $G[D \cup T]$ with the classes D and T distinguished.

3 First Order: the Template and Determination

In this section G is a simple cubic graph on n vertices with the degree-aware preparation, unless the Hadamard preparation is named. The pre-mixer amplitude is

$$\phi(x) = \Big[\prod_v a(x_v)\Big] \exp\Big[-\tfrac{i\gamma}{2} \sum_{uv \in E} z_u z_v\Big], \qquad a(0) = c, \quad a(1) = -is,$$

so that $|\phi(x)|^2 = \mu(x)$; for r tied repetitions the edge phase is applied r times. Write $p_0 = \mu$, $p_1 = \partial_\beta p_G(x;\beta)|_{\beta=0}$ and $\lambda(x) = p_1(x)/p_0(x)$ for the first-order value. Complete proofs are in Appendix A.

3.1 The flip ratio and the template

Let $\phi_j(x) = \phi(x \oplus e_j)$ be the amplitude of the string with bit j flipped, and $R_j(x) = \phi_j(x)/\phi(x)$ the *flip ratio*. Write $S_j(x) = \sum_{v \in N(j)} z_v$; on a cubic graph $S_j = 2k_j(D) - 3$.

Lemma 3.1 (Flip ratio; Lemma B.3)**.** $R_j(x) = -iz_j\, t^{z_j} \exp\big(i\gamma z_j S_j(x)\big)$.

Proof. Flipping bit j changes the preparation factor by $a(1)/a(0) = -it$ when $z_j = +1$ and by $a(0)/a(1) = i/t$ when $z_j = -1$, both equal to $-iz_j t^{z_j}$. It changes the edge phase only on the edges at j, by $\exp(i\gamma z_j \sum_{v \in N(j)} z_v)$. □

Differentiating $(U_M E)^r \phi_{\text{prep}}$ at $\beta = 0$ replaces one mixer factor by $-\frac{i}{2}\Sigma$, $\Sigma = \sum_j \sigma_j$ the mixer generator, and gives $p_1(x) = \mathrm{Im}[\overline{\phi(x)} \sum_{b=1}^{r} (E^{r-b} \Sigma E^b \phi_{\text{prep}})(x)]$, where E is the edge-phase diagonal (Lemma A.1). For $k \in \{0, 1, 2, 3\}$ define the *mixer profiles*

$$h^{X,r}(k) = \sum_{b=1}^{r} \cos\big(b\gamma(2k-3)\big), \qquad h^{Y}(k) = \sin\big(\gamma(2k-3)\big). \tag{8}$$

Theorem 3.2 (Layer template; Theorem A.4)**.** *For every outcome x with defect set D,*

$$\lambda(x) = \frac{1}{t} \sum_{j \notin D} h\big(k_j(D)\big) - t \sum_{j \in D} h\big(k_j(D)\big), \tag{9}$$

with $h = h^{X,r}$ for the X mixer at r tied repetitions and $h = h^Y$ for the Y mixer at one repetition. Under the Hadamard preparation with one X mixer, $p_0 = 2^{-n}$ and

$$p_1(x) = 2^{-n}\Big[\sum_{j \in D} \sin\big(\gamma(2k_j - 3)\big) - \sum_{j \notin D} \sin\big(\gamma(2k_j - 3)\big)\Big]. \tag{10}$$

Proof idea. For the X mixer the term with b edge-phase factors to the right of the generator has, at x, the ratio $R_j^{(b)} = -iz_j t^{z_j} e^{ib\gamma z_j S_j}$ to $\phi(x)$ (Lemma A.2), so $p_1 = \mu \sum_j \sum_b \operatorname{Im} R_j^{(b)}$ with $\operatorname{Im} R_j^{(b)} = -z_j t^{z_j} \cos(b\gamma S_j)$. For the Y mixer $(Y_j\phi)(x) = -iz_j\phi_j(x)$ (Lemma A.3), so $p_1 = -\mu \sum_j z_j t^{z_j} \sin(\gamma S_j)$; under the Hadamard preparation the flip ratio has no preparation factor, so $\operatorname{Im} R_j = z_j \sin(\gamma S_j)$. In each case $S_j = 2k_j - 3$. □

The template says that λ is the linear functional $t^{-1}\langle q^{\text{out}}, h\rangle - t\langle q^{\text{in}}, h\rangle$ of the profile (3).

3.2 Determination for the two mixer axes

The X profiles are even under $k \mapsto 3-k$: $h^{X,r}(0) = h^{X,r}(3) =: h_{\text{e}}$ and $h^{X,r}(1) = h^{X,r}(2) =: h_{\text{o}}$, with $h_{\text{e}} = \sum_{b\le r} \cos 3b\gamma$ and $h_{\text{o}} = \sum_{b\le r} \cos b\gamma$. Hence $\langle q, h\rangle = (h_{\text{e}} - h_{\text{o}})(q_0 + q_3) + h_{\text{o}} \sum_k q_k$ for either count vector, and (9) becomes

$$\lambda(x) = \frac{(h_{\text{e}} - h_{\text{o}})E_{\text{out}}(D) + (n - |D|)h_{\text{o}}}{t} - t\big[(h_{\text{e}} - h_{\text{o}})E_{\text{in}}(D) + |D|h_{\text{o}}\big]. \tag{11}$$

Theorem 3.3 (Layer determination; Theorems A.8 and A.21)**.** *Fix $m \ge 1$ and let $t > 1$.*

(X) For the X-mixer members, suppose $h_{\text{e}} \ne h_{\text{o}}$ and that either (a) $t^2 > n - m$ or (b) t^2 is irrational. Then on the m-defect layer the first-order value determines $(E_{\text{in}}(D), E_{\text{out}}(D))$. Under (a), two outcomes of the layer with different pairs have first-order values at least $|h_{\text{e}} - h_{\text{o}}|\, t^{-1} \min\{1, t^2 - (n-m)\}$ apart.

(Y) For PE-Y1, suppose $\sin\gamma \ne 0$ and that 1, $\sin^2\gamma$, t^2 and $t^2\sin^2\gamma$ are linearly independent over $\mathbb{Q}$. Then on the m-defect layer the first-order value determines the odd readings $(q_3 - q_0,\ q_2 - q_1)$ of the profile, inside and outside D.

In either case, for all sufficiently small $\beta > 0$, the sorted readout on the m-defect block determines the histogram of the readings over all m-subsets of $V(G)$. At the schedule, (b) holds for every n because $t^2 = \tan^2(23/16)$ is transcendental, (a) holds for $n - m < 55.6$, and the independence in (Y) holds, so both parts hold for every n.

Proof. (X) By (11), the values of two m-subsets differ by $(h_{\text{e}} - h_{\text{o}})t^{-1}[(E_{\text{out}} - E'_{\text{out}}) - t^2(E_{\text{in}} - E'_{\text{in}})]$, so they are equal iff $E_{\text{out}} - E'_{\text{out}} = t^2(E_{\text{in}} - E'_{\text{in}})$. If $E_{\text{in}} \ne E'_{\text{in}}$, under (a) the right side has modulus at least $t^2 > n - m \ge |E_{\text{out}} - E'_{\text{out}}|$, and under (b) it is irrational while the left side is an integer; so $E_{\text{in}} = E'_{\text{in}}$ and then $E_{\text{out}} = E'_{\text{out}}$. The gap follows from the same bracket. (Y) The Y value is $t^{-1}[(q_3 - q_0)^{\text{out}} \sin 3\gamma + (q_2 - q_1)^{\text{out}} \sin\gamma] - t[(q_3 - q_0)^{\text{in}} \sin 3\gamma + (q_2 - q_1)^{\text{in}} \sin\gamma]$, so two profiles with equal values give an integer relation among $\sin 3\gamma/t$, $\sin\gamma/t$, $t\sin 3\gamma$, $t\sin\gamma$, which after division by $\sin\gamma/t$ is a rational relation among 1, $\sin^2\gamma$, t^2, $t^2\sin^2\gamma$; independence forces all four reading differences to vanish. For the sorted statement, $p_G(x;\beta) = \mu_m(1 + \beta\lambda(x)) + O(\beta^2)$ on the layer, which occupies the block for small β. At the schedule, t^2 is transcendental by the Lindemann–Weierstrass theorem (Lemma A.7), and the independence is a Laurent-polynomial identity check in $e^{i/8}$ (Lemma A.20). □

Since $h^{X,r}$ is even under $k \mapsto 3-k$ for every r, the first-order value under the X mixer is the function (11) of $(E_{\text{in}}(D), E_{\text{out}}(D))$ at every depth: under the hypotheses of Theorem 3.3(X), PE-X1 and PE-X2 determine the same histogram on every layer and differ at first order only in the weight $h_{\text{e}} - h_{\text{o}}$ and the gap.

4 What the First Three Layers Determine

4.1 The two-defect layer and the pair census

Theorem 4.1 (Two-defect class values; Theorem A.11)**.** *For a degree-prepared member with mixer profile h, every two-defect outcome of class (a,k) has $p_1(x) = \mu_2 f(a,k)$ with $\mu_2 = c^4 s^{2(n-2)}$ and*

$$f(a,k) = \frac{(n-8+2a+k)h(0) + (6-2a-2k)h(1) + k\,h(2)}{t} - 2t\,h(a). \tag{12}$$

The function is affine in the class labels,

$$f(a,k) = f(0,0) + k\,\Delta_k + a\,\Delta_a, \quad \Delta_k = \frac{h(0)-2h(1)+h(2)}{t}, \quad \Delta_a = 2\big(h(0)-h(1)\big)\Big(t+\frac{1}{t}\Big), \tag{13}$$

and takes seven distinct values on the seven classes iff $\Delta_k \neq 0$ and $\Delta_a/\Delta_k \notin \{-2,-1,0,1,2,3\}$; for PE-X1 and PE-X2 this is $h_{\mathrm{e}} \neq h_{\mathrm{o}}$ and $t^2 \neq \frac{1}{2}$, and for PE-Y1 it is $\sin\gamma \neq 0$ and $(1+t^2)\cos 2\gamma/\sin^2\gamma \notin \{-3,\dots,2\}$.

Proof idea. Insert the two-defect profile (6) into (9) and difference in k and in a. Two classes collide iff $(k-k')\Delta_k + (a-a')\Delta_a = 0$, which needs $\Delta_k = 0$ or $\Delta_a/\Delta_k = k-k' \in \{-2,\dots,3\}$; trigonometric identities give the stated forms. □

Corollary 4.2 (Pair-census readout; Corollary A.13)**.** *For each of PE-X1, PE-Y1 and PE-X2 at angles satisfying its injectivity condition and $t > 1$, and for all sufficiently small $\beta > 0$, the sorted readout on ranks $n+2,\dots,K_2(n)$ determines the pair census $(N_{a,k})$, hence C_3, C_4 and $D_\diamond$ by (5). Conversely, two cubic graphs with the same pair census have identical two-defect first-order blocks under every degree-prepared member.*

Proof. For the X-mixer members this is Theorem 3.3 at $m = 2$ with (6); for PE-Y1 the injectivity of (12) replaces the extremal-count separation. The converse is immediate from $p_1 = \mu_2 f(a,k)$. □

The converse bounds the first-order block: the pair census does not determine C_5, C_6, the girth or the spectrum; on the $n = 14$ census 2,623 of the 3,171 pairs that share a pair census differ in C_5.

4.2 The three-defect layer at girth at least five

For a three-set D with e internal edges, $E_{\mathrm{in}} = 3,1,0,0$ for $e = 0,1,2,3$ and $E_{\mathrm{out}} = n - 12 + 2e + q_2^{\mathrm{out}} + 3q_3^{\mathrm{out}}$ (Proposition A.14). For a three-set D with no internal edge let $w(D)$ be the number of its pairs at distance two; for $D = \{a,b,c\}$ with the one internal edge ab let $w(D)$ be the number of the two vertices a,b at distance two from c.

Theorem 4.3 (Three defects at girth at least five; Theorem A.16)**.** *Let G be cubic on n vertices with girth at least five, and write C_5, C_6 for its numbers of five- and six-cycles. The two-defect histogram of $(E_{\mathrm{in}}, E_{\mathrm{out}})$ is $(0, n-6)$, $(2, n-7)$, $(2, n-8)$ with multiplicities $3n/2$, $3n$, $\binom{n}{2} - 9n/2$. The three-defect histogram is the one in Table 1; its entries at $(1, n-8)$ and $(3, n-9)$ are $5C_5$ and $n + 2C_6$. The odd-reading histogram of the three-defect layer is likewise a function of (n, C_5, C_6), with $5C_5$ three-sets of one internal edge and outside readings $(-(n-8), -1)$ and $2C_6$ independent three-sets with outside readings $(-(n-9), 0)$.*

Table 1: The three-defect histogram of $(E_{\text{in}}, E_{\text{out}})$ at girth at least five, by the number e of internal edges and the number w of pairs at distance two.

e	$(E_{\text{in}}, E_{\text{out}})$	multiplicity	e	$(E_{\text{in}}, E_{\text{out}})$	multiplicity
2	$(0, n-8)$	$3n$	0, $w=0$	$(3, n-12)$	$\binom{n}{3} - \frac{9}{2}n^2 + 41n - 5C_5 - 2C_6$
1, $w=0$	$(1, n-10)$	$\frac{3}{2}n(n-14) + 5C_5$	0, $w=1$	$(3, n-11)$	$3n^2 - 48n + 10C_5 + 6C_6$
1, $w=1$	$(1, n-9)$	$12n - 10C_5$	0, $w=2$	$(3, n-10)$	$12n - 5C_5 - 6C_6$
1, $w=2$	$(1, n-8)$	$5C_5$	0, $w=3$	$(3, n-9)$	$n + 2C_6$

Proof idea. At girth at least five two vertices have at most one common neighbour and the six vertices at distance two from a vertex are distinct, so $E_{\text{out}} = n - 12 + 2e + w$. A three-set with the one internal edge ab has $w = 2$ exactly when a, b, c lie on a five-cycle with c opposite ab, a bijection with the pairs (five-cycle, edge of it). An independent three-set has $w = 3$ exactly when it is a neighbourhood $N(v)$ or an alternating triple of a six-cycle, and the sums of w and of $\binom{w}{2}$ over independent three-sets are $3n(n-7)$ and $15n - 5C_5$, the latter because the adjacent pairs among the six second neighbours of a vertex correspond to the five-cycles through it. The totals give the table. □

Corollary 4.4 (Girth from three defects; Corollary A.17)**.** *At the schedule, under each of PE-X1, PE-Y1 and PE-X2, for every n and all sufficiently small $\beta > 0$: the first $K_3(n)$ ranks of the sorted readout determine* $\min(\text{girth}, 7)$ *of every cubic graph on n vertices; they determine C_5 and C_6 when the girth is at least five; and when the girth is at least seven they agree through first order, the first $K_3(n)$ sorted entries of two such graphs differing by $O(\beta^2)$.*

Proof. The two-defect block gives C_3 and C_4 (Corollary 4.2), hence whether the girth is 3, 4 or at least 5; in the last case Theorem 3.3 gives the histogram of Theorem 4.3, hence C_5 and C_6, hence whether the girth is 5, 6 or at least 7. At girth at least seven the pair census and both histograms are functions of n. □

Corollary 4.5 (Complementarity of the mixer axes; Corollary A.19)**.** *The PE-X1 value of an outcome is a function of the extremal counts of its profile and the PE-Y1 value a function of its odd readings; given $|D|$ and n the two readings determine the profile, so at the schedule the two values of one outcome determine its profile for every n. Two sorted readouts give the two value multisets and not the pairing by outcome.*

4.3 The cost of reading the two-defect block

A sampled readout sees the block of Corollary 4.2 through multinomial noise. Fix n, a degree-prepared member at angles satisfying its injectivity condition, and $\beta > 0$. Let Δ be the least difference between the class values $f(a,k)$ of two distinct realizable classes, F the difference between the largest and the smallest class value, $F_{\max}$ the largest $|f(a,k)|$, and L_2 the set of outcomes with two defects. The *first-order window* hypothesis at (n, β) is that for every cubic G on n vertices and every $x \in L_2$ of class (a,k),

$$\left|p_G(x;\beta) - \mu_2(1 + \beta f(a,k))\right| \le \tfrac{1}{4}\beta\mu_2\Delta \quad \text{and} \quad \beta(F_{\max} + \tfrac{1}{4}\Delta) \le \tfrac{1}{2}. \tag{14}$$

Proposition 4.6 (Shots for the pair census; Appendix F)**.** *Assume* (14)*, and let the observer receive N independent samples of p_G as labelled bit strings.*

1. *If $N \geq 51\log(n^2/\delta)/(\beta^2\mu_2\Delta^2)$, then assigning every outcome with two defects to the class whose value $\mu_2(1+\beta f(a,k))$ is nearest to its empirical frequency returns the pair census of G with probability at least $1-\delta$.*

2. *Let G and H be cubic graphs on the same n vertices with different pair censuses, $D = \sum_{x\in L_2}(\sqrt{p_G(x)} - \sqrt{p_H(x)})^2$ and $P = \max\{p_G(L_2), p_H(L_2)\}$. A procedure that uses only the shots with two defects and returns the pair census of every cubic graph on n vertices with probability at least $3/4$ needs $N \geq (1-P)/(4D)$ shots. If G and H differ by one edge switch, then $D \leq \beta^2\mu_2[\frac{1}{8}\binom{n}{2}\Delta^2 + (2n-5)(F+\frac{1}{2}\Delta)^2]$.*

Both parts concern an observer who keeps the bit strings; an observer who sees only the sorted empirical vector must first locate the two-defect block, and we do not treat that model. Both bounds carry the factor $1/(\beta^2\mu_2)$, and $1/\mu_2 = c^{-4}s^{-2(n-2)}$ grows as $s^{-2n} = 1.018^n$ at the schedule. The window shrinks with n; on the $n = 14$ census under PE-X1 it is $\beta \leq 7\times10^{-4}$, where the bounds are $N = 2.4\times10^{14}$ at $\delta = 10^{-2}$ and $N \geq 9.0\times10^6$ on a switch pair, and at the operating angle the same scale is 1.2×10^7 (Appendix F).

5 Placement in the Weisfeiler–Leman Hierarchy

Proposition 5.1 (Order-j locality; Proposition A.24)**.** *Fix a member, its angles, n, and $j \geq 0$. There is a function g_j on patterns with $|T| \leq j$, not depending on G, such that for every cubic G on n vertices and every outcome x with defect set D,*

$$[\beta^j]\,\frac{p_G(x;\beta)}{\mu(x)} = \sum_{T\subseteq V\setminus D,\ |T|\leq j} g_j(\pi(D,T)). \tag{15}$$

Proof idea. Expand each mixer factor as $\sum_{S\subseteq V} c_\beta^{n-|S|}(-is_\beta)^{|S|}\sigma^S$. The term with flip sets $S_1,\dots,S_r$ is $O(\beta^{\sum_b|S_b|})$, and its ratio to $\phi(x)$ is a product of preparation and phase ratios over the flipped vertices, a function of the marked graph on $D\cup\bigcup_b S_b$. The order-j coefficient of the probability collects pairs of such terms with at most j flipped vertices in all; grouping the pairs by $T = (S\cup S')\setminus D$ gives (15). □

Lemma 5.2 (Extension counts; Lemma A.23)**.** *Let $\bar u \in V^k$ have $|U|$ distinct entries and let $0 \leq i \leq k-|U|$. Then the round-i colour of $\bar u$ under folklore k-WL determines, for every $i' \leq i$ and every atomic type ρ, the number of i'-tuples of distinct vertices outside U whose atomic type together with $\bar u$ is ρ. For $i' = 1$ the round-one colour determines these numbers for every $\bar u$.*

The proof is an induction on i: the round-i colour records, for every w, the round-$(i-1)$ colour of $\bar u$ with a repeated entry replaced by w, and the replaced entry is still present elsewhere (Appendix A).

Theorem 5.3 (Weisfeiler–Leman placement; Corollary A.26)**.** *Let G and H be cubic graphs on n vertices. Parts (1) and (2) concern one of the degree-prepared members PE-X1, PE-Y1, PE-X2 at fixed angles with $t > 1$; part (3) concerns PE-QX1.*

1. *If one round of folklore $\max(m,2)$-WL does not distinguish G from H, their first-order m-defect blocks coincide. On cubic graphs the pair census and the multiset of colours after one round of folklore 2-WL determine each other, so at angles satisfying the injectivity condition of Theorem 4.1 the first-order two-defect blocks coincide iff one round of folklore 2-WL does not distinguish G from H.*

2. *If G and H are K-WL equivalent, $K \geq 2$, then for every $m \leq K$ the Taylor coefficients of the m-defect layer agree as multisets jointly through order $K - m$, and the sorted m-defect blocks differ by $O(\beta^{K-m+1})$ as $\beta \to 0$.*

3. *Under PE-QX1 there are no layers. If G and H are K-WL equivalent, the multisets over the outcomes with m defects of the Taylor coefficients through order $K - \min(m, n - m)$ coincide, and no statement about ranks follows.*

Proof idea. For (1), the first-order value is a function of the one-vertex extension counts of D by Theorem 3.2, which the round-one colour of an ordering of D determines by Lemma 5.2; the round-one colour of a pair (u, v) records a and the number of common neighbours, which on a cubic graph is all it records. For (2), pad an ordering of D to a K-tuple by repeating its last entry; Lemma 5.2 with $i = K - m$ gives the pattern counts with $|T| \leq K - m$, Proposition 5.1 turns them into the coefficients, and sorting is nonexpansive (Proposition 2.1). For (3), the global spin flip is a symmetry of PE-QX1, and μ is constant, so the outcomes of different defect counts interleave at every $\beta > 0$. □

Proposition 5.4 (A folklore-2-WL-equivalent cubic pair; Appendix E)**.** *Folklore* 2*-WL is injective on the complete connected cubic censuses at* $n = 14$*,* 16*,* 18 *and* 20*. Let G_0 and G_1 be the Cai–Fürer–Immerman pair over K_4: two nonisomorphic cubic graphs on* 40 *vertices, of girth six, that folklore* 2*-WL does not distinguish.*

1. *The first-order histograms of $(E_{\text{in}}, E_{\text{out}})$ and of the odd readings coincide for G_0 and G_1 on every layer with at most six defects. Hence the first-order m-defect blocks of the two sorted readouts coincide for $m \leq 6$ under every degree-prepared member, and so do the first-order sector multisets under PE-QX1.*

2. *Under PE-X1 at the operating angle, the modal outcome of both graphs is 1^{40}, with $p(1^{40}) = 0.7376$; ranks* 1 *to $K_2(40) = 821$ of both sorted readouts are the outcomes with at most two defects; the* 16,231 *outcomes with at most four defects whose probability exceeds the shell bound P_5 are ranks* 1 *to* 16,231 *of both sorted readouts, on which the two sorted vectors agree to 1.7×10^{-18}; and $p_{G_0}(1^{40}) - p_{G_1}(1^{40}) = 6.7957 \times 10^{-30}$, the sign and the first digits certified by outward-rounded interval arithmetic. In particular $p^{\downarrow}_{G_0} \neq p^{\downarrow}_{G_1}$.*

Three facts make part (2) a separation and not an observation: exact joint refinement shows that the pair is 2-WL equivalent, the shell bound of Lemma B.33 and the computed values put 1^{40} at rank one in both graphs, and interval arithmetic certifies that the two rank-one probabilities differ. Hence the sorted readout at the operating angle is not a function of the folklore-2-WL colouring, and level two does not determine every Taylor coefficient of the sorted readout. Part (1) goes beyond Theorem 5.3; for two and three defects Theorem 4.3 explains it, both graphs having girth six and $C_6 = 16$. The modal difference is a trigonometric polynomial in β with period 2π that is not identically zero, so it vanishes at finitely many angles in each period and the two sorted readouts differ at every other angle at which 1^{40} is modal in both graphs; halving β from 0.1 to 0.0125 divides it by 2^{16} at each step to within 0.07 in the exponent, so its first nonzero Taylor coefficient appears to be the sixteenth, where Theorem 5.3(2) guarantees agreement through the second. The rank-one separation is one no sampling resolves, at 10^{58} shots; the computed ranks carry 99.98% of the probability and the two full sorted readouts are within L_1 distance 3.8×10^{-4}, and the number of shots that distinguishes the two full sorted readouts is not known (Appendix E). For every K the construction over cubic base graphs of larger treewidth gives cubic pairs that K-WL does not distinguish (Section 1.4), whose sorted m-defect blocks agree through order $K - m$

by Theorem 5.3(2); so for every m and j some cubic pair has m-defect blocks that agree through order j.

6 Second Order and the Certified Head

The output purity $M_2(G;\beta) = \sum_x p_G(x;\beta)^2 = \|p_G^{\downarrow}(\beta)\|_2^2$ reads the whole distribution at once; primes denote derivatives in β at $\beta = 0$. $M_2(G;0)$ and $M_2'(G;0)$ do not depend on G (Proposition B.6), and graph structure enters at second order through a sum over vertex pairs grouped by class; proofs are in Appendix B.

Theorem 6.1 (Second-purity closures; Theorems B.17 and B.22, Corollary B.23)**.** *For every simple cubic graph G of order n, under PE-X1,*

$$M_2''(G;0) = A_n + b_3 C_3(G) + b_4 C_4(G) + b_D D_\diamond(G), \tag{16}$$

with A_n, b_3, b_4, b_D depending on n, η, γ and not on G, and with interval enclosures at $n = 14$ and the schedule that exclude zero for b_3, b_4 and b_D. Under PE-QX1, $M_2''(G;0) = A_n^H + 4 \cdot 2^{-n} \sin^6(\gamma)\, D_\diamond(G)$ with A_n^H independent of G.

The degree-aware preparation exposes $(C_3, C_4, D_\diamond)$ at second order, which the pair census already determines at first order; the Hadamard preparation exposes $D_\diamond$ alone, so the purity of depth-one MaxCut QAOA on a cubic graph sees, at second order in β, the $K_4 - e$ count and nothing else.

Corollary 6.2 (Cospectral separation at small mixer angle)**.** *Let G and H be adjacency-cospectral simple cubic graphs with $D_\diamond(G) \neq D_\diamond(H)$. Then for all sufficiently small nonzero β, $p_G^{\downarrow} \neq p_H^{\downarrow}$ under PE-X1 whenever $b_D \neq 0$ and under PE-QX1 whenever $\gamma \notin \pi\mathbb{Z}$, because cospectrality fixes C_3 and C_4 ($\operatorname{tr} A^3 = 6C_3$ and $\operatorname{tr} A^4 = 15n + 8C_4$ on cubic graphs, Proposition B.26), so the purities, which depend only on the multiset of probabilities, differ at second order.*

Proposition 6.3 (Certified head for PE-X1; Proposition B.36)**.** *For PE-X1 at the operating angle and $n \in \{14, 16, 18\}$, the first $K_2(n)$ sorted probabilities are exactly the outcomes with at most two defects.*

The statement reduces to four inequalities among finitely many enumerated quantities, certified at the operating angle by outward-rounded interval arithmetic (Appendix B); it is membership in the head, not order within the block, and on the $n = 14$ census the PE-X1 head is lost between $\beta = 0.125$ and 0.5 by two-flip interference and intact again from 0.6 (Appendix C).

7 Verification on the Censuses

The censuses are generated by `nauty-geng` [10]: 19 connected cubic graphs at $n = 10$, 85 at $n = 12$ and 509 at $n = 14$. On them Theorem 3.2 holds on all 2^n outcomes of every graph for all four members, Theorem 4.1 on every two-defect outcome, the histograms of Theorem 4.3 on all 516 census graphs of girth at least five through $n = 18$, and the Hadamard closure of Theorem 6.1 at $n = 14$ and 16, all to within 10^{-13}. On the $n = 14$ census the three-defect block of PE-X1 determines C_5, C_6 and the girth and leaves 19 pairs unresolved, the four-defect block separates every graph under each degree-prepared member, and at the operating angle every member separates every graph and the six $D_\diamond$-varying cospectral pairs of the $n = 18$ census (Appendices C, D and F). The scripts, notebooks and outputs are at `https://github.com/Luke-J-Miller/Structural-Accessibility-in-Phase-Encoded-Quantum-Graph-Embeddings`.

A Proofs of the First-Order Theorems

This appendix proves Theorems 3.2, 3.3, 4.1, 4.3 and 5.3, Corollaries 4.2, 4.4 and 4.5, Lemma 5.2 and Proposition 5.1 of the body, which appear below as Theorems A.4, A.8 with A.21, A.11 and A.16 and Corollary A.26, Corollaries A.13, A.17 and A.19, Lemma A.23 and Proposition A.24; Proposition A.14 gives the three-defect profiles used in Section 4. It uses the flip-ratio lemma of Appendix B (Lemma B.3).

A.1 First-order readout of the defect layers

Here G is a simple cubic graph on n vertices and the conventions of the preceding sections stand: $c = \cos(\eta/2)$, $s = \sin(\eta/2)$, $t = s/c$, $z_v = (-1)^{x_v}$, $D = D(x)$ the defect set, $\mu(x) = c^{2|D|}s^{2(n-|D|)}$, and $R_i(x) = \phi(x \oplus e_i)/\phi(x)$ the flip ratio of Lemma B.3. Numerical statements refer to two notebooks run on the complete cubic censuses at $n = 10, 12, 14$: `formalism_first_order_pair_census_readout.ipynb`, the first run of record, and `formalism_defect_layer_first_order_readout.ipynb`, the second run of record. For a vertex i write

$$k_i(D) = |N(i) \cap D|, \qquad S_i(x) = \sum_{v \in N(i)} z_v = 2k_i(D) - 3, \tag{17}$$

the second equality because every vertex has exactly three neighbours. The *profile* of D is the pair of count vectors

$$q_k^{\mathrm{in}}(D) = \#\{i \in D : k_i(D) = k\}, \qquad q_k^{\mathrm{out}}(D) = \#\{i \notin D : k_i(D) = k\}, \tag{18}$$

for $k = 0, 1, 2, 3$, with $\sum_k q_k^{\mathrm{in}} = |D|$, $\sum_k q_k^{\mathrm{out}} = n - |D|$, $\sum_k k\, q_k^{\mathrm{in}} = 2e(D)$ and $\sum_k k\, q_k^{\mathrm{out}} = 3|D| - 2e(D)$, where $e(D)$ is the number of edges inside D. The *extremal counts* are

$$E_{\mathrm{in}}(D) = q_0^{\mathrm{in}} + q_3^{\mathrm{in}}, \qquad E_{\mathrm{out}}(D) = q_0^{\mathrm{out}} + q_3^{\mathrm{out}}, \tag{19}$$

the numbers of vertices inside and outside D whose neighbourhood lies entirely inside or entirely outside D.

A.1.1 The first-order Born term under each mixer

Write $p(x;\beta) = |\psi(x;\beta)|^2$ with $\psi(\beta) = (U(\beta)E)^r \phi_{\mathrm{prep}}$, where E is the edge-phase diagonal, $U(\beta) = \bigotimes_j e^{-i\beta\sigma_j/2}$ with $\sigma \in \{X, Y\}$ the mixer generator, and $r \geq 1$ the number of tied repetitions; $\phi = E^r\phi_{\mathrm{prep}}$ is the pre-mixer amplitude and $p_0 = |\phi|^2 = \mu$. Put $p_1 = \partial_\beta p|_{\beta=0}$.

Lemma A.1 (Generator form of the first-order term)**.** *Let $\Sigma = \sum_j \sigma_j$. Then*

$$p_1(x) = \mathrm{Im}\Big[\overline{\phi(x)} \sum_{b=1}^{r} (E^{r-b}\, \Sigma\, E^b \phi_{\mathrm{prep}})(x)\Big]. \tag{20}$$

Proof. $U(0) = I$ and $\partial_\beta U|_0 = -\frac{i}{2}\Sigma$. Differentiating the product $(UE)^r$ at $\beta = 0$ replaces one factor U by $-\frac{i}{2}\Sigma$ and every other factor by I; the factor replaced has b copies of E to its right and $r - b$ to its left, $b = 1, \ldots, r$. Hence $\partial_\beta\psi|_0 = -\frac{i}{2}\sum_b E^{r-b}\Sigma E^b\phi_{\mathrm{prep}}$ and $p_1 = 2\,\mathrm{Re}[\overline{\phi}\,\partial_\beta\psi] = \mathrm{Re}[-i\,\overline{\phi}\sum_b(\cdots)] = \mathrm{Im}[\overline{\phi}\sum_b(\cdots)]$. □

Lemma A.2 (Repetition ratios)**.** *For the R_X preparation, with $\phi^{(b)} = E^b\phi_{\mathrm{prep}}$ and x fixed,*

$$\frac{(E^{r-b}X_i\,\phi^{(b)})(x)}{\phi(x)} = R_i^{(b)}(x) := -iz_i\,t^{z_i}\exp\bigl(ib\gamma z_i S_i(x)\bigr), \qquad b = 1,\dots,r. \tag{21}$$

In particular $R_i^{(1)} = R_i$ of Lemma B.3.

Proof. $(E^{r-b}X_i\phi^{(b)})(x) = E(x)^{r-b}E(x\oplus e_i)^b\phi_{\mathrm{prep}}(x\oplus e_i)$ and $\phi(x) = E(x)^r\phi_{\mathrm{prep}}(x)$, so the quotient is $[E(x\oplus e_i)/E(x)]^b\cdot\phi_{\mathrm{prep}}(x\oplus e_i)/\phi_{\mathrm{prep}}(x)$. The preparation quotient is $-iz_it^{z_i}$ and the single-layer phase quotient is $\exp(i\gamma z_iS_i)$, both by the proof of Lemma B.3; raising the latter to the power b gives (21). □

Lemma A.3 (The Y generator)**.** $(Y_i\phi)(x) = -iz_i\,\phi(x\oplus e_i)$.

Proof. $Y|0\rangle = i|1\rangle$ and $Y|1\rangle = -i|0\rangle$, so $\langle x|Y_i|x\oplus e_i\rangle$ equals i when $x_i = 1$ ($z_i = -1$) and $-i$ when $x_i = 0$ ($z_i = +1$); both are $-iz_i$. □

Theorem A.4 (Layer template)**.** *Let G be a simple cubic graph. For every outcome x with defect set D,*

$$\frac{p_1(x)}{p_0(x)} = \frac{1}{t}\sum_{j\notin D}h\bigl(k_j(D)\bigr) - t\sum_{j\in D}h\bigl(k_j(D)\bigr), \tag{22}$$

with the mixer profile $h:\{0,1,2,3\}\to\mathbb{R}$ given by

$$h^{X,r}(k) = \sum_{b=1}^{r}\cos\bigl(b\gamma(2k-3)\bigr) \qquad \textit{(R_X preparation, X mixer, r tied repetitions),} \tag{23}$$

$$h^{Y,1}(k) = \sin\bigl(\gamma(2k-3)\bigr) \qquad \textit{(R_X preparation, Y mixer, one repetition).} \tag{24}$$

For the Hadamard-prepared X-mixer member (one repetition), $p_0 = 2^{-n}$ and

$$p_1(x) = 2^{-n}\Bigl[\sum_{j\in D}\sin\bigl(\gamma(2k_j-3)\bigr) - \sum_{j\notin D}\sin\bigl(\gamma(2k_j-3)\bigr)\Bigr]. \tag{25}$$

Proof. X mixer, r repetitions. By Lemmas A.1 and A.2, $p_1 = \mu\sum_i\sum_{b=1}^{r}\operatorname{Im}R_i^{(b)}$. Expanding $e^{ib\gamma z_iS_i} = \cos(b\gamma S_i) + iz_i\sin(b\gamma S_i)$ (as $z_i = \pm1$),

$$R_i^{(b)} = t^{z_i}\sin(b\gamma S_i) - iz_i\,t^{z_i}\cos(b\gamma S_i), \qquad \operatorname{Im}R_i^{(b)} = -z_it^{z_i}\cos(b\gamma S_i). \tag{26}$$

A defect ($z_i = +1$) contributes $-t\cos(b\gamma S_i)$ and a non-defect ($z_i = -1$) contributes $t^{-1}\cos(b\gamma S_i)$; with $S_i = 2k_i - 3$ and summing over b this is (22) with (23).

Y mixer. By Lemmas A.1 and A.3 with $r = 1$, $p_1 = \operatorname{Im}[\overline{\phi}\sum_i(-iz_i)\phi_i] = -\mu\sum_i z_i\operatorname{Re}R_i = -\mu\sum_i z_it^{z_i}\sin(\gamma S_i)$ by (26), which is (22) with (24).

Hadamard preparation. By Lemma B.19 at $(\theta,\varphi) = (\pi/2, 0)$, $R_i = \exp(i\gamma z_iS_i)$, so $\operatorname{Im}R_i = z_i\sin(\gamma S_i)$ and $\mu = 2^{-n}$; with the X generator, $p_1 = \mu\sum_i\operatorname{Im}R_i$ gives (25). □

Remark A.5. Equation (22) is exact and holds for every outcome, not only on the two-defect layer. It depends on x only through the profile (18): $p_1/p_0 = t^{-1}\langle q^{\mathrm{out}}, h\rangle - t\langle q^{\mathrm{in}}, h\rangle$. The numerical verification (second run of record, exact generator derivative, no finite differences) reports a worst deviation of 9.9×10^{-14} over all 2^n outcomes of every census graph at $n = 10, 12, 14$ for the four members, with the Qiskit construction checked against the circuit construction of record (worst difference 0) and against Richardson differences (7.3×10^{-10}).

A.1.2 Determination by the X-mixer members: extremal counts

Corollary A.6 (Extremal-count form)**.** *For the X-mixer members the profile $h^{X,r}$ is even under $k \mapsto 3-k$, $h(0) = h(3) =: h_{\mathrm{e}}$ and $h(1) = h(2) =: h_{\mathrm{o}}$, where $h_{\mathrm{e}} = \sum_{b\le r} \cos 3b\gamma$ and $h_{\mathrm{o}} = \sum_{b\le r} \cos b\gamma$. Hence for every outcome with $|D| = m$,*

$$\frac{p_1(x)}{p_0(x)} = \frac{(h_{\mathrm{e}} - h_{\mathrm{o}})E_{\mathrm{out}}(D) + (n-m)h_{\mathrm{o}}}{t} - t\big[(h_{\mathrm{e}} - h_{\mathrm{o}})E_{\mathrm{in}}(D) + m\,h_{\mathrm{o}}\big]. \tag{27}$$

Proof. $\langle q, h\rangle = (q_0+q_3)h_{\mathrm{e}} + (q_1+q_2)h_{\mathrm{o}} = (h_{\mathrm{e}} - h_{\mathrm{o}})(q_0+q_3) + h_{\mathrm{o}}\sum_k q_k$ for either count vector; insert the sizes m and $n-m$. □

Lemma A.7 (Transcendence of the layer ratio)**.** *Let η be a nonzero algebraic number with $\cos\eta \neq -1$. Then $t^2 = \tan^2(\eta/2)$ is transcendental. At the schedule, $\eta = 23/8$, so t^2 is transcendental and in particular irrational; and $\gamma = 2$ is not in $\frac{\pi}{2}\mathbb{Z}$, so $h_{\mathrm{e}} \neq h_{\mathrm{o}}$ for $r = 1$.*

Proof. By the Lindemann–Weierstrass theorem $e^{i\eta}$ is transcendental, because $i\eta$ is a nonzero algebraic number. If $\cos\eta$ were algebraic, then $\sin\eta = \pm\sqrt{1-\cos^2\eta}$ would be algebraic and so would $e^{i\eta} = \cos\eta + i\sin\eta$; hence $\cos\eta$ is transcendental. Then $t^2 = (1-\cos\eta)/(1+\cos\eta)$ is transcendental, since $\cos\eta = (1-t^2)/(1+t^2)$ would otherwise be algebraic. The statement about γ holds because π is irrational, so $2/\pi \notin \frac{1}{2}\mathbb{Z}$; and $h_{\mathrm{e}} - h_{\mathrm{o}} = \cos 3\gamma - \cos\gamma = -4\cos\gamma\sin^2\gamma$ vanishes only for $\gamma \in \frac{\pi}{2}\mathbb{Z}$. □

Theorem A.8 (Layer determination, X-mixer members)**.** *Fix $m \ge 1$ and suppose $t > 1$, $h_{\mathrm{e}} \neq h_{\mathrm{o}}$ and that either (a) $t^2 > n-m$ or (b) t^2 is irrational. Then on the m-defect layer the first-order value $p_1(x)/p_0(x)$ determines the pair $(E_{\mathrm{in}}(D), E_{\mathrm{out}}(D))$. Under (a), two outcomes of the layer with different pairs have first-order values that differ by at least $|h_{\mathrm{e}} - h_{\mathrm{o}}|\,t^{-1}\min\{1,\, t^2 - (n-m)\}$. Consequently, for all sufficiently small $\beta > 0$, the sorted readout restricted to the ranks $K_{m-1}(n)+1, \dots, K_m(n)$ determines the histogram of $(E_{\mathrm{in}}, E_{\mathrm{out}})$ over all m-subsets of $V(G)$, where $K_m(n) = \sum_{i\le m}\binom{n}{i}$. At the schedule, (b) holds for every n by Lemma A.7, and (a) holds for $n - m < 55.6$.*

Proof. By (27) two m-subsets with extremal counts $(E_{\mathrm{in}}, E_{\mathrm{out}})$ and $(E'_{\mathrm{in}}, E'_{\mathrm{out}})$ have first-order values that differ by $(h_{\mathrm{e}} - h_{\mathrm{o}})\,t^{-1}\big[(E_{\mathrm{out}} - E'_{\mathrm{out}}) - t^2(E_{\mathrm{in}} - E'_{\mathrm{in}})\big]$, so the values are equal iff $(E_{\mathrm{out}} - E'_{\mathrm{out}}) = t^2(E_{\mathrm{in}} - E'_{\mathrm{in}})$, because $h_{\mathrm{e}} \neq h_{\mathrm{o}}$ and $t \neq 0$. Suppose $E_{\mathrm{in}} \neq E'_{\mathrm{in}}$. Under (a) the right side has modulus at least $t^2 > n-m \ge |E_{\mathrm{out}} - E'_{\mathrm{out}}|$, a contradiction; under (b) the right side is irrational and the left side is an integer, a contradiction. So $E_{\mathrm{in}} = E'_{\mathrm{in}}$ and then $E_{\mathrm{out}} = E'_{\mathrm{out}}$. For the gap under (a): if $E_{\mathrm{in}} = E'_{\mathrm{in}}$ and the pairs differ, the bracket is a nonzero integer; if $E_{\mathrm{in}} \neq E'_{\mathrm{in}}$, the bracket has modulus at least $t^2 - (n-m)$. For the last statement, on the m-defect layer $p(x;\beta) = \mu_m(1 + \beta\,p_1(x)/p_0(x)) + O(\beta^2)$ with $\mu_m = c^{2m}s^{2(n-m)}$ common to the layer, and $\mu_m/\mu_{m-1} = t^{-2} < 1$ is fixed, so for β small enough the layers occupy consecutive rank blocks and the block of layer m, sorted, is the multiset of $\mu_m(1+\beta\,p_1/p_0)$ up to $O(\beta^2)$; the multiset of first-order values is read off, and by the first statement it is the histogram of $(E_{\mathrm{in}}, E_{\mathrm{out}})$. □

Remark A.9. At the schedule $t^2 = \tan^2(23/16) = 55.616$, so hypothesis (a) holds for every layer whenever $n \le 56$ and the gap is $|h_{\mathrm{e}} - h_{\mathrm{o}}|/t$ whenever $n - m \le 54$; $h_{\mathrm{e}} - h_{\mathrm{o}} = \cos 3\gamma - \cos\gamma = 1.376$ for $r = 1$ and $(\cos 3\gamma + \cos 6\gamma) - (\cos\gamma + \cos 2\gamma) = 2.874$ for $r = 2$, so the gaps are 0.1846 and 0.3854. For $r = 1$ the condition $h_{\mathrm{e}} \neq h_{\mathrm{o}}$ is $\gamma \notin \frac{\pi}{2}\mathbb{Z}$. Hypothesis (b) gives determination for every n but no gap bound: the smallest difference between distinct first-order values on a layer is positive at each n and may shrink as n grows, so the range of β in which the sorted readout resolves the histogram shrinks with it. The one-defect layer has a single profile on a regular graph ($E_{\mathrm{in}} = 1$, $E_{\mathrm{out}} = n-4$), so it carries no information; the first informative layer is $m = 2$.

A.1.3 The two-defect layer: the pair census for the three degree-prepared members

Lemma A.10 (Two-defect profiles)**.** *For $D = \{u, v\}$ with $a = \mathbf{1}[uv \in E]$ and $k = |N(u) \cap N(v)|$, $q^{\text{in}} = (2(1-a), 2a, 0, 0)$ and $q^{\text{out}} = (n-8+2a+k,\ 6-2a-2k,\ k,\ 0)$; in particular $E_{\text{in}} = 2(1-a)$ and $E_{\text{out}} = n-8+2a+k$. The seven classes $(a,k) \in \{0,1\} \times \{0,1,2,3\} \setminus \{(1,3)\}$ exhaust the realizable profiles.*

Proof. Each defect has a defect neighbours. Outside D, the common neighbours have $k_j = 2$; the other $3-a-k$ neighbours of u and of v have $k_j = 1$; the rest have $k_j = 0$. Adjacent vertices in a cubic graph share at most two neighbours. □

Theorem A.11 (Two-defect first-order class values)**.** *For a degree-prepared member with profile h (one of (23), (24)), every two-defect outcome of class (a,k) has $p_1(x) = \mu_2 f(a,k)$, $\mu_2 = c^4 s^{2(n-2)}$, with*

$$f(a,k) = \frac{(n-8+2a+k)h(0) + (6-2a-2k)h(1) + k\,h(2)}{t} - 2t\,h(a). \tag{28}$$

It is affine in the class labels,

$$f(a,k) = f(0,0) + k\,\Delta_k + a\,\Delta_a, \qquad \Delta_k = \frac{h(0) - 2h(1) + h(2)}{t}, \tag{29}$$

$$\Delta_a = 2\big(h(0) - h(1)\big)\Big(t + \frac{1}{t}\Big), \tag{30}$$

and takes seven distinct values on the seven classes iff $\Delta_k \neq 0$ and $\Delta_a/\Delta_k \notin \{-2,-1,0,1,2,3\}$. Explicitly, with $A = n-8+2a+k$, $B = 2(1-a)$, $h_{\text{e}} = \cos 3\gamma + \cos 6\gamma$ and $h_{\text{o}} = \cos\gamma + \cos 2\gamma$,

$$f_{\text{X1}}(a,k) = \frac{A\cos 3\gamma + (n-2-A)\cos\gamma}{t} - t\big[B\cos 3\gamma + (2-B)\cos\gamma\big], \tag{31}$$

$$f_{\text{Y1}}(a,k) = \frac{(3k+2a-6)\sin\gamma - A\sin 3\gamma}{t} + 2t\big[a\sin\gamma + (1-a)\sin 3\gamma\big], \tag{32}$$

$$f_{\text{X2}}(a,k) = \frac{A\,h_{\text{e}} + (n-2-A)\,h_{\text{o}}}{t} - t\big[B\,h_{\text{e}} + (2-B)\,h_{\text{o}}\big], \tag{33}$$

with increments

$$\textit{PE-X1:}\quad \Delta_k = \frac{\cos 3\gamma - \cos\gamma}{t} = -\frac{4\cos\gamma\sin^2\gamma}{t}, \qquad \frac{\Delta_a}{\Delta_k} = 2(1+t^2), \tag{34}$$

$$\textit{PE-Y1:}\quad \Delta_k = \frac{3\sin\gamma - \sin 3\gamma}{t} = \frac{4\sin^3\gamma}{t}, \qquad \frac{\Delta_a}{\Delta_k} = -\frac{(1+t^2)\cos 2\gamma}{\sin^2\gamma}, \tag{35}$$

$$\textit{PE-X2:}\quad \Delta_k = \frac{h_{\text{e}} - h_{\text{o}}}{t}, \qquad \frac{\Delta_a}{\Delta_k} = 2(1+t^2). \tag{36}$$

Hence PE-X1 is injective iff $\gamma \notin \frac{\pi}{2}\mathbb{Z}$ and $t^2 \neq \frac{1}{2}$; PE-X2 iff $h_{\text{e}} \neq h_{\text{o}}$ and $t^2 \neq \frac{1}{2}$; PE-Y1 iff $\sin\gamma \neq 0$ and $(1+t^2)\cos 2\gamma/\sin^2\gamma \notin \{-3,\dots,2\}$.

Proof. Insert Lemma A.10 into (22): the inside sum is $2h(a)$ and the outside sum is $\langle q^{\text{out}}, h\rangle$, giving (28). Differencing (28) in k gives $[h(0) - 2h(1) + h(2)]/t$; differencing in a gives $[2h(0) - 2h(1)]/t - 2t[h(1) - h(0)]$, which is Δ_a. Two classes $(a,k) \neq (a',k')$ collide iff $(k-k')\Delta_k + (a-a')\Delta_a = 0$; for $a = a'$ this needs $\Delta_k = 0$, and for $a \neq a'$ it needs $\Delta_a/\Delta_k = k - k'$, where $k \in \{0,1,2,3\}$ is the label of the nonadjacent class and $k' \in \{0,1,2\}$ that of the adjacent one, i.e. a value in $\{-2,\dots,3\}$. The member-specific forms follow from (23)–(24): for the X-mixer members $h(1) = h(2)$ so $\Delta_k = (h(0) - h(1))/t$ and $\Delta_a/\Delta_k = 2t(t + 1/t) = 2(1+t^2)$, which lies in $\{-2,\dots,3\}$

only for $t^2 \in \{-1/2, 0, 1/2\}$; for the Y-mixer member $h(2) = -h(1)$ and $h(0) = -h(3)$, so $\Delta_k = (-\sin 3\gamma + 3\sin\gamma)/t = 4\sin^3\gamma/t$, $\Delta_a = 2(\sin\gamma - \sin 3\gamma)(t + 1/t) = -4\sin\gamma\cos 2\gamma\,(t + 1/t)$, and the ratio is as stated. The trigonometric identities used are $\cos 3\gamma - \cos\gamma = -4\cos\gamma\sin^2\gamma$, $\sin 3\gamma = 3\sin\gamma - 4\sin^3\gamma$ and $1 - 2\sin^2\gamma = \cos 2\gamma$. □

Remark A.12 (Values at the schedule). With $(\eta, \gamma) = (23/8, 2)$: $t = 7.45760$, $t^2 = 55.6158$. The increments are $\Delta_k = 0.18455$, $\Delta_a = 20.897$ (PE-X1); $\Delta_k = 0.40325$, $\Delta_a = 18.049$ (PE-Y1); $\Delta_k = 0.38535$, $\Delta_a = 43.634$ (PE-X2). The ratios Δ_a/Δ_k are 113.2, 44.76 and 113.2, so all three members are injective on the seven classes. The base values at $n = 14$ are $f(0,0) = -13.8834$ (PE-X1), -4.6743 (PE-Y1) and -26.3166 (PE-X2), and $f(0,0)$ grows with n at the rates $\cos 3\gamma/t = 0.1288$, $-\sin 3\gamma/t = 0.0375$ and $(\cos 3\gamma + \cos 6\gamma)/t = 0.2419$ per vertex. The first run of record reproduces (31) exactly on all 46,319 two-defect outcomes of the 509 graphs at $n = 14$ (worst deviation 5.3×10^{-15}) and the empirical PE-Y1 and PE-X2 class centres at $n = 10, 12, 14$ (within-class spread $\le 6.5 \times 10^{-10}$) agree with (32)–(33) to the four printed decimals in all 63 cases.

Corollary A.13 (Pair-census readout)**.** *For each of PE-X1, PE-Y1, PE-X2 at angles satisfying the member's injectivity condition and $t > 1$, and for all sufficiently small $\beta > 0$, the sorted readout on ranks $n + 2, \dots, K_2(n)$ determines the pair census $(N_{a,k})$ and hence $C_3 = \frac{1}{3}\sum_k kN_{1,k}$, $C_4 = \frac{1}{2}\sum_{a,k}\binom{k}{2}N_{a,k}$ and $D_\diamond = \sum_k \binom{k}{2}N_{1,k}$. Conversely, two cubic graphs with the same pair census have identical two-defect first-order blocks under every degree-prepared member.*

Proof. Theorem A.8 with $m = 2$ gives the histogram of $(E_{\text{in}}, E_{\text{out}}) = (2(1-a), n - 8 + 2a + k)$, which is in bijection with (a, k); for PE-Y1 the same argument runs with the injectivity of (32) in place of the extremal-count separation. The converse is immediate from $p_1 = \mu_2 f(a, k)$. □

A.1.4 The three-defect layer

Proposition A.14 (Three-defect profiles)**.** *Let $|D| = 3$ with $e = e(D) \in \{0, 1, 2, 3\}$ internal edges. The in-profile is determined by e,*

$$q^{\text{in}} = (3,0,0,0),\ (1,2,0,0),\ (0,2,1,0),\ (0,0,3,0) \quad \text{for } e = 0, 1, 2, 3, \tag{37}$$

so $E_{\text{in}} = 3, 1, 0, 0$. The out-profile satisfies $q_1^{\text{out}} + 2q_2^{\text{out}} + 3q_3^{\text{out}} = 9 - 2e$ and $\sum_k q_k^{\text{out}} = n - 3$, whence

$$E_{\text{out}} = n - 12 + 2e + q_2^{\text{out}} + 3q_3^{\text{out}}. \tag{38}$$

Under PE-X1 the first-order value of the layer is therefore

$$\frac{p_1}{p_0} = \frac{(\cos 3\gamma - \cos\gamma)(n - 12 + 2e + q_2^{\text{out}} + 3q_3^{\text{out}}) + (n-3)\cos\gamma}{t} - t\big[(\cos 3\gamma - \cos\gamma)E_{\text{in}}(e) + 3\cos\gamma\big], \tag{39}$$

and under PE-Y1

$$\frac{p_1}{p_0} = \frac{(q_3^{\text{out}} - q_0^{\text{out}})\sin 3\gamma + (q_2^{\text{out}} - q_1^{\text{out}})\sin\gamma}{t} - t\,\sigma(e), \tag{40}$$

where $\sigma(e) = -3\sin 3\gamma,\ -\sin 3\gamma - 2\sin\gamma,\ -\sin\gamma,\ 3\sin\gamma$ for $e = 0, 1, 2, 3$. In particular PE-X1 does not separate $e = 2$ from $e = 3$ ($E_{\text{in}} = 0$ in both cases) and reads the out-profile only through $q_2^{\text{out}} + 3q_3^{\text{out}}$, while PE-Y1 separates the four values of e whenever $(n-3)(|\sin 3\gamma| + |\sin\gamma|)/t$ is smaller than half the least gap among $t\,\sigma(0), \dots, t\,\sigma(3)$, and reads the out-profile through $(q_3^{\text{out}} - q_0^{\text{out}},\ q_2^{\text{out}} - q_1^{\text{out}})$.

Proof. The in-profiles are read off the four graphs on three vertices (a vertex of a triangle has two neighbours inside D, not three). The two constraints are the degree sum of D and the size of the complement; eliminating q_1^{out} between them gives (38). Equations (39) and (40) are (27) and (22) specialized, using $\sin(\gamma(2k-3)) = -\sin 3\gamma, -\sin\gamma, \sin\gamma, \sin 3\gamma$ for $k = 0, 1, 2, 3$. For the separation of e under PE-Y1, the out-sum in (40) is bounded by $(n-3)(|\sin 3\gamma| + |\sin\gamma|)/t$ in modulus, so distinct e cannot collide when the inside terms are farther apart than twice this bound. □

Remark A.15. At the schedule $t\,\sigma(e) = 6.25, -11.48, -6.78, 20.34$ for $e = 0, 1, 2, 3$, with least gap 4.70 (between $e = 1$ and $e = 2$), while the out-sum of (40) is at most $(n-3) \cdot 0.1594$, i.e. 1.75 at $n = 14$; the magnitude criterion of Proposition A.14 is therefore met for $n \le 17$, and the PE-Y1 value alone determines e on the three-defect layer at every order run; Theorem A.21 below removes the bound, since the four in-profiles have distinct odd readings $(q_3 - q_0, q_2 - q_1)^{\mathrm{in}} = (-3, 0), (-1, -2), (0, -1), (0, 3)$. On the complete censuses at $n = 10, 12, 14$ (second run of record) every realized three-defect profile is resolved by the pair of values (PE-X1, PE-Y1): 26 profiles at $n = 14$ (eleven with $e = 0$, eight with $e = 1$, five with $e = 2$, two with $e = 3$), of which PE-X1 alone distinguishes 20 and PE-Y1 alone 14. The same holds on the four-defect layer (78 profiles at $n = 14$; 35 and 42). On the census the first-order sorted readout restricted to layers $|D| \le 3$ of PE-X1 determines C_5, C_6 and the girth in addition to the pair census (19 graph pairs share it, none differing in those quantities) and not the adjacency spectrum (18 of the 19 differ); restricted to layers $|D| \le 4$ it is injective on all 509 graphs under each of PE-X1, PE-Y1 and PE-X2 separately. These are census facts and carry no statement beyond $n = 14$; the girth part of the statement holds at every order by Corollary A.17 below.

A.1.5 The three-defect layer at girth at least five

The three-defect histogram of $(E_{\mathrm{in}}, E_{\mathrm{out}})$ is not a subgraph count in general. On cubic graphs of girth at least five it is one. Write g for the girth and C_5, C_6 for the numbers of five- and six-cycles. For a three-set D with $e(D) = 0$ let $w(D)$ be the number of its pairs at distance two, and for $D = \{a, b, c\}$ with the one internal edge ab let $w(D)$ be the number of the two vertices a, b at distance two from c.

Theorem A.16 (Three defects at girth at least five)**.** *Let G be a simple cubic graph on n vertices with $g \ge 5$.*

(a) The two-defect histogram of $(E_{\mathrm{in}}, E_{\mathrm{out}})$ has the entries $(0, n-6)$ with multiplicity $3n/2$, $(2, n-7)$ with multiplicity $3n$ and $(2, n-8)$ with multiplicity $\binom{n}{2} - 9n/2$.

(b) The three-defect histogram of $(E_{\mathrm{in}}, E_{\mathrm{out}})$ is the one in Table 1. In particular its entries at $(1, n-8)$ and at $(3, n-9)$ are $5C_5$ and $n + 2C_6$.

(c) The odd-reading histogram of the three-defect layer is also a function of (n, C_5, C_6): the three-sets with two internal edges all have outside odd readings $(q_3 - q_0, q_2 - q_1)^{\mathrm{out}} = (-(n-8), -5)$; those with one internal edge and outside readings $(-(n-10) - w,\ 3w - 7)$ number m_w, the $(1, n - 10 + w)$ entry of Table 1, for $w = 0, 1, 2$; and the independent three-sets with outside readings $(-(n-12) - w,\ 3w - 9)$ number n_0, $n_1 + n$, n_2 and $2C_6$ for $w = 0, 1, 2, 3$, where n_w is the $(3, n - 12 + w)$ entry of the table. In particular $5C_5$ three-sets have one internal edge and outside readings $(-(n-8), -1)$, and $2C_6$ independent three-sets have outside readings $(-(n-9), 0)$.

Consequently the two-defect histogram depends on n alone; the three-defect histograms of the extremal counts and of the odd readings are functions of (n, C_5, C_6) and each determines C_5 and C_6; and at $g \geq 7$ both depend on n alone.

Proof. Three consequences of $g \geq 5$ are used throughout: there are no triangles and no four-cycles; two distinct vertices have at most one common neighbour, and none if they are adjacent; and for every vertex v the six vertices at distance two from v (two further neighbours of each neighbour) are distinct, are not v, and are not adjacent to v.

(a) For $D = \{u, v\}$ the out-profile is $q^{\text{out}} = (n - 8 + 2a + k,\ 6 - 2a - 2k,\ k,\ 0)$ (Lemma A.10), so $(E_{\text{in}}, E_{\text{out}}) = (2(1 - a),\ n - 8 + 2a + k)$. An adjacent pair has $k = 0$ and there are $3n/2$ of them; a pair at distance two has $k = 1$ and there are $3n$ of them, the number of two-edge paths; every other pair has $k = 0$.

(b) By Proposition A.14, $E_{\text{in}} = 3, 1, 0$ for $e = 0, 1, 2$, there is no three-set with $e = 3$, and $E_{\text{out}} = n - 12 + 2e + q_2^{\text{out}} + 3q_3^{\text{out}}$.

Two internal edges. Let $D = \{a, b, c\}$ with $ab, bc \in E$. No outside vertex is adjacent to two of a, b, c: adjacent to a and b, or to b and c, it would close a triangle, and adjacent to a and c it would be a second common neighbour of a and c besides b. So $q_2^{\text{out}} = q_3^{\text{out}} = 0$ and $(E_{\text{in}}, E_{\text{out}}) = (0, n - 8)$. There are $3n$ such sets, one per two-edge path.

One internal edge. Let $D = \{a, b, c\}$ with $ab \in E$ and c adjacent to neither a nor b. No outside vertex is adjacent to all of D or to both a and b (triangle), so $q_3^{\text{out}} = 0$ and q_2^{out} counts the outside vertices adjacent to c and to exactly one of a, b, which is $w(D)$: a pair at distance two has exactly one common neighbour, and the common neighbour of a, c differs from that of b, c, since a vertex adjacent to a and b would close a triangle. Hence $(E_{\text{in}}, E_{\text{out}}) = (1, n - 10 + w)$ with $w \in \{0, 1, 2\}$. The number of such D is $\frac{3n}{2}(n - 6)$: for each edge ab the vertex c ranges over the complement of $\{a, b\} \cup N(a) \cup N(b)$, a set of $n - 6$ vertices because $N(a) \setminus \{b\}$ and $N(b) \setminus \{a\}$ are disjoint, and each such D has exactly one internal edge. If $w = 2$, let x be the common neighbour of a, c and y that of b, c; then $x \neq y$, and $a\,x\,c\,y\,b$ is a five-cycle with c opposite the edge ab. Conversely, a five-cycle and one of its edges ab determine the opposite vertex c, which is at distance two from a and from b along the cycle and adjacent to neither (a chord of a five-cycle closes a triangle or a four-cycle). The two maps are inverse, since x and y are determined by D as unique common neighbours; so $\#\{D : e = 1, w = 2\} = 5C_5$. For the sum of w over the sets with $e = 1$, fix an edge ab. The vertices at distance two from a are its six second neighbours; two of them are the further neighbours of b, and the other four are not adjacent to b (a vertex c at distance two from a through $x \neq b$ and adjacent to b would lie on the four-cycle $a\,x\,c\,b$). So exactly four admissible c are at distance two from a, four at distance two from b, and $\sum_{D:e=1} w(D) = \frac{3n}{2} \cdot 8 = 12n$. Writing m_w for the number of sets with $e = 1$ and the given w, this gives $m_2 = 5C_5$, $m_1 = 12n - 10C_5$ and $m_0 = \frac{3n}{2}(n - 6) - m_1 - m_2 = \frac{3}{2}n(n - 14) + 5C_5$.

No internal edge. Let $D = \{a, b, c\}$ be independent. If $D = N(v)$ for some v, then $q_3^{\text{out}} = 1$ and $q_2^{\text{out}} = 0$ (a second vertex adjacent to two of D would lie on a four-cycle with v), and $w(D) = 3$; the n neighbourhoods are distinct, since $N(v) = N(v')$ with $v \neq v'$ gives a four-cycle. Otherwise $q_3^{\text{out}} = 0$ and $q_2^{\text{out}} = w(D)$, the outside vertices adjacent to exactly two of D being the common neighbours of the pairs at distance two, one per pair and distinct for distinct pairs. In both cases $E_{\text{out}} = n - 12 + w$, so $(E_{\text{in}}, E_{\text{out}}) = (3, n - 12 + w)$ with $w \in \{0, 1, 2, 3\}$. If $w = 3$ and D is not a neighbourhood, the common neighbours x of a, b, y of b, c and z of c, a are distinct, and $a\,x\,b\,y\,c\,z$ is a six-cycle. Conversely a six-cycle has two alternating triples, each independent (a chord between alternating vertices closes a triangle) with all three pairs at distance two and with no vertex adjacent to all three (such a vertex would close a four-cycle); the maps are inverse because the common neighbours are unique. Hence $n_3 := \#\{D : e = 0, w = 3\} = n + 2C_6$. Next,

$\sum_{D:e=0} w(D) = 3n(n-7)$: there are $3n$ pairs at distance two, and for each the third vertex ranges over the complement of $\{a,c\} \cup N(a) \cup N(c)$, which has $n-7$ elements since $N(a)$ and $N(c)$ share exactly one vertex. Further, $\sum_{D:e=0} \binom{w(D)}{2} = 15n - 5C_5$: a pair of distance-two pairs inside D shares one vertex a, so the sum counts the pairs $\{b,c\}$ of distinct second neighbours of a with $b \not\sim c$, and of the $\binom{6}{2} = 15$ pairs of second neighbours of each a the adjacent ones correspond bijectively to the five-cycles through a (if $b \sim c$ with x the common neighbour of a, b and y that of a, c, then $x \neq y$, as $x = y$ closes a triangle, and $a\,x\,b\,c\,y$ is a five-cycle; conversely a five-cycle through a contains exactly one adjacent pair of second neighbours of a). With n_w the number of independent sets with the given w, the three sums and the total $\binom{n}{3} - 3n - \frac{3n}{2}(n-6)$ of independent three-sets give $n_2 = 15n - 5C_5 - 3n_3 = 12n - 5C_5 - 6C_6$, $n_1 = 3n(n-7) - 2n_2 - 3n_3 = 3n^2 - 48n + 10C_5 + 6C_6$ and $n_0 = \binom{n}{3} - \frac{9}{2}n^2 + 41n - 5C_5 - 2C_6$.

(c) Each count vector satisfies $\sum_k q_k = n-3$ and $\sum_k k\,q_k = 9-2e$. For $e = 2$, $q^{\text{out}} = (n-8, 5, 0, 0)$. For $e = 1$, $q^{\text{out}} = (n-10+w,\ 7-2w,\ w,\ 0)$, whose odd readings are $(-(n-10)-w,\ 3w-7)$, injective in w. For $e = 0$ and D not a neighbourhood, $q^{\text{out}} = (n-12+w,\ 9-2w,\ w,\ 0)$ with readings $(-(n-12)-w,\ 3w-9)$; for $D = N(v)$, $q^{\text{out}} = (n-10, 6, 0, 1)$ with readings $(-(n-11), -6)$, the same as for $w = 1$. The multiplicities follow from (b), the n neighbourhoods moving from the $w = 3$ entry to the $w = 1$ entry.

The consequences: (a) is a function of n; the table and (c) are functions of (n, C_5, C_6); the entries $5C_5$ and $n + 2C_6$ of the table, and the entries $5C_5$ and $2C_6$ of (c), return C_5 and C_6; and $g \geq 7$ means $C_5 = C_6 = 0$. □

Corollary A.17 (Girth from three defects)**.** *At the schedule, under each of PE-X1, PE-Y1 and PE-X2, for every n and all sufficiently small $\beta > 0$, the sorted readout restricted to the first $K_3(n)$ ranks determines* $\min(g, 7)$ *for every simple cubic graph on n vertices; it determines C_5 and C_6 when $g \geq 5$; and when $g \geq 7$ it agrees through first order between graphs, the first $K_3(n)$ sorted entries of two such graphs differing by $O(\beta^2)$.*

Proof. The two-defect block determines the pair census (Corollary A.13), hence C_3 and C_4 by (5), hence whether $g = 3$, $g = 4$ or $g \geq 5$. In the last case the three-defect block determines the extremal-count histogram under PE-X1 and PE-X2 (Theorem A.8, hypothesis (b), which holds at the schedule for every n) and the odd-reading histogram under PE-Y1 (Theorem A.21), hence C_5 and C_6 by Theorem A.16; and $g = 5$ iff $C_5 > 0$, $g = 6$ iff $C_5 = 0 < C_6$, $g \geq 7$ otherwise. When $g \geq 7$ the pair census is $N_{1,0} = 3n/2$, $N_{0,1} = 3n$, $N_{0,0} = \binom{n}{2} - 9n/2$, and the three-defect histograms are those of Theorem A.16 with $C_5 = C_6 = 0$; so the first-order values on the three layers, which are functions of the readings, form multisets that depend on n alone, and so do the first $K_3(n)$ sorted entries up to $O(\beta^2)$. □

Remark A.18. Theorem A.16 says why the census fact of the preceding remark holds and where it stops. On the $n = 14$ census the layers with at most three defects determine C_5, C_6 and the girth; the girth part holds at every order, and the C_5, C_6 part holds at every order on the girth-five class, where the three-defect histogram is a subgraph count, and not on the whole census, where it is not. The theorem also says where the three-defect block stops on that class: two cubic graphs of girth at least five on n vertices with the same C_5 and C_6 have the same first-order readout on the first $K_3(n)$ ranks. At $n = 14$ the nine such graphs take nine values of (C_5, C_6), so the 19 pairs that the PE-X1 layers ≤ 3 leave unresolved all have girth at most four; at $n = 16$ the 49 such graphs take 23 values, and the 46 pairs that share one are first-order indistinguishable through three defects. The smallest cubic graphs of girth seven have 24 vertices, so the statement at $g \geq 7$ is empty below that order. The closed forms were checked by exhaustive enumeration of the two- and three-subsets on every connected cubic graph of girth at least five in the censuses at $n = 10, 12, 14, 16, 18$ (1, 2, 9, 49

and 455 graphs, 516 in all), on 25 random connected cubic graphs of girth five with $20 \le n \le 40$, and on the McGee graph ($n = 24$, $g = 7$) and the Tutte–Coxeter graph ($n = 30$, $g = 8$), with no failure; and on the whole censuses at $n \le 16$ the pair census and the three-defect histogram together never take the same value on two graphs with different $\min(g, 7)$ (`probe_girth5_histograms.py`).

A.1.6 Mixer-axis complementarity

Corollary A.19 (Complementarity)**.** *For the single-repetition members, $h^{X,1}(k) = \cos(\gamma(2k-3))$ is even and $h^{Y,1}(k) = \sin(\gamma(2k-3))$ is odd under $k \mapsto 3-k$. Consequently the PE-X1 value of an outcome is a function of the* even *readings $(E_{\mathrm{in}}, E_{\mathrm{out}})$ of its profile and the PE-Y1 value a function of the* odd *readings $(q_3-q_0,\, q_2-q_1)^{\mathrm{in}}$, $(q_3-q_0,\, q_2-q_1)^{\mathrm{out}}$. Given $|D|$ and n, the even and odd readings together determine the profile* (18)*: for either count vector, q_0+q_3, q_1+q_2 (from the even reading and the size) and q_3-q_0, q_2-q_1 (odd reading) are four independent linear forms in $(q_0, \dots, q_3)$.*

Proof. The parity statements are $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$ with $\theta = \gamma(2k-3)$, $2(3-k)-3 = -(2k-3)$. Then $\langle q, h^X\rangle = (q_0+q_3)\cos 3\gamma + (q_1+q_2)\cos\gamma$ and $\langle q, h^Y\rangle = (q_3-q_0)\sin 3\gamma + (q_2-q_1)\sin\gamma$. The four forms have the invertible matrix of ± 1 entries $\left(\begin{smallmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{smallmatrix}\right)$. □

The corollary says what the two mixers *read*. What their single real values *determine* is Theorem A.8 for the X value, which separates its two integer readings by the magnitude or the irrationality argument, and the following theorem for the Y value, which packs four integers into one real number with the coefficients $\{\sin 3\gamma/t, \sin\gamma/t, t\sin 3\gamma, t\sin\gamma\}$.

Lemma A.20 (Rational independence at the schedule)**.** *At $\eta = 23/8$ and $\gamma = 2$ the four numbers 1, $\sin^2\gamma$, t^2 and $t^2\sin^2\gamma$ are linearly independent over $\mathbb{Q}$.*

Proof. Put $u = e^{i/8}$, which is transcendental by the Lindemann–Weierstrass theorem. Then $\cos 2\gamma = \cos 4 = (u^{32}+u^{-32})/2$, $\sin^2\gamma = (1-\cos 2\gamma)/2$, $\cos\eta = (u^{23}+u^{-23})/2$ and $t^2 = (1-\cos\eta)/(1+\cos\eta)$. A relation $a + b\sin^2\gamma + ct^2 + dt^2\sin^2\gamma = 0$ with rational a, b, c, d is, after multiplication by $1+\cos\eta$, the vanishing of

$$(a + b\sin^2\gamma)\big(2 + u^{23} + u^{-23}\big) + (c + d\sin^2\gamma)\big(2 - u^{23} - u^{-23}\big),$$

a Laurent polynomial in u with rational coefficients, which vanishes at a transcendental number only if it is identically zero. Its coefficients are $2a+b+2c+d$ at u^0, $(a+b/2)-(c+d/2)$ at $u^{\pm 23}$, $-(b+d)/2$ at $u^{\pm 32}$, and $(d-b)/4$ at $u^{\pm 55}$ and $u^{\pm 9}$. The last two give $b = d = 0$, and then the first two give $a = c = 0$. The same computation, run as an exact rank computation on the coefficient matrix, gives rank four (`probe_y_independence.py`); a PSLQ search finds no integer relation with coefficients up to 10^6. □

Theorem A.21 (Layer determination, Y-mixer member)**.** *Suppose $t > 1$, $\sin\gamma \ne 0$ and that 1, $\sin^2\gamma$, t^2 and $t^2\sin^2\gamma$ are linearly independent over $\mathbb{Q}$. Then on every m-defect layer the PE-Y1 first-order value $p_1(x)/p_0(x)$ determines the odd readings $(q_3-q_0,\, q_2-q_1)^{\mathrm{in}}$ and $(q_3-q_0,\, q_2-q_1)^{\mathrm{out}}$ of the profile of D. Consequently, for all sufficiently small $\beta > 0$, the sorted readout restricted to the ranks $K_{m-1}(n)+1, \dots, K_m(n)$ determines the histogram of the odd readings over all m-subsets of $V(G)$. At the schedule the hypotheses hold for every n by Lemma A.20, and together with Theorem A.8 the PE-X1 value and the PE-Y1 value of one outcome determine its profile for every n.*

Proof. By Theorem A.4 with h^Y and the proof of Corollary A.19,

$$\frac{p_1(x)}{p_0(x)} = \frac{(q_3^{\text{out}} - q_0^{\text{out}})\sin 3\gamma + (q_2^{\text{out}} - q_1^{\text{out}})\sin\gamma}{t} - t\big[(q_3^{\text{in}} - q_0^{\text{in}})\sin 3\gamma + (q_2^{\text{in}} - q_1^{\text{in}})\sin\gamma\big].$$

Two profiles with equal values give integers a_1, a_2, a_3, a_4, the differences of their four readings, with $a_1 \sin 3\gamma/t + a_2 \sin\gamma/t - a_3 t \sin 3\gamma - a_4 t \sin\gamma = 0$. Divide by $\sin\gamma/t \neq 0$ and use $\sin 3\gamma/\sin\gamma = 3 - 4\sin^2\gamma$:

$$(3a_1 + a_2) - 4a_1 \sin^2\gamma - (3a_3 + a_4)t^2 + 4a_3\, t^2 \sin^2\gamma = 0.$$

Independence forces $a_1 = 0$, then $a_2 = 0$, then $a_3 = 0$, then $a_4 = 0$, so the readings agree. The sorted statement follows as in the proof of Theorem A.8. The last statement is Corollary A.19: the even readings are $(q_0 + q_3, q_1 + q_2)$ given the sizes, and with the odd readings they determine the profile. □

Remark A.22. Theorem A.21 gives no gap bound. The smallest difference between distinct Y values on a layer is positive at every n and may shrink with n, as under hypothesis (b) of Theorem A.8; the explicit gap of hypothesis (a) has no analogue here. On the censuses at $n = 10, 12, 14$ the realized profiles of the layers $|D| \leq 4$ are resolved by the pair of values (second run of record; Appendix D), in agreement with the theorem.

A.1.7 Placement of each order relative to the Weisfeiler–Leman refinement

This subsection places the Taylor expansion of the Born probabilities in β, order by order, relative to the folklore Weisfeiler–Leman refinement. Fix $k \geq 2$. For $\bar{u} = (u_1, \ldots, u_k) \in V^k$ and $w \in V$ write $\bar{u}[w/i]$ for the tuple with u_i replaced by w, and write $U(\bar{u})$ for the set of entries of $\bar{u}$. The *atomic type* $c_0(\bar{u})$ records, for every pair of positions $i < i'$, whether $u_i = u_{i'}$ and whether $u_i u_{i'} \in E$; for a longer tuple $(\bar{u}, \bar{w})$ the atomic type $c_0(\bar{u}, \bar{w})$ is defined in the same way on all positions. One round of folklore k-WL is

$$c_{\tau+1}(\bar{u}) = \Big(c_\tau(\bar{u}),\ \{\!\{(c_\tau(\bar{u}[w/1]), \ldots, c_\tau(\bar{u}[w/k])) : w \in V\}\!\}\Big), \qquad \tau \geq 0. \tag{41}$$

Colours are nested structures built from atomic types and are therefore comparable across graphs. Two graphs G and H on n vertices are *k-WL equivalent* if for every $\tau \geq 0$ the multisets $\{\!\{c_\tau(\bar{u}) : \bar{u} \in V(G)^k\}\!\}$ and $\{\!\{c_\tau(\bar{v}) : \bar{v} \in V(H)^k\}\!\}$ coincide, and they are *not distinguished by one round* if the two multisets coincide at $\tau = 1$. For $D \subseteq V$ and $T \subseteq V \setminus D$ the *pattern* $\pi(D, T)$ is the isomorphism type of the induced subgraph $G[D \cup T]$ with the two vertex classes D and T distinguished, that is, the class of $(G[D \cup T]; D, T)$ under bijections that preserve adjacency and each class.

Lemma A.23 (Extension counts)**.** *Let $\bar{u} \in V^k$ and $0 \leq i \leq k - |U(\bar{u})|$. Then $c_i(\bar{u})$ determines, for every $i' \leq i$ and every atomic type ρ of $(k + i')$-tuples, the number*

$$N_\rho(\bar{u}) = \#\{\bar{w} \in (V \setminus U(\bar{u}))^{i'} \text{ with distinct entries} :\ c_0(\bar{u}, \bar{w}) = \rho\}. \tag{42}$$

Moreover, for $i' = 1$ the colour $c_1(\bar{u})$ determines $N_\rho(\bar{u})$ for every $\bar{u} \in V^k$, with no condition on $k - |U(\bar{u})|$.

Proof. Induction on i. For $i = 0$ only $i' = 0$ arises and $N_\rho(\bar{u}) = \mathbf{1}[c_0(\bar{u}) = \rho]$. Let $1 \leq i \leq k - |U(\bar{u})|$ and assume the statement for $i - 1$. Since $k > |U(\bar{u})|$, some position ℓ carries an entry that also occurs at another position ℓ'; fix such a pair, read off $c_0(\bar{u})$, which is part of $c_i(\bar{u})$. By (41), $c_i(\bar{u})$ determines the multiset $\{\!\{c_{i-1}(\bar{u}[w/\ell]) : w \in V\}\!\}$ of ℓ-th components of the recorded k-tuples.

The atomic type $c_0(\bar u[w/\ell])$, part of $c_{i-1}(\bar u[w/\ell])$, shows whether w equals an entry at a position other than ℓ, which is whether $w \in U(\bar u)$, because every element of $U(\bar u)$ occurs at a position other than ℓ. Hence $c_i(\bar u)$ determines $M = \{\{c_{i-1}(\bar u[w/\ell]) : w \in V \setminus U(\bar u)\}\}$. For $w \notin U(\bar u)$ the tuple $\bar u[w/\ell]$ has entry set $U(\bar u) \cup \{w\}$, of size $|U(\bar u)| + 1 \le k - (i-1)$, so by the induction hypothesis $c_{i-1}(\bar u[w/\ell])$ determines, for every $i'' \le i-1$ and every atomic type ρ' of $(k+i'')$-tuples, the number of distinct-entry $\bar w' \in (V \setminus (U(\bar u) \cup \{w\}))^{i''}$ with $c_0(\bar u[w/\ell], \bar w') = \rho'$. Given ℓ and ℓ', the atomic types $c_0(\bar u, w, \bar w')$ and $c_0(\bar u[w/\ell], \bar w')$ determine each other: the first is obtained from the second by reinserting at position ℓ an entry equal to the entry at position ℓ', whose equalities and adjacencies are those recorded for position ℓ'. Therefore, for $i' = i''+1 \le i$ and an atomic type ρ of $(k+i')$-tuples with ρ' the corresponding type,

$$N_\rho(\bar u) = \sum_{w \in V \setminus U(\bar u)} \#\{\bar w' :\ c_0(\bar u[w/\ell], \bar w') = \rho'\},$$

and each summand is a function of $c_{i-1}(\bar u[w/\ell])$, so the sum is a function of M. For $i' \le i-1$ the count is determined by $c_{i-1}(\bar u)$, which is part of $c_i(\bar u)$.

For the last statement, $c_1(\bar u)$ determines the multiset over $w \in V$ of $(c_0(\bar u[w/1]), \dots, c_0(\bar u[w/k]))$. Since $k \ge 2$, $c_0(\bar u[w/1])$ records the equalities and adjacencies of w with $u_2, \dots, u_k$ and $c_0(\bar u[w/2])$ those with u_1, so the k-tuple of atomic types together with $c_0(\bar u)$ determines $c_0(\bar u, w)$, and in particular whether $w \in U(\bar u)$. □

Proposition A.24 (Order-j locality)**.** *Fix n, the angles η and γ, a mixer axis $\sigma \in \{X, Y\}$, the number r of tied repetitions, either preparation, and $j \ge 0$. There is a function g_j on patterns with $|T| \le j$, not depending on G, such that for every cubic G on n vertices and every outcome x with defect set D,*

$$[\beta^j]\, \frac{p(x;\beta)}{\mu(x)} = \sum_{\substack{T \subseteq V \setminus D \\ |T| \le j}} g_j(\pi(D,T)). \tag{43}$$

Proof. Write $c_\beta = \cos(\beta/2)$, $s_\beta = \sin(\beta/2)$ and $\sigma^S = \prod_{v \in S} \sigma_v$ for $S \subseteq V$. Each mixer factor expands as $U(\beta) = \prod_v (c_\beta I - i s_\beta \sigma_v) = \sum_{S \subseteq V} c_\beta^{n-|S|} (-i s_\beta)^{|S|} \sigma^S$, so that

$$\psi(x) = \sum_{S_1, \dots, S_r \subseteq V} \Big[\prod_{b=1}^{r} c_\beta^{n-|S_b|} (-i s_\beta)^{|S_b|}\Big] (\sigma^{S_r} E\, \sigma^{S_{r-1}} E \cdots \sigma^{S_1} E\, \phi_{\mathrm{prep}})(x), \tag{44}$$

with S_1 the flip set of the mixer applied first. Put $U_b = S_r \triangle S_{r-1} \triangle \cdots \triangle S_b$ and $x_b = x \oplus e_{U_b}$ for $b = 1, \dots, r$, and $x_{r+1} = x$. Since $(\sigma_X^S \chi)(y) = \chi(y \oplus e_S)$, $(\sigma_Y^S \chi)(y) = \prod_{v \in S} (-i z_v(y))\, \chi(y \oplus e_S)$ and E is diagonal,

$$(\sigma^{S_r} E \cdots \sigma^{S_1} E\, \phi_{\mathrm{prep}})(x) = \omega \prod_{b=1}^{r} E(x_b)\ \phi_{\mathrm{prep}}(x_1), \qquad \omega = \begin{cases} 1, & \sigma = X, \\ \prod_{b=1}^{r} \prod_{v \in S_b} (-i z_v(x_{b+1})), & \sigma = Y. \end{cases}$$

Divide by $\phi(x) = E(x)^r \phi_{\mathrm{prep}}(x)$ and call the quotient $F_{S_1 \cdots S_r}(x)$. It is a product of three kinds of factor. The preparation ratio $\phi_{\mathrm{prep}}(x \oplus e_{U_1})/\phi_{\mathrm{prep}}(x) = \prod_{u \in U_1} (-i z_u t^{z_u})$, equal to 1 under Hadamard preparation, depends on U_1 and on which of its elements lie in D. The phase ratios are

$$\frac{E(x \oplus e_U)}{E(x)} = \exp\Big[i\gamma \sum_{u \in U} \sum_{v \in N(u) \setminus U} z_u z_v\Big], \qquad \sum_{v \in N(u) \setminus U} z_u z_v = z_u \Big[2\,|N(u) \cap (D \setminus U)| + |N(u) \cap U| - 3\Big],$$

because $z_v = +1$ on D and -1 off D and every vertex has three neighbours. The sign factor ω depends on $z_v(x_{b+1}) = z_v(x)(-1)^{[v\in U_{b+1}]}$ for $v \in S_b$. Every quantity named here is a function of the membership of the vertices of $S = S_1 \cup \cdots \cup S_r$ in D and in each S_b, and of the adjacencies between vertices of S and vertices of $D \cup S$. Hence $F_{S_1\cdots S_r}(x)$ is a function of the marked graph $(G[D \cup S]; D, S_1, \ldots, S_r)$, invariant under isomorphisms of marked graphs.

Now $p(x;\beta)/\mu(x) = |\psi(x)/\phi(x)|^2$ is the double sum

$$\sum_{(S_b),(S'_b)} \Big[\prod_b c_\beta^{n-|S_b|}(-is_\beta)^{|S_b|}\Big]\Big[\prod_b c_\beta^{n-|S'_b|}(is_\beta)^{|S'_b|}\Big] F_{S_1\cdots S_r}(x)\,\overline{F_{S'_1\cdots S'_r}(x)},$$

and the term indexed by $(S_b), (S'_b)$ carries the factor $c_\beta^{2rn-N}s_\beta^N$ with $N = \sum_b(|S_b| + |S'_b|)$, whose Taylor coefficients vanish below order N. Writing $\alpha_j(N) = [\beta^j]\, c_\beta^{2rn-N}s_\beta^N$, a number that depends on n and r only,

$$[\beta^j]\,\frac{p(x;\beta)}{\mu(x)} = \sum_{N\le j}\alpha_j(N) \sum_{\substack{(S_b),(S'_b)\\ \sum_b(|S_b|+|S'_b|)=N}} (-i)^{\sum_b|S_b|}\, i^{\sum_b|S'_b|}\, F_{S_1\cdots S_r}(x)\,\overline{F_{S'_1\cdots S'_r}(x)}.$$

Group the terms by $T = (S \cup S') \setminus D$ with $S' = \bigcup_b S'_b$; then $|T| \le N \le j$, and the inner sum over all $(S_b), (S'_b)$ with this T is a sum over tuples of subsets of $D \cup T$ of a function of the marked graph $(G[D \cup T]; D, S_1, \ldots, S_r, S'_1, \ldots, S'_r)$. An isomorphism $(G[D \cup T]; D, T) \cong (G'[D' \cup T']; D', T')$ maps the index set of this sum bijectively onto the corresponding index set for (D', T') and preserves every summand, so the inner sum is a function $g_j(\pi(D, T))$ of the pattern alone, and (43) follows. □

Remark A.25. At $j = 1$ the only terms have a single flipped vertex u in one mixer factor of ψ or of $\overline{\psi}$, with $\alpha_1(1) = \frac{1}{2}$; they give $\sum_u\sum_b \operatorname{Im} R_u^{(b)}(x)$ and recover Theorem A.4, and the pattern $\pi(D, \{u\})$ for $u \notin D$ is $k_u(D)$ together with the pattern of D. The proof uses the cubic hypothesis only through the term -3 in the phase ratio; for d-regular graphs the proposition holds with $-d$ in its place, and so does Corollary A.26(ii) below, whose proof depends on the degree only through the proposition and through the layer structure $\mu(x) = c^{2|D|}s^{2(n-|D|)}$, which the degree-aware preparation gives every regular graph. We have not run the regular-degree case.

Corollary A.26 (Weisfeiler–Leman placement)**.** *Let G and H be cubic graphs on n vertices. Parts (i) and (ii) concern one of the degree-prepared members PE-X1, PE-Y1, PE-X2 at fixed angles with $t > 1$; part (iii) concerns PE-QX1.*

(i) If G and H are not distinguished by one round of folklore k-WL with $k = \max(m, 2)$, then $\{\{p_1(x)/p_0(x) : |D(x)| = m\}\}$ is the same multiset for G and H. On a cubic graph the multiset of colours after one round of folklore 2-WL and the pair census $(N_{a,k})$ determine each other. Hence, at angles satisfying the injectivity condition of Theorem A.11 for the member, the first-order two-defect blocks of G and H coincide iff one round of folklore 2-WL does not distinguish them; at degenerate angles the block is a coarsening of the round-one colouring and only the forward implication holds.

(ii) If G and H are K-WL equivalent, $K \ge 2$, then for every $0 \le m \le K$ the multisets

$$\Big\{\Big\{\big([\beta^j]\,p(x;\beta)/\mu_m\big)_{0\le j\le K-m} : |D(x)| = m\Big\}\Big\}$$

coincide for G and H, and consequently

$$\big\|\mathrm{block}_m(G;\beta) - \mathrm{block}_m(H;\beta)\big\|_\infty = O(\beta^{K-m+1}) \qquad (\beta \to 0), \tag{45}$$

where $\mathrm{block}_m(G;\beta)$ *is the vector of ranks* $K_{m-1}(n)+1,\dots,K_m(n)$ *of the sorted readout* $p_G^{\downarrow}(\beta)$. *In particular the first* $K_m(n)$ *ranks of the two sorted readouts differ by* $O(\beta^{K-m+1})$.

(iii) Under PE-QX1, $\mu(x)=2^{-n}$ *for every outcome and there are no layers. If* G *and* H *are* K*-WL equivalent then for every* m *the multisets, over the outcomes with* m *defects, of the coefficient vectors* $([\beta^j]\,p(x;\beta))_{0\le j\le K-m'}$ *with* $m'=\min(m,n-m)$ *coincide; no statement about the ranks of the sorted readout follows, because the sorted readout of PE-QX1 pools the outcomes of every defect count.*

Proof. (i) By Theorem A.4, $p_1(x)/p_0(x)$ is a function of the profile of D, that is, of the numbers $k_u(D)$ for $u\in D$, which are read off $c_0(\bar d)$ for an ordering $\bar d$ of D, and of the counts $\#\{w\notin D : |N(w)\cap D|=\kappa\}$, $\kappa=0,\dots,3$, each of which is a sum of the one-vertex extension counts $N_\rho(\bar d)$ over the atomic types ρ of $(\bar d,w)$ with $w\notin D$ and κ adjacencies to D. For $m\ge 2$ take $\bar d\in V^m$ with distinct entries; for $m=1$ take $\bar d=(d,d)\in V^2$. By the last statement of Lemma A.23, $c_1(\bar d)$ determines these counts, hence $p_1(x)/p_0(x)$. The multiset over the m-defect outcomes is, up to the constant multiplicity $m!$, the multiset over the k-tuples whose atomic type is that of $\bar d$, and the atomic type is part of the colour; so it is determined by the multiset of round-one colours of all k-tuples, which is the same for G and H.

For the pair census, let $u\neq v$. The colour $c_1(u,v)$ consists of $c_0(u,v)$, which records $a=\mathbf{1}[uv\in E]$, and the multiset over w of $(c_0(w,v),c_0(u,w))$, which records the numbers of $w\notin\{u,v\}$ adjacent to both of u,v, to exactly one, and to neither; on a cubic graph these are k, $2(3-a-k)$ and $n-8+2a+k$. Hence $c_1(u,v)$ is a function of (a,k) and determines (a,k), while the n diagonal tuples (u,u) share one colour. The multiset of round-one colours of V^2 is therefore the diagonal colour with multiplicity n together with the colour of class (a,k) with multiplicity $2N_{a,k}$ for each class, and it and the pair census determine each other. For the last sentence of (i): under the injectivity condition the first-order two-defect block determines the pair census (Corollary A.13), hence the round-one colour multiset; without it, distinct classes may share a value and the block determines only a coarsening of the census.

(ii) Let $1\le m\le K$, let $\bar d=(d_1,\dots,d_m)$ be an ordering of D, and let $\bar d^{+}\in V^K$ be $\bar d$ followed by $K-m$ copies of d_m. Its entry set is D, so Lemma A.23 applies with $i=K-m$: the colour $c_{K-m}(\bar d^{+})$ determines, for every $i'\le K-m$, the number of distinct-entry i'-tuples $\bar w$ outside D of each atomic type $c_0(\bar d^{+},\bar w)$. The atomic type of $(\bar d^{+},\bar w)$ determines the pattern $\pi(D,\{w_1,\dots,w_{i'}\})$, and each set T of size i' arises from exactly $i'!$ tuples; hence $c_{K-m}(\bar d^{+})$ determines the number of $T\subseteq V\setminus D$ of each pattern with $|T|\le K-m$, and by Proposition A.24 the coefficients $[\beta^j]\,p(x;\beta)/\mu_m$ for all $j\le K-m$. For $m=0$ write

$$[\beta^j]\,\frac{p(1^n;\beta)}{\mu_0}=g_j(\pi(\emptyset,\emptyset))+\sum_{v\in V}\sum_{\substack{T\ni v\\ |T|\le j}}\frac{g_j(\pi(\emptyset,T))}{|T|},$$

and note that the inner sum is a function of the numbers of distinct-entry tuples $\bar w$ of length at most $j-1\le K-1$ outside $\{v\}$ by atomic type of $((v,\dots,v),\bar w)$, which $c_{K-1}((v,\dots,v))$ determines by Lemma A.23, the tuple $(v,\dots,v)\in V^K$ having one entry.

The multiset over the ordered m-tuples $\bar d$ of distinct vertices of $c_{K-m}(\bar d^{+})$ is the restriction of $\{\{c_{K-m}(\bar u):\bar u\in V^K\}\}$ to the tuples whose atomic type, part of the colour, is "first m entries distinct, remaining entries equal to the m-th", and each D corresponds to $m!$ such tuples; for $m=0$ it is the restriction to the constant tuples. K-WL equivalence makes these multisets equal for G and H, which proves the first assertion.

For (45), choose $\beta_0(n)>0$ such that for $0<\beta<\beta_0$ the ℓ-defect outcomes occupy ranks $K_{\ell-1}(n)+1,\dots,K_\ell(n)$ of the sorted readout of every cubic graph on n vertices; this is possible

because $p(x;\beta) = \mu_{|D|}(1 + O(\beta))$ with a constant depending on n only (ψ is a trigonometric polynomial in β whose derivatives are bounded in terms of n) and $\mu_{\ell+1}/\mu_\ell = t^{-2} < 1$. Let $\tilde{p}(x;\beta) = \mu_m \sum_{j \le K-m} \beta^j \, [\beta^j]\, p(x;\beta)/\mu_m$ be the truncated expansion on the m-defect layer; then $|p(x;\beta) - \tilde{p}(x;\beta)| \le C\beta^{K-m+1}$ on the layer with C depending on n. The multisets of $\tilde{p}$ over the layer coincide for G and H by the first assertion, so their sorted vectors coincide, and since sorting is nonexpansive in ℓ_∞,

$$\|\mathrm{block}_m(G) - \mathrm{block}_m(H)\|_\infty \le \|\operatorname{sort} p_G - \operatorname{sort} \tilde{p}_G\|_\infty + \|\operatorname{sort} \tilde{p}_H - \operatorname{sort} p_H\|_\infty \le 2C\beta^{K-m+1}.$$

The statement about the first $K_m(n)$ ranks collects the blocks $\ell \le m$, each with error $O(\beta^{K-\ell+1})$.

(iii) Proposition A.24 holds for PE-QX1 with preparation ratio 1, so the argument of (ii) applies to the outcomes with m defects and gives the coefficient multisets through order $K - m$. Under Hadamard preparation the global spin flip $x \mapsto \bar{x}$ preserves the prepared state, the edge phase and the X mixer, so $p(x;\beta) = p(\bar{x};\beta)$ and the multiset over the outcomes with m defects equals the multiset over those with $n-m$ defects; applying the argument to whichever of D and its complement is smaller gives the order $K - \min(m, n-m)$. Since μ is constant, the outcomes of different defect counts are interleaved in the sorted readout at every $\beta > 0$, and no rank statement follows. □

Remark A.27 (What the corollary does and does not say). Parts (i), (ii) and (iii) are upper bounds on the level of the hierarchy that captures a given order: order j of layer m is captured by $(m+j)$-WL, and order one by one round of $\max(m, 2)$-WL. No part asserts that the level is needed, and no part places the readout at a fixed angle, which is the full series, at any level. Part (ii) bounds from below the order at which K-WL-equivalent cubic graphs can first part in the small-angle expansion; it is consistent with such graphs being separated at a finite angle and does not imply it. The pair-census statement of part (i) is exact at nondegenerate angles: on cubic graphs the first-order two-defect block is the first round of folklore 2-WL and nothing more, and the stable 2-WL colouring refines it, strictly so on the $n = 14$ census, where the pair census has 99 classes and 2-WL is injective on all 509 graphs. Part (iii) is about sector-indexed multisets and not about ranks: the Hadamard member has no blocks to compare. Numerically, Proposition A.24 was checked by fitting one linear function of the pattern counts to the exact Taylor coefficients across the complete censuses at $n = 10$ and $n = 12$, for the X and Y mixers at one repetition, on layers $m \le 3$ and orders $j \le 3$ ($j \le 2$ at $n = 12$): the largest residual is 1.3×10^{-12} against coefficient spreads between 0.03 and 237. Corollary A.26 was checked at $n = 10$ by refining all graphs of the census jointly under folklore k-WL for $(m, j, k) \in \{(2,1,2), (3,1,3), (1,2,3), (2,2,4), (3,1,4)\}$ and asserting equal coefficient vectors on equal colours: no violation.

A.1.8 Remarks for the finite-angle sections

Remark A.28 (Slopes and the operating angle). The class value $f(a,k)$ is the logarithmic slope of the two-defect probability at $\beta = 0$. At the schedule the non-adjacent classes of PE-X1 have $f \approx -14$, so βf is of order one already at $\beta \approx 0.07$; the small-β statement of Corollary A.13 is exact only for $\beta \lesssim 10^{-3}$ at $n = 14$ (first run of record: exact pair-census recovery by rounding at $\beta = 0.001$, not at 0.002), while the (a,k) block structure and head membership persist to $\beta = 0.1$ by the full computation. The second run of record maps the deformation on the $n = 14$ census: under PE-X1 the class means of p/μ_2 of the non-adjacent pairs fall from 0.75 at $\beta = 0.02$ to 0.007 at $\beta = 0.2$ and pass through their minimum near $\beta = 0.3$, below the largest three-defect value from $\beta \approx 0.125$ to ≈ 0.5, and the mean top-K_2 overlap with the $\beta = 0$ anchor follows, $106 \to 70.3 \to 65.9 \to 42.0 \to 52.2 \to 67.3 \to 67.6 \to 104.2 \to 106$ over $\beta = 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.5, 0.6$; this reproduces, to the printed decimal, the sequence $106, 106, 106, 65.9, 67.6, 106$ at $\beta = 0, 0.05, 0.1, 0.2, 0.4, 0.8$ that

an independent angle transect of the same census had recorded. The adjacent classes ($f \approx +7$) stay between 2.0 and 3.4 and never leave the head. Under PE-Y1 ($f(0,k) \approx -4.7$) the head stays intact at every angle of the grid. These are grid-resolved observations, not theorems.

Remark A.29 (Higher purity moments). $M_q''(G;0)$ closes exactly on $(1, C_3, C_4, D_\diamond)$ for $q = 2, \dots, 6$ (first run of record, worst residual 2×10^{-14}, $n = 10, 12, 14$), but the 3×3 coefficient matrix over $q = 2, 3, 4$ has singular values 3.7×10^{-5}, 1.0×10^{-9}, 9.3×10^{-14} at $n = 14$ and the $q \geq 4$ coefficients of C_4 and $D_\diamond$ sit below fit resolution. The moment ladder therefore adds nothing resolvable to the second-purity closure; the determination statements are carried by Theorems A.8 and A.11.

B Second-Order Theory and Proofs

B.1 Statements and continuation to the operating angle

With $r = c^2$, $u = s^2$ and $L = r^2 + u^2$,

Proposition B.1 (First-order purity response; Proposition B.6)**.** *For every simple cubic graph G of order n, $M_2'(G;0) = 2nL^{n-4}\Lambda(\eta,\gamma)$, where Λ is an explicit four-spin sum that does not depend on G.*

By Lemma 3.1 each vertex contributes a term that depends on its own spin and the three spins of its neighbours, and the zero-mixer measure factorizes over vertices; every vertex of every cubic graph contributes the same amount. Graph structure enters when two such stars interact, at second order. Differentiating twice gives a sum over unordered vertex pairs grouped by class,

$$M_2''(G;0) = B_{n,3} + \sum_{a,k} N_{a,k}(G)\, F_{a,k}, \tag{46}$$

where $B_{n,3}$ collects graph-independent single-root terms and $F_{a,k}$ is the contribution of a pair of class (a,k). Put $v_a = F_{a,1} - F_{a,0}$ and $\delta_a = F_{a,2} - 2F_{a,1} + F_{a,0}$. The coefficients of Theorem 6.1 are

$$b_3 = 3(v_1 - v_0), \quad b_4 = 2\delta_0, \quad b_D = \delta_1 - \delta_0, \quad A_n = B_{n,3} + \tfrac{3n}{2}F_{1,0} + \left(\tbinom{n}{2} - \tfrac{3n}{2}\right)F_{0,0} + 3nv_0.$$

For nonadjacent pairs the third difference $F_{0,3} - 3F_{0,2} + 3F_{0,1} - F_{0,0}$ vanishes identically (Proposition B.14), so $F_{0,k}$ is quadratic in k on $k \in \{0,1,2,3\}$; adjacent pairs have $k \in \{0,1,2\}$ only, so their kernel is quadratic on its range. Expanding both kernels in the basis $1, k, \binom{k}{2}$ and applying (5) to (46) gives (16). The coefficients scale exactly as L^{n-14} in n.

For the Hadamard member, under a general product preparation $\cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle$ the flip ratio is $e^{i\varphi z_j} t^{z_j} e^{i\gamma z_j S_j}$ (Lemma B.19). At $\varphi = 0$ and $t = 1$ the nonadjacent-pair kernel vanishes identically under the root-spin involution, so $F_{0,k} = 0$ for all k and $b_3 = b_4 = 0$. The adjacent-pair kernel reduces through (5) to the single coordinate $D_\diamond$, and its Laurent numerator in $z = e^{i\gamma}$ is $-(z^2-1)^6/16z^6$, which gives the closed form of Theorem 6.1. At $\gamma = 2$ the coefficient is $b_D^H = 1.3800\text{e}{-4}$ at $n = 14$ and $3.4500\text{e}{-5}$ at $n = 16$. The vanishing is about the equality $r = u$ of the two basis-outcome magnitudes, not about uniformity across vertices: on a regular graph PE-X1 is also uniform across vertices but has $r \neq u$.

For a fixed-order cubic population $\mathcal{G}_n$ write $\overline{f} = |\mathcal{G}_n|^{-1}\sum_{H\in\mathcal{G}_n} f(H)$, and put $q_X(G) = b_3C_3 + b_4C_4 + b_D D_\diamond$ and $q_H(G) = b_D^H D_\diamond$.

Proposition B.2 (Centered purity continuation)**.** *With $q = q_X$ for PE-X1 and $q = q_H$ for PE-QX1, as $\beta \to 0$,*

$$\frac{2}{\beta^2}\left[M_2(G;\beta) - \overline{M_2}(\beta)\right] = q(G) - \overline{q} + O(\beta).$$

Proof. $M_2(G;0)$ and $M_2'(G;0)$ are graph-independent on a fixed-order cubic family (Proposition B.1). Taylor expansion through second order followed by population centering cancels them and the graph-independent constant of the closure. □

At $\beta = 0.1$ on the $n = 14$ census the PE-X1 bases C_3, (C_3, C_4) and $(C_3, C_4, D_\diamond)$ give mean held-out R^2 of 0.999270, 0.999865 and 0.999999902 for the centered purity, and the three-term basis is at least this accurate at every tested angle in $\{0.005, 0.01, 0.02, 0.05, 0.075, 0.1\}$. The fitted C_4 coefficient at $\beta = 0.1$ has the opposite sign from b_4, so the basis continues to the operating angle but the second-derivative coefficients do not transfer numerically. For PE-QX1 the $D_\diamond$-only score decreases from 0.999914 at $\beta = 0.005$ to 0.944076 at $\beta = 0.1$; adding C_3 and C_4 raises it to 0.994124, and the fitted $D_\diamond$ coefficient 1.2911e−4 is within 6.4% of b_D^H.

The higher moments $M_q = \sum_x p_G(x)^q$ also close at second order: $M_q''(G;0)$ is an exact affine function of $(C_3, C_4, D_\diamond)$ for $q = 2, \dots, 6$ on the censuses at $n = 10, 12, 14$ (worst residual 2×10^{-14}). The closure adds nothing resolvable: the 3×3 coefficient matrix over $q = 2, 3, 4$ has singular values 3.7×10^{-5}, 1.0×10^{-9} and 9.3×10^{-14} at $n = 14$, and the $q \geq 4$ coefficients of C_4 and $D_\diamond$ lie below fit resolution (Appendix D, Table 7).

B.2 Proofs

This part proves Proposition B.1 above, Proposition 6.3 and Theorem 6.1 of the body, and the closed-walk identities used by Corollary 6.2, which appear below as Proposition B.6, Proposition B.36, Theorem B.17, Theorem B.22 with Corollary B.23, and Proposition B.26 with Corollary B.27. Its first subsection fixes the conventions, which are those of the body.

B.2.1 Conventions

Let G be a simple cubic graph on n vertices. Write $c = \cos(\eta/2)$, $s = \sin(\eta/2)$, $t = s/c$, $r = c^2$, $u = s^2$ and $L = r^2 + u^2 = c^4 + s^4$. For a computational basis string x put $z_v = (-1)^{x_v}$. The pre-mixer amplitude is

$$\phi(x) = \Big[\prod_v a(x_v)\Big] \exp\Big[-\tfrac{i\gamma}{2} \sum_{uv \in E} z_u z_v\Big], \qquad a(0) = c, \quad a(1) = -is. \tag{47}$$

Defects are the zero bits, $D(x) = \{i : x_i = 0\}$ and $\ell = |D(x)|$; throughout $t > 1$, so that the all-ones string is modal at $\beta = 0$. Then $\mu(x) = |\phi(x)|^2 = c^{2\ell} s^{2(n-\ell)}$ and

$$\mu(x)^2 = \prod_v w(z_v), \qquad w(+1) = r^2, \quad w(-1) = u^2. \tag{48}$$

Equation (48) is a product over *vertices* and carries no graph dependence, because the graph enters (47) only through a phase of unit modulus. This is used repeatedly.

Write $\phi_i(x) = \phi(x \oplus e_i)$, $R_i = \phi_i/\phi$, and $H = \sum_i X_i$. Put

$$\pi_\pm = r^2 e^{\pm i\gamma} + u^2 e^{\mp i\gamma}, \qquad \pi_\pm^{(2)} = r^2 e^{\pm 2i\gamma} + u^2 e^{\mp 2i\gamma}, \qquad P_\pm = \pi_\pm / L, \tag{49}$$

and $S(q) = r^2 e^{i\gamma q} + u^2 e^{-i\gamma q}$, so $S(0) = L$, $S(\pm 1) = \pi_\pm$ and $S(\pm 2) = \pi_\pm^{(2)}$.

B.2.2 The flip ratio

Lemma B.3. *For every vertex i and every x,*

$$R_i(x) = -iz_i\, t^{z_i} \exp\Big(i\gamma z_i \sum_{v\in N(i)} z_v\Big). \tag{50}$$

Proof. The ratio factors into an encoder part and an entangler part. Flipping bit i replaces $a(x_i)$ by $a(1-x_i)$ and leaves every other vertex factor unchanged. If $x_i = 0$, so $z_i = +1$, the ratio is $a(1)/a(0) = -it$; if $x_i = 1$, so $z_i = -1$, it is $a(0)/a(1) = +i/t$. Both equal $-iz_it^{z_i}$. For the entangler, flipping i sends $z_i \mapsto -z_i$ and fixes all other spins, so only edges incident to i change, by $\sum_{v\in N(i)}[(-z_i)z_v - z_iz_v] = -2z_i\sum_{v\in N(i)} z_v$. Multiplying by $-i\gamma/2$ gives the stated exponent. □

Corollary B.4. *R_i depends only on z_i and on the spins of $N(i)$. Edges among the neighbours of i, edges from those neighbours to other vertices, and edges among vertices at distance at least two from i are not incident to i, so their phases are identical in ϕ_i and ϕ and cancel in the ratio.*

Corollary B.5. *If i and j are* nonadjacent*, $R_{ij} = R_iR_j$.*

Proof. The encoder factors act on distinct vertices. The edge sets incident to i and to j are disjoint, since $ij \notin E$; a common neighbour w carries the two distinct edges iw and jw, each altered by one flip only. For adjacent roots the edge ij is incident to both and the factorization acquires a correction, which is why the nonadjacent case is treated separately. □

B.2.3 The first-order purity response

Proposition B.6. *For every simple cubic graph G of order n,*

$$M_2'(G;0) = 2n\, L^{\,n-4}\,\Lambda(\eta,\gamma), \qquad \Lambda = \sum_{\sigma,\xi_1,\xi_2,\xi_3\in\{\pm1\}} w(\sigma)\prod_{q=1}^{3} w(\xi_q)\ \mathrm{Im}\Big[-i\sigma t^{\sigma} e^{i\gamma\sigma(\xi_1+\xi_2+\xi_3)}\Big]. \tag{51}$$

In particular $M_2'(G;0)$ does not depend on G.

Proof. From $M_2(\beta) = \sum_x p(x;\beta)^2$ we get $M_2'(0) = 2\sum_x p_0p_1$ with $p_0 = \mu$ and $p_1 = \mathrm{Im}(\overline{\phi}\,H\phi)$. Since $(H\phi)(x) = \sum_i \phi_i(x)$, $\overline{\phi}\,H\phi = \mu\sum_i R_i$, hence

$$M_2'(0) = 2\sum_x \mu(x)^2 \sum_i \mathrm{Im}\, R_i(x) = 2\sum_i S_i, \qquad S_i = \sum_x \mu(x)^2\,\mathrm{Im}\, R_i(x). \tag{52}$$

By Lemma B.3 and Corollary B.4, the summand of S_i depends on x only through z_i and the three spins of $N(i)$, and in particular edges among those neighbours do not enter. In a simple cubic graph the four vertices $\{i\}\cup N(i)$ are distinct. By (48) the measure factorizes over vertices, so the remaining $n-4$ spins sum to $(r^2+u^2)^{n-4} = L^{\,n-4}$ and the residual four-spin sum is exactly Λ. Every vertex of every simple cubic graph on n vertices therefore yields the same $S_i = L^{\,n-4}\Lambda$. □

Remark B.7. Graph structure cannot enter before two stars interact, which happens first at order β^2. This is why the motif content of the purity expansion begins at second order.

B.2.4 The third-difference identity

Let i, j be nonadjacent with codegree $k = |N(i) \cap N(j)|$, so $k \in \{0, 1, 2, 3\}$ on a cubic graph. The pair kernel is

$$F_{0,k} = \sum_x \Big[4p_i p_j + 2p_0 \operatorname{Re}(\phi_i \overline{\phi_j}) - 2p_0 \operatorname{Re}(\overline{\phi}\, \phi_{ij}) \Big], \qquad p_i = \operatorname{Im}(\overline{\phi}\, \phi_i). \tag{53}$$

The local motif has $2 + k + 2(3 - k) = 8 - k$ vertices, so $F_{0,k}$ carries a spectator factor L^{n-8+k} whose exponent depends on k.

Lemma B.8. $F_{0,k} = 4 \sum_x \mu(x)^2 [\operatorname{Re}(R_i \overline{R_j}) - \operatorname{Re}(R_i R_j)]$.

Proof. $p_i = \mu \operatorname{Im} R_i$, and by Corollary B.5 $\phi_i \overline{\phi_j} = \mu R_i \overline{R_j}$ and $\overline{\phi}\, \phi_{ij} = \mu R_i R_j$. The integrand of (53) becomes $\mu^2[4 \operatorname{Im} R_i \operatorname{Im} R_j + 2 \operatorname{Re}(R_i \overline{R_j}) - 2 \operatorname{Re}(R_i R_j)]$, and $4 \operatorname{Im} A \operatorname{Im} B = 2 \operatorname{Re}(A\overline{B}) - 2 \operatorname{Re}(AB)$ doubles the last two terms. □

Lemma B.9. *Write $\alpha_\sigma = -i\sigma t^\sigma$. For fixed root spins $\sigma = z_i$, $\tau = z_j$, the contribution of that sector to $F_{0,k}$ is*

$$\mathcal{K}_{\sigma,\tau}(k) = 4\, w(\sigma) w(\tau)\, L^{n-8+k} \operatorname{Re}\Big[\alpha_\sigma \overline{\alpha_\tau}\, \pi_\sigma^{3-k} \pi_{-\tau}^{3-k} S(\sigma - \tau)^k - \alpha_\sigma \alpha_\tau\, \pi_\sigma^{3-k} \pi_\tau^{3-k} S(\sigma + \tau)^k \Big], \tag{54}$$

and $F_{0,k} = \sum_{\sigma,\tau \in \{\pm 1\}} \mathcal{K}_{\sigma,\tau}(k)$.

Proof. By Lemma B.3, $R_i = \alpha_\sigma \exp(i\gamma\sigma \sum_{v \in N(i)} z_v)$ and likewise for j. In $R_i \overline{R_j}$ each neighbour spin occurs once with an integer coefficient: a private neighbour of i with coefficient σ, a private neighbour of j with $-\tau$, a shared neighbour with $\sigma - \tau$, and a spectator with 0. Because (48) is a product measure and, by Corollary B.4, no edge fails to cancel unless incident to a root, the sum over the non-root spins factorizes, contributing $\sum_\zeta w(\zeta) e^{i\gamma q \zeta} = S(q)$ per vertex with coefficient q. There are $3 - k$ private neighbours on each root, k shared, and $n - 8 + k$ spectators, giving π_σ^{3-k}, $\pi_{-\tau}^{3-k}$, $S(\sigma - \tau)^k$ and L^{n-8+k}. The term $R_i R_j$ is identical except that private neighbours of j carry $+\tau$ and shared neighbours carry $\sigma + \tau$. The roots contribute $w(\sigma)w(\tau)$ and the α factors, and the overall 4 is from Lemma B.8. □

Lemma B.10. *Put $A_k = \operatorname{Re}(\pi_+^{6-2k} (\pi_+^{(2)})^k)$ and $B_k = (\pi_+ \pi_-)^{3-k} L^k$, both real. Then*

$$F_{0,k} = 4 L^{n-8+k}\, ru(r - u)^2\, (A_k + B_k). \tag{55}$$

Proof. Both α products are real: $\alpha_\sigma \overline{\alpha_\tau} = \sigma\tau\, t^{\sigma+\tau}$ and $\alpha_\sigma \alpha_\tau = -\sigma\tau\, t^{\sigma+\tau}$. In the aligned sector $(+,+)$ the first term of Lemma B.9 is $t^2 (\pi_+ \pi_-)^{3-k} L^k = t^2 B_k$, already real, and the second contributes $t^2 A_k$ after taking the real part, giving $4r^4 t^2 (A_k + B_k)$. The sector $(-,-)$ is its conjugate with weight u^4 and factor t^{-2}. In the anti-aligned sector $(+,-)$ we have $\alpha_+ \overline{\alpha_-} = -1$ and $\alpha_+ \alpha_- = +1$, and the real part is $-(A_k + B_k)$, giving $-4r^2 u^2 (A_k + B_k)$; the sector $(-,+)$ gives exactly the same value. Summing the four and using $t^2 = u/r$,

$$r^4 t^2 + u^4 t^{-2} - 2r^2 u^2 = r^3 u + r u^3 - 2r^2 u^2 = ru(r - u)^2. \tag{56}$$

□

Lemma B.11. *With $b_1 = L\pi_+^{(2)}/\pi_+^2$, $b_2 = \overline{b_1}$, $b_3 = L^2/(\pi_+ \pi_-)$, $\kappa = 2ru(r-u)^2/L^2$, $\Gamma_1 = \kappa P_+^6$, $\Gamma_2 = \kappa P_-^6$ and $\Gamma_3 = 2\kappa (P_+ P_-)^3$,*

$$F_{0,k} = L^n (\Gamma_1 b_1^k + \Gamma_2 b_2^k + \Gamma_3 b_3^k). \tag{57}$$

Proof. Expand against (55). Using $\kappa = 2ru(r-u)^2/L^2$,

$$L^n \Gamma_1 b_1^k = L^n \kappa \frac{\pi_+^6}{L^6} \cdot \frac{L^k (\pi_+^{(2)})^k}{\pi_+^{2k}} = 2ru(r-u)^2 L^{n-8+k} \pi_+^{6-2k} (\pi_+^{(2)})^k, \tag{58}$$

and its conjugate for Γ_2, so the two sum to $4ru(r-u)^2 L^{n-8+k} A_k$. Similarly $L^n \Gamma_3 b_3^k = 4ru(r-u)^2 L^{n-8+2k} (\pi_+\pi_-)^{3-k} = 4ru(r-u)^2 L^{n-8+k} B_k$. □

Remark B.12. The multiplicity two on Γ_3 enters once, in Lemma B.10, as the two anti-aligned root-spin configurations $(+,-)$ and $(-,+)$ contributing identically. It is not an independent assumption.

Lemma B.13. *With $\Delta = \cos 2\gamma - 1$, $L\pi_+^{(2)} - \pi_+^2 = 2r^2u^2\Delta$ and $L^2 - \pi_+\pi_- = -2r^2u^2\Delta$, hence*

$$b_1 - 1 = \frac{2r^2u^2\Delta}{\pi_+^2}, \qquad b_2 - 1 = \frac{2r^2u^2\Delta}{\pi_-^2}, \qquad b_3 - 1 = \frac{-2r^2u^2\Delta}{\pi_+\pi_-}. \tag{59}$$

Proof. $L\pi_+^{(2)} = r^4 e^{2i\gamma} + u^4 e^{-2i\gamma} + r^2u^2(e^{2i\gamma} + e^{-2i\gamma})$ while $\pi_+^2 = r^4 e^{2i\gamma} + u^4 e^{-2i\gamma} + 2r^2u^2$, and the difference is $r^2u^2(e^{2i\gamma} + e^{-2i\gamma} - 2) = 2r^2u^2\Delta$. Also $\pi_+\pi_- = r^4 + u^4 + r^2u^2(e^{2i\gamma} + e^{-2i\gamma})$ and $L^2 = r^4 + u^4 + 2r^2u^2$. □

Proposition B.14. *For nonadjacent roots in a simple cubic graph, $F_{0,3} - 3F_{0,2} + 3F_{0,1} - F_{0,0} = 0$; equivalently $F_{0,k}$ is exactly quadratic in k over the admissible range $k \in \{0,1,2,3\}$.*

Proof. The third forward difference of b^k at $k=0$ is $(b-1)^3$, so by (57) the third difference of $F_{0,k}$ is $L^n \sum_j \Gamma_j (b_j - 1)^3$. By Lemma B.13 every Laurent factor $\pi_\pm^6$ in Γ_j cancels the $\pi_\pm^{-6}$ in $(b_j-1)^3$:

$$\Gamma_1(b_1-1)^3 = \Gamma_2(b_2-1)^3 = \frac{8\kappa r^6 u^6 \Delta^3}{L^6}, \qquad \Gamma_3(b_3-1)^3 = -\frac{16\kappa r^6 u^6 \Delta^3}{L^6}, \tag{60}$$

and the sum is $8\kappa r^6 u^6 \Delta^3 L^{-6}[1+1-2] = 0$. □

Remark B.15 (Removable singularities). Lemmas B.11 and B.13 are stated where $\pi_+\pi_- \neq 0$. Since $\pi_+\pi_- = r^4 + u^4 + 2r^2u^2\cos 2\gamma$ and $r/u + u/r \geq 2$, this vanishes only if $r = u$ and $\gamma \in \frac{\pi}{2} + \pi\mathbb{Z}$. By Lemma B.8, $F_{0,k}$ is a finite spin sum, hence a trigonometric polynomial in γ for each fixed η; a trigonometric polynomial vanishing off a finite set vanishes identically, so Proposition B.14 extends to those points by continuity. At $r = u$ one has $\kappa = 0$, all Γ_j vanish and $F_{0,k}$ is independent of k.

B.2.5 The cubic second-purity motif closure

Lemma B.16. *Let $H = \sum_i X_i$. Then*

$$M_2''(G;0) = B_{n,3} + \sum_{a,k} N_{a,k}(G)\, F_{a,k}, \tag{61}$$

where $N_{a,k}(G)$ counts unordered vertex pairs with adjacency a and codegree k, the kernel $F_{a,k}$ is (53)*, and*

$$B_{n,3} = 2\sum_x \mu(x)^2 \Big[\sum_i (\operatorname{Im} R_i)^2 + \tfrac{1}{2} \sum_i |R_i|^2 - \tfrac{n}{2} \Big] \tag{62}$$

is graph independent on the cubic class.

Proof. Expanding $\psi = e^{-i\beta H/2}\phi$ to second order gives $p = p_0 + \beta p_1 + \beta^2 p_2 + O(\beta^3)$ with $p_0 = \mu$, $p_1 = \mathrm{Im}(\overline{\phi} H\phi)$ and $p_2 = \frac{1}{2}(|H\phi|^2 - \mathrm{Re}(\overline{\phi} H^2\phi))$, whence $M_2''(0) = 2\sum_x (p_1^2 + p_0 p_2)$. Since $H\phi = \sum_i \phi_i$ and $H^2\phi = \sum_{i,j} \phi_{ij}$ with $\phi_{ii} = \phi$,

$$p_1 = \mu \sum_i \mathrm{Im}\, R_i, \qquad p_0 p_2 = \tfrac{1}{2}\mu^2 \Big[\sum_{i,j} \mathrm{Re}(R_i \overline{R_j}) - n - \sum_{i \neq j} \mathrm{Re}\, R_{ij} \Big]. \tag{63}$$

Separating the diagonal $i = j$ from the off-diagonal terms collects all single-root contributions into $B_{n,3}$ and leaves

$$\sum_{i \neq j} \mu^2 \Big[2\,\mathrm{Im}\, R_i\, \mathrm{Im}\, R_j + \mathrm{Re}(R_i \overline{R_j}) - \mathrm{Re}\, R_{ij} \Big] = \sum_{i<j} \mu^2 \Big[4\,\mathrm{Im}\, R_i\, \mathrm{Im}\, R_j + 2\,\mathrm{Re}(R_i \overline{R_j}) - 2\,\mathrm{Re}\, R_{ij} \Big], \tag{64}$$

each summand being symmetric in i and j. Summing over x gives exactly $\sum_{i<j} F_{a_{ij},k_{ij}}$.

That $F_{a,k}$ depends on the pair only through (a,k) follows from Corollary B.4: no edge fails to cancel from R_i or R_j unless incident to a root, and μ^2 is a product measure over vertices by (48), so the summand is determined by the local configuration of the two roots, which on a cubic graph is fixed by adjacency and codegree. The same two facts make each single-root term depend only on a vertex and its three neighbours, so $B_{n,3}$ is graph independent by the argument of Proposition B.6. □

Theorem B.17 (Cubic second-purity motif closure)**.** *For every simple cubic graph G of order n, and for angles in the domain of the kernel formulas,*

$$M_2''(G;0) = A_n + b_3 C_3(G) + b_4 C_4(G) + b_D D_\diamond(G), \tag{65}$$

where the coefficients depend on (n, η, γ) but not on G. With

$$v_a = F_{a,1} - F_{a,0}, \qquad \delta_a = F_{a,2} - 2F_{a,1} + F_{a,0}, \tag{66}$$

one has

$$b_3 = 3(v_1 - v_0), \qquad b_4 = 2\delta_0, \qquad b_D = \delta_1 - \delta_0. \tag{67}$$

Proof. Start from Lemma B.16. For nonadjacent pairs, Proposition B.14 states that $F_{0,k}$ is exactly quadratic in k on the admissible range, so $F_{0,k} = F_{0,0} + kv_0 + \binom{k}{2}\delta_0$ for $k \in \{0,1,2,3\}$. An adjacent pair in a cubic graph has $k \in \{0,1,2\}$, three points, so the quadratic through them gives $F_{1,k} = F_{1,0} + kv_1 + \binom{k}{2}\delta_1$ identically on that range. Substituting both into (61),

$$\sum_{a,k} N_{a,k} F_{a,k} = \sum_a F_{a,0} \sum_k N_{a,k} + \sum_a v_a \sum_k k N_{a,k} + \sum_a \delta_a \sum_k \binom{k}{2} N_{a,k}. \tag{68}$$

On a cubic graph the pair census satisfies

$$\sum_k N_{1,k} = \tfrac{3n}{2}, \qquad \sum_k N_{0,k} = \binom{n}{2} - \tfrac{3n}{2}, \qquad \sum_{a,k} k N_{a,k} = \sum_v \binom{3}{2} = 3n, \tag{69}$$

together with the incidence identities $\sum_k k N_{1,k} = 3C_3$, $\sum_{a,k} \binom{k}{2} N_{a,k} = 2C_4$ and $\sum_k \binom{k}{2} N_{1,k} = D_\diamond$. Hence $\sum_k k N_{0,k} = 3n - 3C_3$ and $\sum_k \binom{k}{2} N_{0,k} = 2C_4 - D_\diamond$, and collecting terms gives

$$M_2''(G;0) = A_n + 3(v_1 - v_0)C_3 + 2\delta_0 C_4 + (\delta_1 - \delta_0) D_\diamond, \tag{70}$$

with

$$A_n = B_{n,3} + \tfrac{3n}{2} F_{1,0} + \Big(\binom{n}{2} - \tfrac{3n}{2} \Big) F_{0,0} + 3n v_0, \tag{71}$$

which involves no property of G beyond its order. □

Remark B.18. The theorem is stated for angles in the domain of the kernel formulas. Exceptional angles at which a coefficient vanishes are permitted; at isolated zeros of a carrier denominator the coefficients are defined by continuous extension of the finite spin sum, as in the remark following Proposition B.14. Nonvanishing at the schedule is a separate certified statement.

B.2.6 Phase-general preparation and the two specializations

The preceding results assume the R_X preparation. The family study requires the Hadamard-prepared member, so this section carries the derivation at general product preparation

$$|\chi(\theta,\varphi)\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi}\sin(\theta/2)|1\rangle, \tag{72}$$

which covers $R_X(\eta)|0\rangle = |\chi(\eta,-\pi/2)\rangle$ and $|+\rangle = |\chi(\pi/2,0)\rangle$. Write $c=\cos(\theta/2)$, $s=\sin(\theta/2)$ and $t=s/c$.

Lemma B.19 (Phase-general flip ratio)**.** $R_i(x) = e^{i\varphi z_i} t^{z_i} \exp\bigl(i\gamma z_i \sum_{v\in N(i)} z_v\bigr)$.

Proof. The encoder factor is $a(1)/a(0) = e^{i\varphi}t$ when $z_i=+1$ and $a(0)/a(1) = e^{-i\varphi}t^{-1}$ when $z_i=-1$, both equal to $e^{i\varphi z_i}t^{z_i}$. The entangler factor is that of Lemma B.3, since it does not involve the preparation. Setting $\varphi=-\pi/2$ recovers $-iz_it^{z_i}$, because $(-i)^{z_i} = -iz_i$ for $z_i=\pm1$. □

Corollary B.20. *With $\alpha_\sigma = e^{i\varphi\sigma}t^\sigma$,*

$$\alpha_\sigma\overline{\alpha_\tau} = e^{i\varphi(\sigma-\tau)}t^{\sigma+\tau}, \qquad \alpha_\sigma\alpha_\tau = e^{i\varphi(\sigma+\tau)}t^{\sigma+\tau}. \tag{73}$$

At $\varphi=-\pi/2$ these differ by a sign in every spin sector; at $\varphi=0$ they are equal. The preparation phase, not the amplitude magnitude, distinguishes the specializations.

Lemma B.21 (Three-exponent reduction)**.** *For any vertex pair,*

$$F_{a,k} = \sum_x \mu^2\Bigl[4\,\mathrm{Re}(R_i\overline{R_j}) - 2\,\mathrm{Re}(R_iR_j) - 2\,\mathrm{Re}\,R_{ij}\Bigr]. \tag{74}$$

Writing $\sigma=z_i$, $\tau=z_j$ and $\Sigma_{\mathrm{sh}},\Sigma_i,\Sigma_j$ for the shared and private neighbour spin sums, the three exponents of $e^{i\gamma\cdot}$ are

$$A = (\sigma-\tau)\Sigma_{\mathrm{sh}} + \sigma\Sigma_i - \tau\Sigma_j, \qquad C = (\sigma+\tau)\Sigma_{\mathrm{sh}} + \sigma\Sigma_i + \tau\Sigma_j, \tag{75}$$

with R_iR_j carrying $B = C + 2\sigma\tau\mathbf{1}[ij\in E]$ and R_{ij} carrying C.

Proof. Substitute $p_i = \mu\,\mathrm{Im}\,R_i$ into (53) and apply $4\,\mathrm{Im}\,A\,\mathrm{Im}\,B = 2\,\mathrm{Re}(A\overline{B}) - 2\,\mathrm{Re}(AB)$. For a nonadjacent pair Corollary B.5 gives $R_{ij}=R_iR_j$, so $B=C$, the last two terms merge and Lemma B.8 is recovered. For an adjacent pair Corollary B.5 fails and the three terms remain distinct; the $\sigma\tau$ contributions of the edge ij cancel out of A and out of C. □

Theorem B.22 (Hadamard vanishing)**.** *For the preparation $|\chi(\pi/2,0)\rangle$ the nonadjacent pair kernel vanishes identically: $F_{0,k}=0$ for every codegree k and every γ.*

Proof. At $\theta=\pi/2$ the measure is uniform, $w(+1)=w(-1)=\frac14$, so μ^2 is constant; at $\varphi=0$ and $t=1$, Corollary B.20 makes both prefactors equal to one. For a nonadjacent pair $B=C$, so by Lemma B.21 the integrand is $4\mu^2[\cos(\gamma A)-\cos(\gamma C)]$.

Flipping the single root spin z_j preserves the uniform measure and the unit prefactors, and sends $A\mapsto C$ and $C\mapsto A$. Summing over all spin assignments therefore gives $\sum\cos(\gamma A) = \sum\cos(\gamma C)$, so the kernel vanishes. For an adjacent pair the involution still exchanges A and C, but $B = C + 2\sigma\tau$ is unpaired and the residue does not cancel. □

Corollary B.23 (Diamond-only closure)**.** *For the Hadamard-prepared, single-layer X-mixer circuit on simple cubic graphs,*

$$M_2''(G;0) = A_n + b_D\, D_\diamond(G), \qquad b_D = 4\sin^6\gamma \cdot 2^{-n}, \tag{76}$$

and $b_3 = b_4 = 0$. In particular $b_D \neq 0$ for every $\gamma \notin \pi\mathbb{Z}$.

Proof. Theorem B.22 gives $F_{0,k} \equiv 0$, hence $v_0 = \delta_0 = 0$ and therefore $b_4 = 2\delta_0 = 0$. Direct evaluation of the adjacent kernels gives $F_{1,0} = F_{1,1}$, so $v_1 = 0$ and $b_3 = 3(v_1 - v_0) = 0$. The remaining coefficient is $b_D = \delta_1$. Carrying out the spin sum for the adjacent motifs in Laurent form with $z = e^{i\gamma}$ gives numerator $-(z^2-1)^6$ over $16z^6$; since $(z^2-1)/z = 2i\sin\gamma$ and $(2i)^6 = -64$, this equals $4\sin^6\gamma$, and the spectator factor $L^{n-m} = 2^{-n}\cdot 2^m$ contributes 2^{-n} after normalization. □

Remark B.24. Corollary B.23 is a statement about the β^2 Taylor coefficient at $\beta = 0$. It is not a finite-angle result, and the finite-angle head certificate of Proposition B.36 is specific to the R_X preparation at the schedule and the operating angle and does not transfer. The size of b_D carries no implication for finite-shot recovery: uniform preparation gives base purity 2^{-n} rather than L^n, so the estimation cost differs by orders of magnitude in the opposite direction.

Remark B.25. Under $|\chi(\pi/2, -\pi/2)\rangle$, that is $R_X(\pi/2)$, the same construction gives $b_4 = b_D = 0$ with $b_3 \neq 0$: the closure is triangle-only. The two preparations have identical amplitude magnitudes and opposite surviving coordinates, which is the content of Corollary B.20.

B.2.7 The cubic trace identities

Proposition B.26. *For every simple cubic graph on n vertices,*

$$\mathrm{tr}(A^3) = 6C_3, \quad \mathrm{tr}(A^4) = 15n + 8C_4, \quad \mathrm{tr}(A^6) = 87n + 6C_3 + 96C_4 + 12(C_6 + D_\diamond), \tag{77}$$

where C_4 counts all simple four-cycles including chorded ones and $D_\diamond = \sum_k \binom{k}{2} N_{1,k}$ is the pair-incidence diamond count.

Proof. $\mathrm{tr}(A^m)$ counts closed m-walks. Grouping walks by their support and then by isomorphism type gives $\mathrm{tr}(A^m) = \sum_H w_H^{(m)} N_H(G)$, where $N_H(G)$ counts subgraphs of G isomorphic to H and $w_H^{(m)}$ is the number of closed m-walks in H using every edge of H, an integer depending only on H.

Structure. A closed m-walk is an Eulerian circuit of the multigraph of its traversals, so that *multigraph* has all even degrees; the underlying simple support need not, as P_4 shows. The support is connected with at most m edges and at most m vertices. Consequently every bridge carries even multiplicity, so a tree support needs $\sum_e m(e) \geq 2|E(H)|$ and trees with more than three edges are excluded; and around a cycle the even-degree condition forces all multiplicities to share a parity, so a five-cycle would need either all even, summing to at least ten, or all odd, summing to an odd number. Neither equals six, which is why no C_5 term appears.

Enumeration. Over the finite set of connected graphs with at most six edges and six vertices, exactly ten admit a closed 6-walk of full support, with $w_H^{(6)}$ equal to $2, 12, 24, 12, 6, 48, 36, 12, 24, 12$ for $K_2, P_3, C_3, K_{1,3}, P_4, C_4$, the diamond, the tadpole, the bowtie and C_6.

Counts. On a cubic graph $N_{K_2} = 3n/2$, $N_{P_3} = \sum_v \binom{3}{2} = 3n$ and $N_{K_{1,3}} = \sum_v \binom{3}{3} = n$. The bowtie needs a degree-four vertex, so $N = 0$. For P_4, count directed three-edge paths by their middle edge: for an ordered edge (v, w) the ends may be chosen in $(d_v - 1)(d_w - 1) = 4$ ways minus the κ_{vw} choices making them coincide; summing over both orientations and using $\sum_{vw\in E} \kappa_{vw} = 3C_3$ gives $12n - 6C_3$ directed paths, hence $N_{P_4} = 6n - 3C_3$. For the tadpole, each vertex of a four-cycle

has exactly one neighbour off the cycle unless it is an endpoint of a chord, and chords of four-cycles correspond bijectively to diamonds, so $N_{\text{tadpole}} = 4C_4 - 2D_\diamond$. Substituting and collecting gives the three identities. □

Corollary B.27 (Fifth moment)**.** *For every simple cubic graph,* $\operatorname{tr}(A^5) = 60C_3 + 10C_5$.

Proof. A closed 5-walk traverses its support with all multiplicities summing to five, an odd number, so by the parity argument of the proof above the support contains an odd cycle, and with at most five edges that cycle is a triangle or a five-cycle. A five-cycle support uses each edge once, and there are $w^{(5)}_{C_5} = 10$ such walks per five-cycle (five starting vertices, two directions). A triangle support with all five traversals on the triangle: every closed 5-walk in K_3 uses all three edges, since a closed walk of odd length cannot live on a bipartite subgraph, so $w^{(5)}_{C_3} = \operatorname{tr}(A^5_{K_3}) = 2^5 - 1 - 1 = 30$. The only other support is a triangle with one pendant edge, the pendant edge traversed twice: the walk is the triangle with an excursion to the pendant vertex inserted at its attachment vertex, and as a sequence of five steps it has five starting points and two directions, so $w^{(5)} = 10$. On a cubic graph each triangle vertex has exactly one neighbour off the triangle, so the number of such supports is $3C_3$. Hence $\operatorname{tr}(A^5) = 30C_3 + 10 \cdot 3C_3 + 10C_5$. □

Remark B.28. The conventions are load bearing: a different reading of C_4 or $D_\diamond$ changes the tadpole coefficient and hence the sixth identity.

Corollary B.29. *On cubic graphs the spectrum determines C_3, then C_4, then the sum $C_6 + D_\diamond$. It does not follow that no spectral function determines the split; that is established by exhibited cospectral pairs whose diamond counts differ, since cospectrality equalizes every moment.*

B.2.8 The finite-angle head reduction

Fix n and angles with $0 < \eta < \pi$, $c_\eta < s_\eta$ and $0 < \beta < \pi/2$, where $c_\eta = \cos(\eta/2)$, $s_\eta = \sin(\eta/2)$, $c_\beta = \cos(\beta/2)$ and $s_\beta = \sin(\beta/2)$. Write ψ for the post-mixer amplitude and $p = |\psi|^2$.

Lemma B.30. *Writing $R_X(\beta) = c_\beta I - i s_\beta X$ and expanding the tensor mixer,*

$$\psi(x) = \sum_{S \subseteq [n]} c_\beta^{n-|S|} (-i s_\beta)^{|S|} \phi(x \oplus e_S). \tag{78}$$

Define

$$Z_1(x) = \sum_i R_i(x), \qquad Z_2(x) = \sum_{i<j} R_{ij}(x), \qquad \mathcal{R}_3(x) = \sum_{|S|\geq 3} c_\beta^{n-|S|} (-i s_\beta)^{|S|} \phi(x \oplus e_S). \tag{79}$$

Retaining the terms with $|S| \leq 2$ in (78) gives the two-flip truncation

$$\psi = \phi\, c_\beta^{n-2} \big[c_\beta^2 - i s_\beta c_\beta Z_1 - s_\beta^2 Z_2\big] + \mathcal{R}_3, \tag{80}$$

using $c_\beta^2 \tan(\beta/2) = s_\beta c_\beta$ and $c_\beta^2 \tan^2(\beta/2) = s_\beta^2$.

Definition B.31 (All-path enclosures)**.** For x with ℓ defects let $d_\ell = c_\beta^n c_\eta^\ell s_\eta^{n-\ell}$ be the modulus of the zero-flip term of (78) and let o_ℓ be the sum of the moduli of all terms with $|S| \geq 1$. By the triangle inequality $\underline{p}_\ell \leq p(x) \leq \overline{p}_\ell$ with $\underline{p}_\ell = \max(0, d_\ell - o_\ell)^2$ and $\overline{p}_\ell = (d_\ell + o_\ell)^2$. Both bounds are independent of G, because $|\phi|$ depends only on the defect count.

Definition B.32 (Profile extrema)**.** For a defect set with local profile P, let known(P) be the modulus of the bracket in (80), computed with the profile-determined part of Z_2; let unknown(P) be the bound $U_{\ell,e}$ on the residual edge sum of Z_2; and let rem(P) be the bound on $|\mathcal{R}_3|$. Put

$$\underline{A}_2 = \min_{P\in\mathcal{P}_2} \big(\text{known} - \text{unknown} - \text{rem}\big)_+, \qquad \overline{A}_3 = \max_{P\in\mathcal{P}_3} \big(\text{known} + \text{unknown} + \text{rem}\big), \tag{81}$$

where $\mathcal{P}_2$ and $\mathcal{P}_3$ are the enumerated two- and three-defect profile sets of Lemma B.35. Explicitly, for a profile $P = (q^{\text{in}}, q^{\text{out}})$ of a defect set with ℓ elements and e internal edges, Lemma B.3 gives

$$\begin{aligned} Z_1(P) &= \sum_{k=0}^{3} \Big[q_k^{\text{in}}\,(-it)\,e^{i\gamma(2k-3)} + q_k^{\text{out}}\,\tfrac{i}{t}\,e^{-i\gamma(2k-3)} \Big], \\ Q(P) &= \sum_i R_i^2 = \sum_{k=0}^{3} \Big[-q_k^{\text{in}}\,t^2 e^{2i\gamma(2k-3)} - q_k^{\text{out}}\,t^{-2} e^{-2i\gamma(2k-3)} \Big], \\ Z_2(P) &= \tfrac{1}{2}\big(Z_1(P)^2 - Q(P)\big) + \sum_{ij\in E} R_i R_j \big(e^{-2i\gamma z_i z_j} - 1\big), \end{aligned} \tag{82}$$

the last sum being the residual edge sum: for adjacent roots i, j the edge ij changes sign under each single flip and not under the double flip, so $R_{ij} = R_i R_j e^{-2i\gamma z_i z_j}$. Its modulus is at most

$$U_{\ell,e} = 2|\sin\gamma| \Big[e\,t^2 + (3\ell - 2e) + \big(\tfrac{3n}{2} - 3\ell + e\big) t^{-2} \Big], \tag{83}$$

because $|e^{-2i\gamma z_i z_j} - 1| = 2|\sin\gamma|$, $|R_i| = t$ on a defect and t^{-1} off it, and a cubic graph has e edges inside the defect set, $3\ell - 2e$ edges leaving it and $3n/2 - 3\ell + e$ edges outside it. With $|\phi(x)| = c_\eta^{\ell} s_\eta^{n-\ell}$ the three terms are

$$\begin{aligned} \text{known}(P) &= c_\eta^{\ell} s_\eta^{n-\ell} c_\beta^{n-2} \Big| c_\beta^2 - i s_\beta c_\beta Z_1(P) - \tfrac{1}{2} s_\beta^2 \big(Z_1(P)^2 - Q(P)\big) \Big|, \\ \text{unknown}(P) &= c_\eta^{\ell} s_\eta^{n-\ell} c_\beta^{n-2} s_\beta^2\, U_{\ell,e}, \\ \text{rem}(P) &= \sum_{\substack{0\le a\le \ell,\ 0\le b\le n-\ell \\ a+b\ge 3}} \binom{\ell}{a} \binom{n-\ell}{b}\, c_\beta^{n-a-b} s_\beta^{a+b}\, c_\eta^{\ell-a+b} s_\eta^{n-\ell+a-b}, \end{aligned} \tag{84}$$

where rem bounds $|\mathcal{R}_3|$ by the triangle inequality, since a flip set S with a defect positions and b nondefect positions produces a string with $\ell - a + b$ defects; it depends on P only through ℓ and equals $P_\ell^{1/2}$ of Lemma B.33 minus the terms with $a + b \le 2$. In the same way $o_\ell = P_\ell^{1/2} - d_\ell$ in Definition B.31.

Lemma B.33. *For every simple cubic graph and every x with ℓ defects, $p(x) \le P_\ell$, where $P_\ell^{1/2} = (c_\beta c_\eta + s_\beta s_\eta)^{\ell} (c_\beta s_\eta + s_\beta c_\eta)^{n-\ell}$.*

Proof. Take absolute values in Lemma B.30. Since $|\phi|$ depends only on the defect count, the bound factorizes over vertices: a defect position contributes $c_\beta c_\eta$ unflipped and $s_\beta s_\eta$ flipped, a nondefect position $c_\beta s_\eta$ and $s_\beta c_\eta$. Multiplying the per-position sums gives $P_\ell^{1/2}$ exactly. □

Lemma B.34. *$P_{\ell+1} = \rho P_\ell$ with $\rho = \big[(c_\beta c_\eta + s_\beta s_\eta)/(c_\beta s_\eta + s_\beta c_\eta)\big]^2$, and $\rho < 1$ on the stated domain, so $P_4 \ge P_\ell$ for all $\ell \ge 4$.*

Proof. The ratio is immediate from the product form. For the inequality, $(c_\beta c_\eta + s_\beta s_\eta) - (c_\beta s_\eta + s_\beta c_\eta) = (c_\eta - s_\eta)(c_\beta - s_\beta)$. On $0 < \eta < \pi$ both c_η and s_η are nonnegative and $c_\eta < s_\eta$ makes the first factor negative; on $0 < \beta < \pi/2$ both c_β and s_β are positive with $s_\beta < c_\beta$, making the second positive. Both numerator and denominator are positive on this domain, and the numerator is the smaller. □

Lemma B.35. *For a two-defect set $\{u, v\}$ with $a = \mathbf{1}[uv \in E]$ and $k = |N(u) \cap N(v)|$, the outside counts satisfy $q_2 = k$, $q_3 = 0$, $q_1 = 6 - 2a - 2k$ and $q_0 = n - 2 - q_1 - q_2$. For a three-defect set with e internal edges, $q_1 + 2q_2 + 3q_3 = 9 - 2e$ and $q_0 = n - 3 - q_1 - q_2 - q_3$, and the induced graph on three vertices is determined up to isomorphism by e. Hence every realized profile lies in the enumerated set. Minimizing a lower bound and maximizing an upper bound over a superset can only weaken them, so admitting degree-feasible profiles that no cubic graph realizes is safe.*

Proposition B.36. *If the four quantities $\underline{p}_0 - \overline{p}_1$, $\underline{p}_1 - \overline{p}_2$, $\underline{A}_2^2 - \overline{A}_3^2$ and $\underline{A}_2^2 - P_4$ are positive, where $\underline{A}_2$ and $\overline{A}_3$ are the extrema of the two-flip bounds over the enumerated profile sets and $\underline{p}_\ell, \overline{p}_\ell$ are the all-path enclosures, then for* every *simple cubic graph of order n the three uniform layer inequalities hold, and the first $K_2(n) = 1 + n + \binom{n}{2}$ sorted probabilities are exactly the outcomes with at most two defects.*

Proof. For $\ell \le 2$ the all-path enclosures bound $p(x)$ by the triangle inequality applied to Lemma B.30, giving the first two orderings. For $\ell = 2$ and $\ell = 3$, Lemma B.35 places the profile in the enumerated set, so the extrema over that superset bound $p(x)$, giving the third. For $\ell \ge 4$, Lemmas B.33 and B.34 give $p(x) \le P_\ell \le P_4$, giving the fourth. Together the last two yield $\min_{|D|=2} p > \max_{|D|\ge 3} p$. □

Remark B.37 (The certificate). Proposition B.36 reduces a statement quantified over all cubic graphs to four inequalities among finitely many explicitly enumerated quantities. Their positivity at the schedule and the operating angle is established by outward-rounded interval arithmetic in the notebook `aaai-ef1-finite-angle-head-certificate.ipynb`, with the angles entered as the exact rationals $\eta = 23/8$, $\gamma = 2$, $\beta = 1/10$ and the precision raised from 128 to 256 bits until every sign is stable. The lower endpoints of the certified enclosures of the four quantities are

n	$\underline{p}_0 - \overline{p}_1$	$\underline{p}_1 - \overline{p}_2$	$\underline{A}_2^2 - \overline{A}_3^2$	$\underline{A}_2^2 - P_4$
14	5.815842×10^{-1}	2.394771×10^{-3}	1.236783×10^{-5}	3.583952×10^{-5}
16	5.385203×10^{-1}	2.015152×10^{-3}	1.142480×10^{-5}	3.456613×10^{-5}
18	4.980061×10^{-1}	1.668305×10^{-3}	1.051854×10^{-5}	3.333646×10^{-5}

and the ratio of Lemma B.34 is enclosed at $\rho = 0.0334545 < 1$. On sampled graphs (200 at $n = 14$, 100 at $n = 16$, 50 at $n = 18$) the observed margins exceed these lower endpoints by factors between 1.54 and 3.26, and against the simulation pipeline of record at $n = 18$ the sorted vectors agree to 1.7×10^{-15} with the smallest observed boundary gap 3.17×10^{-5} above the certified 1.05×10^{-5}.

C The Finite-Angle Map

Theorems 3.3 and 4.1 are statements as $\beta \to 0$, and Proposition 6.3 is a certificate at one angle. This appendix measures what lies between, on the complete cubic census at $n = 14$, by exact simulation of the four members at seventeen angles from 0 to 1. Everything here is a grid-resolved measurement, not a theorem.

The first-order window. Exact recovery of the pair census by rounding the two-defect block to the nearest first-order value $\mu_2(1 + \beta f(a, k))$ succeeds on all 509 graphs at $\beta = 0.001$ for PE-X1, at $\beta = 0.001$ and 0.002 for PE-Y1, and at no tested angle for PE-X2, whose slopes are twice those of PE-X1. The strict first-order window is of order 10^{-3}. The (a, k) block structure persists much further: the ratio of the worst within-class spread to the smallest between-class gap of the two-defect values is 0.057 at $\beta = 0.05$ and 0.786 at $\beta = 0.1$ for PE-X1, and 0.019 and 0.057 for PE-Y1. At the

Table 2: Mean over the 509 cubic graphs at $n = 14$ of the number of at-most-two-defect outcomes among the top $K_2(14) = 106$ sorted ranks, as a function of the mixer angle.

β	0.02	0.05	0.075	0.1	0.125	0.15	0.2	0.25	0.3	0.35	0.4	0.5	0.6	0.7	0.8	1.0
PE-X1	106.0	106.0	106.0	106.0	104.2	70.3	65.9	42.0	52.2	67.3	67.6	104.2	106.0	106.0	106.0	106.0
PE-Y1	106.0	106.0	106.0	106.0	106.0	106.0	106.0	106.0	106.0	106.0	106.0	106.0	106.0	106.0	106.0	106.0
PE-X2	106.0	106.0	104.2	67.6	67.6	67.6	104.5	106.0	106.0	106.0	106.0	106.0	85.0	83.9	0.0	0.0

operating angle the PE-X1 classes are at the edge of resolution and the PE-Y1 classes are cleanly separated, although the two members carry the same first-order information; the difference is the size of the slopes.

Head membership. Table 2 gives the mean, over the census, of the number of outcomes with at most two defects among the top $K_2(14) = 106$ ranks. For PE-X1 the head is intact through $\beta = 0.1$, as the certificate requires, is lost between 0.125 and 0.5, and is intact again from 0.6. For PE-Y1 it is intact at every angle. For PE-X2 it is lost from 0.075, regained at 0.25 to 0.5, and lost for good from 0.6. The PE-X1 values at $\beta = 0.2$ and 0.4, namely 65.9 and 67.6, are the values an independent angle transect of the same census had recorded without explanation. The one-defect head ($K_1 = 15$) is intact at every angle for all three members except at $\beta = 0.25$ for PE-X1, where only the modal outcome remains, and at $\beta \geq 0.8$ for PE-X2.

The mechanism. Figure 1 shows, per pair class, the mean over the census of p/μ_2 on the two-defect layer, together with the mean largest three-defect value; Table 3 gives the numbers. Under PE-X1 the four non-adjacent classes, whose first-order slopes are $f(0, k) \approx -14$, fall from 0.75 at $\beta = 0.02$ to 0.0073 at $\beta = 0.2$ and reach their minimum near $\beta = 0.3$, while the largest three-defect value rises from 0.018 to 0.24 at $\beta = 0.2$ and 0.54 at $\beta = 0.3$. The non-adjacent classes sit below the top of the three-defect layer between $\beta \approx 0.125$ and ≈ 0.5, and the outcomes that leave the head are theirs. The three adjacent classes ($f(1, k) \approx +7$) stay between 2.0 and 3.4 and never leave. The mechanism is the two-flip interference of Lemma B.30: the amplitude of a two-defect outcome is the direct term plus two one-defect flip terms of relative size $t \tan(\beta/2)$ carrying the phases $e^{i\gamma z_j S_j}$, and for a non-adjacent pair both phases contribute $\cos 3\gamma = 0.96$ with a negative sign, so the terms cancel the direct amplitude near $t \tan(\beta/2) \approx 1$. Under PE-Y1 the same construction carries $\sin 3\gamma = -0.28$ in place of $\cos 3\gamma$, the non-adjacent classes never fall below 0.76, and the head stays intact. Under PE-X2 the slopes double and the loss comes earlier. The class order also changes with angle: for PE-X1 the three adjacent classes reverse order at $\beta = 0.125$, the non-adjacent classes reverse between 0.3 and 0.35, and the adjacent block drops below the non-adjacent block at $\beta = 0.6$. None of these crossing angles is solved for; each is located between grid points.

D Census Facts in Full

D.1 Censuses and simulation

The cubic censuses are generated by `nauty-geng` [10] with the options `-c -d3 -D3` and used in generator order. Their digests (SHA-256 of the newline-joined graph6 list, first sixteen hexadecimal digits) are `604443fc6194281c` ($n = 10$, 19 graphs), `fc515329ef2ce9e9` ($n = 12$, 85 graphs), `6dc9ef9eb030ceff` ($n = 14$, 509 graphs). The 4-regular census at $n = 12$ (1,544 graphs) and the cubic census at $n = 16$ (4,060 graphs) are generated the same way; the cubic census at $n = 18$ (41,301 graphs) is used only classically, to identify cospectral groups. The four irregular strata at

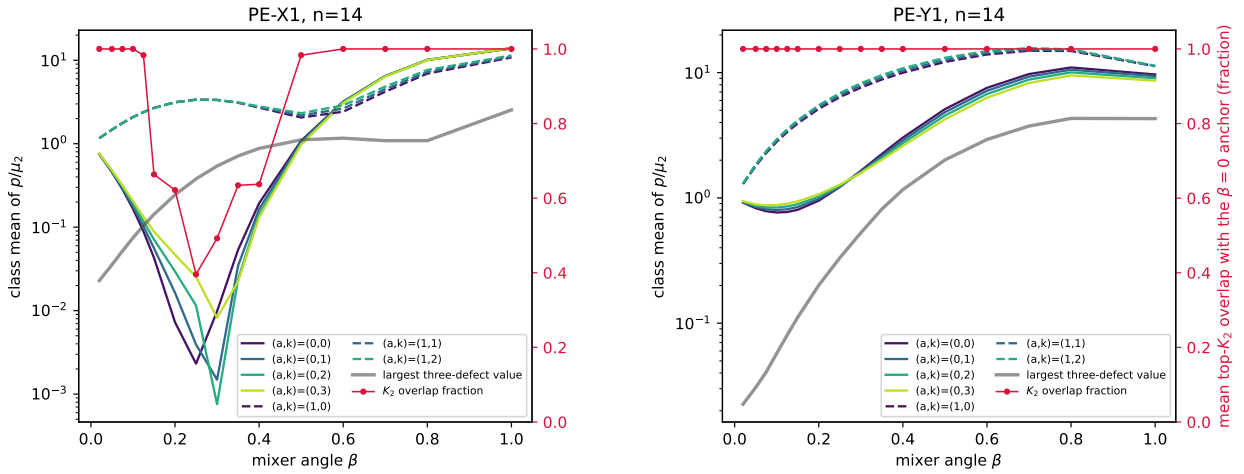


Figure 1: Finite-angle deformation of the two-defect block on the complete cubic census at $n = 14$. For PE-X1 (left) and PE-Y1 (right): the mean over graphs of p/μ_2 in each pair class (a, k) (solid: non-adjacent pairs, dashed: adjacent pairs), the mean largest three-defect value (grey), and on the right axis the mean overlap of the top $K_2(14) = 106$ ranks with the outcomes of at most two defects. Under PE-X1 the non-adjacent classes fall through the three-defect layer between $\beta \approx 0.125$ and ≈ 0.5; under PE-Y1 the head is intact at every angle.

$n = 14$ were generated with seed 0, a 600-candidate budget and 400 retained graphs per stratum. Every motif count is certified by two routes: $C_3 = \mathrm{tr}(A^3)/6$, $C_4 = (\mathrm{tr}(A^4) - 15n)/8$ and $D_\diamond$ by explicit enumeration of 4-subsets against the pair-census identities (5); C_5 and C_6 by simple-cycle enumeration against $\mathrm{tr}(A^5) = 60C_3 + 10C_5$.

Probabilities are computed from the Qiskit statevector of the circuit construction of record and, independently, from a dense tensor simulator that applies the same gates as explicit 2×2 matrices and a diagonal phase vector; the two agree on unsorted probabilities to 3.2×10^{-15}. The first derivative at $\beta = 0$ is computed exactly as $p_1 = \mathrm{Im}[\overline{\phi} \sum_b E^{r-b} \Sigma E^b \phi_{\mathrm{prep}}]$ by index gathers on the 2^n vector, and agrees with Richardson-extrapolated central differences of the Qiskit pipeline (steps 10^{-3} and 2×10^{-3}) to 7.3×10^{-10}. Relabelling invariance of the sorted readout holds to 1.1×10^{-15} on sampled permutations, and the $\beta = 0$ collapse to one vector per order holds to 2.3×10^{-17}. The leading sorted probabilities of the first census graph at $n = 14$, 0.898974008 for PE-X1 and 0.000142976 for PE-QX1, match the values recorded in the original study.

D.2 Three-defect profiles under the two mixer axes

Figure 2 places the realized three-defect profiles of the $n = 14$ census by their PE-X1 and PE-Y1 values. The values are paired by outcome; the figure is a statement about one outcome under two members, and the pairing is not available from two sorted readouts.

D.3 The determination matrix in full

Table 5 completes Table 4. Views are the layer-prefix histograms of each member, the PE-QX1 histogram over all outcomes, the pair census, the power-trace tuple $(\mathrm{tr}\, A^k)_{k \le n}$ as the spectrum, and folklore 2-WL run jointly over the census with one colour dictionary per round. A view that keys each outcome by its PE-X1 and PE-Y1 values together, which pairs values by outcome and is not available from two sorted readouts, coincides with the PE-X1 view at prefixes ≤ 3 and ≤ 4.

Table 3: PE-X1 at $n = 14$: mean over the census of p/μ_2 in the classes $(0,0)$, $(0,3)$ and $(1,0)$ of the two-defect layer, and mean largest three-defect value, by mixer angle.

β	0.02	0.05	0.075	0.1	0.125	0.15	0.2	0.25
$(0,0)$	0.748	0.457	0.285	0.166	0.089	0.043	0.0073	0.0023
$(0,3)$	0.758	0.481	0.319	0.206	0.133	0.089	0.047	0.025
$(1,0)$	1.16	1.47	1.77	2.08	2.39	2.68	3.15	3.38
largest 3-defect	0.023	0.036	0.052	0.075	0.104	0.142	0.244	0.379
β	0.3	0.35	0.4	0.5	0.6	0.7	0.8	1.0
$(0,0)$	0.0098	0.054	0.194	1.08	3.20	6.46	10.07	13.86
$(0,3)$	0.0082	0.022	0.132	0.97	3.08	6.38	10.06	13.98
$(1,0)$	3.35	3.09	2.71	2.06	2.42	4.20	6.92	10.80
largest 3-defect	0.540	0.712	0.877	1.11	1.17	1.09	1.09	2.54

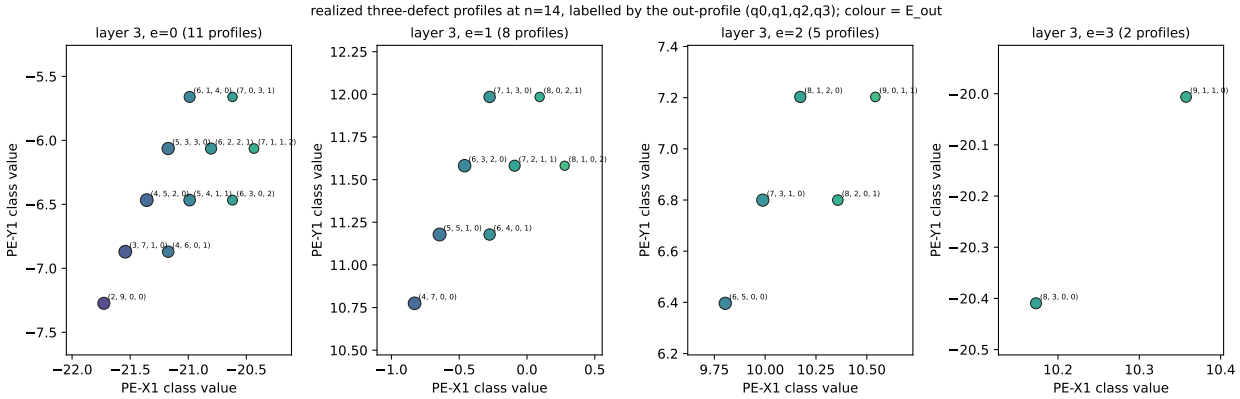


Figure 2: Realized three-defect profiles on the complete cubic census at $n = 14$, one panel per number e of edges inside the defect set, placed by their PE-X1 value (horizontal) and PE-Y1 value (vertical) and labelled by the out-profile $(q_0^{\text{out}}, q_1^{\text{out}}, q_2^{\text{out}}, q_3^{\text{out}})$. PE-X1 separates the columns, which are the values of $E_{\text{out}} = q_0^{\text{out}} + q_3^{\text{out}}$; PE-Y1 separates the rows, which are the values of $(q_3^{\text{out}} - q_0^{\text{out}}, q_2^{\text{out}} - q_1^{\text{out}})$. Points sharing a column are PE-X1 collisions resolved by PE-Y1 and conversely. PE-X1 does not separate the $e = 2$ panel from the $e = 3$ panel; PE-Y1 places them 27 units apart. Marker size grows with the multiplicity of the profile over the census.

D.4 Full first-order readouts and the Hadamard member

Beyond the layer prefixes, the complete first-order sorted readout of PE-X1 is distinct on every graph at the three orders, with smallest pairwise L_1 distance 3.7×10^{-5}, 5.7×10^{-7} and 2.5×10^{-7} at $n = 10, 12, 14$; the separation is carried by the layers with three or more defects and is correspondingly small. The PE-QX1 first-order readout, which has no layers and reads the odd profile of every outcome with unit weights (10), is also distinct on every graph, with smallest pairwise distance 6.9×10^{-2}, 2.8×10^{-2} and 1.2×10^{-2}.

D.5 Exact cospectral witnesses at $n = 18$

Cospectral nonisomorphic graphs are the standard obstruction to spectrum-determined representations [19]. The complete cubic census at $n = 18$ (41,301 graphs) contains 471 adjacency-cospectral groups with 958 members. Exactly six groups vary in $D_\diamond$, all pairs; within each pair $D_\diamond$ differs by one and C_6 by one in the opposite direction, as Proposition B.26 requires. At the operating angle

Table 4: Determination on the cubic census at $n = 14$ (509 graphs). "Classes" is the number of distinct values of the view and "equal pairs" the number of graph pairs it does not separate; each remaining column counts, among those pairs, the ones that differ in the column quantity, so a zero says the view determines the quantity on the census.

View	classes	equal pairs	C_3	C_4	$D_\diamond$	C_5	C_6	girth	spectrum
layers ≤ 2, any degree-prepared member (the pair census)	99	3171	0	0	0	2623	2713	8	3169
layers ≤ 3, PE-X1	493	19	0	0	0	0	0	0	18
layers ≤ 3, PE-Y1	488	24	0	0	0	0	5	0	23
layers ≤ 4, PE-X1, PE-Y1 or PE-X2	509	0	0	0	0	0	0	0	0
PE-QX1 first-order histogram	509	0	0	0	0	0	0	0	0
adjacency spectrum	506	3	0	0	3	0	3	0	—
folklore 2-WL	509	0	0	0	0	0	0	0	0

Table 5: Full determination matrix on the cubic census at $n = 14$. For a row view V and a column W, the entry is the number of graph pairs equal under V and different under W. The PE-X2 views coincide with the PE-X1 views at every prefix.

View	classes	pairs	C_3	C_4	$D_\diamond$	C_5	C_6	girth	spectrum	2-WL	X1 ≤ 3	X1 ≤ 4
pair census / layers ≤ 2	99	3171	0	0	0	2623	2713	8	3169	3171	3152	3171
PE-X1, layers ≤ 3	493	19	0	0	0	0	0	0	18	19	0	19
PE-Y1, layers ≤ 3	488	24	0	0	0	0	5	0	23	24	5	24
PE-X1, layers ≤ 4	509	0	0	0	0	0	0	0	0	0	0	0
PE-Y1, layers ≤ 4	509	0	0	0	0	0	0	0	0	0	0	0
PE-QX1, all outcomes	509	0	0	0	0	0	0	0	0	0	0	0
spectrum	506	3	0	0	3	0	3	0	—	3	2	3
folklore 2-WL	509	0	0	0	0	0	0	0	0	—	0	0

every member separates all six pairs in the full sorted readout, by L_1 distances between 1.59e−5 and 1.63e−5 for PE-X1, 8.14e−5 and 8.30e−5 for PE-Y1, 5.95e−5 and 5.99e−5 for PE-X2, and 1.09e−2 and 1.30e−2 for PE-QX1, each exceeding the worst relabelling discrepancy by more than two orders of magnitude; leave-one-group-out probes on the top 100 ranks order the $D_\diamond$ direction correctly for all six pairs under every member. This is Corollary 6.2 observed at the operating angle. Folklore 2-WL also separates and orders all six, so the witnesses do not rank the circuit readouts against it; they establish that each measured readout carries graph-dependent information absent from the adjacency spectrum on these pairs, and nothing beyond that. Table 6 lists the pairs by graph6 string. Within every pair C_3, C_4 and C_5 agree and the power-trace tuples $(\operatorname{tr} A^k)_{k\leq 18}$ coincide; the certificates were recomputed from the graph6 strings alone.

D.6 Purity moment ladder

Table 7 gives the coefficients of the exact affine closure $M_q''(G;0) = A^{(q)} + b_3^{(q)} C_3 + b_4^{(q)} C_4 + b_D^{(q)} D_\diamond$ fitted on the cubic census at $n = 14$ for PE-X1, and the Hadamard closed form $b_D^{(q)} = 2q(q-1)2^{n(1-q)} \sin^6 \gamma$ with $b_3^{(q)} = b_4^{(q)} = 0$ for PE-QX1; the $q = 2$ row reproduces the certified coefficients of Theorem 6.1 to relative deviations below 6×10^{-9}, the fitted Hadamard values match the closed form to twelve digits, and the residual of the closure is below 2×10^{-14} at every q. Entries marked with an asterisk lie below the resolution of the fit.

Table 6: The six $D_\diamond$-varying adjacency-cospectral pairs at $n = 18$ (group numbers in the census enumeration of cospectral groups).

Group	graph6 (member 1; member 2)	C_3	C_4	C_5	$(D_\diamond, C_6)$: member 1; member 2
257	`Q???C@?GCoW_S_KCCa?Y??w?I_?`; `Q???C@?K?WSAKOE_P_AK??w@?g?`	2	2	4	$(1,6)$; $(0,7)$
401	`Q???C@?K?WEAg_q?I_?EO@g?H_?`; `Q???C@?K?WCQk?E_HG?F?~G@CG?`	2	1	6	$(1,5)$; $(0,6)$
438	`Q??CA?_CCOGgR?IOAD?H_GcAQ??`; `Q??CA?_CCCWOIAHOI_?p?AS?Co?`	2	1	6	$(0,6)$; $(1,5)$
451	`Q??CA?_CCOP@Q_POEO?h?HC?@o?`; `Q??CA?_C?WP@B_p?KG?b??k?oO?`	3	1	8	$(1,8)$; $(0,9)$
461	`Q??CA?_C?WPOo_HAGc@a?AS?Oo?`; `Q??CA?_C?WOPq?M?Cc?H_CS@B??`	4	1	3	$(1,4)$; $(0,5)$
468	`Q??CA?_C?WP@B_p?CW?b?OK?oO?`; `Q??CA?_CCCWOQAL?BC@A_BO?Co?`	2	1	9	$(0,11)$; $(1,10)$

Table 7: Second-derivative closure coefficients of the purity moments M_q at $n = 14$ and the schedule.

q	PE-X1 $b_3^{(q)}$	PE-X1 $b_4^{(q)}$	PE-X1 $b_D^{(q)}$	PE-QX1 $b_D^{(q)}$
2	3.7305e−5	3.3437e−8	3.9910e−8	1.3800e−4
3	7.4968e−7	1.9590e−11	1.5589e−11	2.5268e−8
4	1.3972e−8	8.6e−15*	6.0e−15*	3.0845e−12
5	2.4468e−10	*	*	3.1377e−16
6	4.1126e−12	*	*	2.8727e−20

E A Folklore-2-WL-Equivalent Cubic Pair at the Operating Angle

Theorem 5.3 bounds, order by order, the level of the Weisfeiler–Leman hierarchy that captures the small-angle expansion. This appendix asks the converse: is the sorted readout at the operating angle, which is the full series, a function of the folklore-2-WL colouring? On the censuses the question is empty, because 2-WL is injective there. The cubic pairs it does not separate that we know are the Cai–Fürer–Immerman pairs over cubic base graphs, and for the smallest of them we compute the readout exactly. Everything here is computed; the one inequality that carries the conclusion is certified by interval arithmetic. The runs of record are `probe_census_2fwl.py` and `probe_cfi_operating_angle.py`.

E.1 Folklore 2-WL on the cubic censuses

Every connected cubic graph of the `nauty-geng` censuses at $n = 14$, 16, 18 and 20 receives a canonical folklore-2-WL colouring: the colour of a pair (u, v) after a round is a 64-bit hash of its previous colour and of the sorted multiset, over w, of the hashed pairs $(c(w, v), c(u, w))$, from the atomic types. The hashes are built from structure alone, so two graphs with different stable colour multisets are distinguished by 2-WL, and equal multisets mark a candidate pair, which would then be confirmed or refuted by exact joint refinement of the two graphs with one colour dictionary and no hashing. Table 8 gives the outcome: no candidate arose at any of the four orders, so folklore 2-WL is injective on all 556,359 graphs, and no two of them test Theorem 5.3 in either direction.

E.2 The construction over cubic base graphs

Given a base graph B, the Cai–Fürer–Immerman graph has, for every base vertex v of degree d, the 2^{d-1} inner vertices (v, S) with S an even subset of the edges at v, and the $2d$ pins (v, e, i) with $e \ni v$ and $i \in \{0, 1\}$; the inner vertex (v, S) is adjacent to the pin $(v, e, 1)$ if $e \in S$ and to $(v, e, 0)$ otherwise, and across every base edge $e = uv$ the pins are joined $(u, e, i) \sim (v, e, i)$. The twisted graph joins them $(u, e, i) \sim (v, e, 1 - i)$ on one base edge; for a connected base graph, twisting any other single edge gives an isomorphic graph, and the untwisted and twisted graphs are not

Table 8: Folklore 2-WL on the complete connected cubic censuses. "Digest" is the first sixteen hexadecimal digits of the SHA-256 of the newline-joined graph6 list in generator order; "classes" is the number of distinct stable colour multisets; every partition stabilized by round five of the ten rounds run.

n	graphs	digest	classes	candidate pairs
14	509	`6dc9ef9eb030ceff`	509	0
16	4,060	`ab04e0acea28fba5`	4,060	0
18	41,301	`2a75c1152823c3ef`	41,301	0
20	510,489	`2d74024d1f14c330`	510,489	0

Table 9: Cai–Fürer–Immerman pairs over cubic base graphs: order, whether colour refinement (1-WL) and exact folklore 2-WL distinguish the pair, the girth, and the layers on which the first-order histograms of the two members were compared and found equal.

base	n	1-WL	folklore 2-WL	girth	C_6	equal first-order histograms
K_4	40	no	no	6	16	layers $2, \dots, 6$
prism	60	no	no	6	24	layers $2, 3, 4$
$K_{3,3}$	60	no	no	6	24	layers $2, 3, 4$
cube	80	no	not run	6	—	layers $2, 3$
Petersen	100	no	not run	6	—	layers $2, 3$

isomorphic [3], which was also checked directly for the five pairs below. Over a cubic base every inner vertex has three pins and every pin has two inner neighbours and one cross neighbour, so both graphs are cubic, on $10|V(B)|$ vertices.

Table 9 lists the pairs over the five smallest cubic base graphs. Colour refinement, run jointly on the disjoint union to its stable partition, does not distinguish any of the five pairs, and exact folklore 2-WL, run the same way, does not distinguish the three pairs on at most 60 vertices; for the pairs on 80 and 100 vertices it was not run. Every graph in the table has girth six. The first-order histograms of the extremal counts $(E_{\mathrm{in}}, E_{\mathrm{out}})$ and of the odd readings, computed by enumerating every m-subset, coincide for the two members of each pair on every layer computed: $m \le 6$ over K_4 (3,838,380 six-subsets), $m \le 4$ over the prism and $K_{3,3}$, and $m \le 3$ over the cube and the Petersen graph. By Theorem 3.2 the first-order value of an outcome is a function of its profile, and by Corollary 4.5 the profile is a function of the two readings given $|D|$ and n, so equal histograms give equal first-order m-defect blocks under every degree-prepared member and equal first-order sector multisets under PE-QX1. For the layers with two and three defects this is Theorem 4.3: both graphs of each pair have girth six, no five-cycle and the same number of six-cycles (16 over K_4, 24 over the prism and over $K_{3,3}$). For the deeper layers it is a computed fact; Theorem 5.3 gives, for 2-WL-equivalent graphs, order $2 - m$ on layer m and nothing at $m \ge 3$.

E.3 Exact evaluation at the operating angle

Let G_0 and G_1 be the pair over K_4, on the same 40 labelled vertices; they differ in the four cross edges at the twisted base edge. A Born probability of PE-X1 is a two-state partition function: $\langle x|U_M(\beta)$ is a product over qubits and $U_G(\gamma)$ is diagonal, so

$$\langle x|U_M(\beta)\,U_G(\gamma)\,U_{\mathrm{prep}}|0^n\rangle = \sum_{y\in\{0,1\}^n} \prod_{v} [R_X(\beta)]_{x_v y_v}\, a(y_v) \prod_{uv\in E} e^{-i\gamma z_u(y) z_v(y)/2}, \tag{85}$$

with $a(0) = c$ and $a(1) = -is$. The sum over y is contracted exactly by variable elimination along a greedy minimum-fill order; on these graphs the largest eliminated neighbourhood has nine vertices,

Table 10: PE-X1 at the operating angle on the pair over K_4: for each defect layer, the number of outcomes, the largest and smallest probability (the smallest four-defect value is 1.595107×10^{-9} in G_0 and 1.595179×10^{-9} in G_1), and the largest absolute difference and the L_1 distance between the sorted probability vectors of the two graphs.

layer	outcomes	largest value	smallest value	max |sorted difference|	L_1
0	1	7.376481×10^{-1}	7.376481×10^{-1}	0	0
1	40	5.030555×10^{-3}	5.030555×10^{-3}	1.7×10^{-18}	3.5×10^{-18}
2	780	4.300637×10^{-4}	3.430366×10^{-5}	1.6×10^{-19}	2.0×10^{-18}
3	9,880	7.473502×10^{-6}	2.339256×10^{-7}	1.2×10^{-18}	5.1×10^{-16}
4	91,390	2.542280×10^{-7}	1.5951×10^{-9}	7.1×10^{-14}	2.0×10^{-11}

so no intermediate factor exceeds 2^9 entries, and the cost of one outcome is milliseconds. On every outcome of a random cubic graph at $n = 10$ the contraction agrees with the dense statevector to 1.6×10^{-15}. We computed every outcome with at most four defects of both graphs, 102,091 outcomes each, in double precision.

Ranks. Lemma B.33 bounds every outcome with ℓ defects of every cubic graph by P_ℓ, and Lemma B.34 makes P_ℓ decreasing; at $n = 40$ and the operating angle $P_3 = 2.84 \times 10^{-5}$, $P_4 = 9.49 \times 10^{-7}$ and $P_5 = 3.17 \times 10^{-8}$. In both graphs the smallest value with at most two defects, 3.430×10^{-5}, exceeds P_3, so the outcomes with at most two defects are ranks 1 to $K_2(40) = 821$ of the sorted readout: the conclusion of Proposition 6.3, obtained for these two graphs from the computed values and the shell bound rather than from the certificate. The three- and four-defect layers interleave at this angle: the largest four-defect value, 2.542×10^{-7}, exceeds the smallest three-defect value, 2.339×10^{-7}; the largest four-defect outcome has rank 3,398, and 7,304 of the 9,880 three-defect values lie below it. Every value with at most three defects exceeds P_5, and exactly 5,530 four-defect values do, in each graph; so the 16,231 computed outcomes above P_5 are ranks 1 to 16,231 of both sorted readouts, whatever the uncomputed outcomes with five or more defects do below them. The modal outcome is 1^{40} in both, with $p(1^{40}) = 0.73765$ against a one-defect value of 5.031×10^{-3}. Table 10 compares the two sorted readouts layer by layer. Over the first 16,231 ranks the two sorted vectors agree to 1.7×10^{-18} in absolute value (the largest difference is at rank 22, a one-defect outcome), which is 2×10^{-18} relative to the modal probability and at the rounding error of the computation; the four-defect values above P_5, sorted within the layer, agree to 1.6×10^{-22}, and the tail of the layer, which the sorted readout interleaves with the five-defect outcomes, to 7×10^{-14}.

Pointwise. With the shared labels, the probabilities of the two graphs differ pointwise, above relative 10^{-15}, on 45 of the 780 two-defect outcomes and on 2,316 of the 9,880 three-defect outcomes; on the zero- and one-defect layers no outcome differs at this precision. The defects of the 45 two-defect outcomes lie among 18 vertices around the four cross edges that differ, and the 45 values, sorted within each graph, agree to 1.6×10^{-19}: the twist moves probability among outcomes near it, in pairs that trade values, and the sorted readout does not see the exchange.

The modal outcome at high precision. Recomputing $p(1^{40})$ by the same elimination in 50, 80 and 120 decimal digits gives

$$p_{G_0}(1^{40}) - p_{G_1}(1^{40}) = 6.79574045255756 \times 10^{-30}$$

at all three precisions, and an outward-rounded interval evaluation of (85) at 60 digits, with the gate entries entered as intervals around the exact rationals $\eta = 23/8$, $\gamma = 2$, $\beta = 1/10$, encloses the difference in $[6.795740452557595431827913253 \times 10^{-30}, 6.795740452557595431827913261 \times 10^{-30}]$, which excludes zero. A two-defect outcome far from the twist differs at the same scale (1.76×10^{-30} at the defect set $\{18, 22\}$); the two-defect outcomes $\{0, 17\}$ and $\{0, 19\}$ next to it differ by $\pm 1.7017 \times 10^{-9}$, which is the exchange seen above.

Other angles. The modal difference $p_{G_0}(1^{40}; \beta) - p_{G_1}(1^{40}; \beta)$ is a trigonometric polynomial in β with period 2π that is not identically zero, so it vanishes at finitely many angles in each period, and the two sorted readouts differ at every other angle at which 1^{40} is the modal outcome of both graphs, which holds near $\beta = 0$ and at $\beta = 1/10$. At $\beta = 0.1$, 0.05, 0.025 and 0.0125 the difference is 6.80×10^{-30}, 9.88×10^{-35}, 1.45×10^{-39} and 2.15×10^{-44} at 90 digits, so each halving of β divides it by $2^{16.07}$, $2^{16.06}$ and $2^{16.04}$ in turn, which indicates that the first nonzero Taylor coefficient of the modal difference is the sixteenth; part (2) of Theorem 5.3 guarantees agreement of the modal coefficients of a 2-WL-equivalent pair through the second only. The scaling, the tail mass and the L_1 bound are computed by `probe_cfi_modal_scaling.py` from the saved outcomes of the run of record.

E.4 What the pair shows

The sorted readouts of G_0 and G_1 at the operating angle are different vectors: their first entries differ by 6.8×10^{-30}. Hence the sorted readout at the operating angle is not a function of the folklore-2-WL colouring, and level two does not determine every Taylor coefficient of the sorted readout, since otherwise the readouts of a 2-WL-equivalent pair would agree at every angle. The computed outcomes, those with at most four defects, carry 0.99981 of the probability in each graph, so the uncomputed tail has mass 1.89×10^{-4}, and by Proposition 2.1 the two full sorted readouts are within L_1 distance $2.0 \times 10^{-11} + 2 \times 1.89 \times 10^{-4} = 3.8 \times 10^{-4}$ of each other. The rank-one separation is one no experiment can see: resolving a difference δ in a probability p needs of order $p/\delta^2 \approx 10^{58}$ shots, and the computed ranks agree to 10^{-18}; whether the uncomputed tail separates the two readouts by more than that is not known. The two graphs differ by a global parity and agree on every local count that the first-order theory reads; what the first-order theory reads is what Theorem 5.3 places, and the parity is seen only by the full series. For every K the construction over cubic base graphs of larger treewidth gives cubic pairs that K-WL does not distinguish [3], and part (2) of Theorem 5.3 makes their sorted m-defect blocks agree through order $K - m$; so for every m and j there are cubic pairs whose m-defect blocks agree through order j. Nothing is proved about the layers with more than K defects of a K-WL-equivalent pair. Whether the separation at a finite angle persists for the pairs over larger bases, whether it is ever larger than this, and whether some fixed level of the hierarchy determines the readout at the operating angle, are open.

F Accessibility Experiments

The theorems say what the sorted readout determines. A classical learner receives the sorted readout, or a sampled estimate of it, and the practical question is what it can recover. We separate three levels.

Definition F.1 (Accessibility levels)**.** Fix a measured quantum feature map with invariant readout, a graph population, and a structural target τ. The map has *representational content* on the population if its ideal invariant vectors distinguish the graphs concerned. The target τ is *linearly*

accessible if it is recoverable out of sample by a linear probe from the ideal vector under a frozen probe protocol. It is *statistically accessible* if that recovery survives finite multinomial sampling under a specified invariantization and estimator.

Representational content is settled on the populations in hand by Section 7: every member is injective on them, so every target is represented and the content lies at the other two levels. Two further populations enter here: the complete connected 4-regular census at $n = 12$ (1,544 graphs) as a second regularity regime, and four fixed-degree irregular strata at $n = 14$ of 400 graphs each. Statistical accessibility of the pair census is the subject of Proposition 4.6, proved first.

Proof of Proposition 4.6. (1) Fix $x \in L_2$ of class (a, k), write $p = p_G(x; \beta)$ and $\hat{p} = N_x/N$ with $N_x \sim \mathrm{Bin}(N, p)$. By (14), $p \le \mu_2(1 + \beta F_{\max} + \frac{1}{4}\beta\Delta) \le \frac{3}{2}\mu_2$, and $\beta\Delta \le \beta F \le 2\beta F_{\max} \le 1$. Bernstein's inequality with $\epsilon = \frac{1}{4}\beta\mu_2\Delta$ gives

$$\Pr[|\hat{p} - p| \ge \epsilon] \le 2\exp\Big(-\frac{N\epsilon^2}{2(p(1-p) + \epsilon/3)}\Big) \le 2\exp\Big(-\frac{3N\beta^2\mu_2\Delta^2}{152}\Big),$$

because $p(1-p) + \epsilon/3 \le \frac{3}{2}\mu_2 + \frac{1}{12}\mu_2$. On the event that $|\hat{p} - p| < \epsilon$ for every $x \in L_2$, every x is assigned to its class: $\hat{p}$ is within $\epsilon + \frac{1}{4}\beta\mu_2\Delta = \frac{1}{2}\beta\mu_2\Delta$ of the value $\mu_2(1 + \beta f(a, k))$ of its class, and the value of every other class is at least $\beta\mu_2\Delta$ from that value, hence more than $\frac{1}{2}\beta\mu_2\Delta$ from $\hat{p}$. The union bound over the $\binom{n}{2} < n^2/2$ outcomes of L_2 bounds the failure probability by $n^2\exp(-3N\beta^2\mu_2\Delta^2/152)$, which is at most δ when $N \ge \frac{152}{3}\log(n^2/\delta)/(\beta^2\mu_2\Delta^2)$.

(2) A procedure of the stated kind is a function of the N samples after every outcome outside L_2 is replaced by one symbol $\perp$, so its input is N independent samples of $\tilde{p}_G$, the distribution on $L_2 \cup \{\perp\}$ with $\tilde{p}_G(\perp) = 1 - p_G(L_2)$. Run on G it returns the pair census of G with probability at least $3/4$; run on H it returns that census with probability at most $1/4$; so the total variation distance between $\tilde{p}_G^{\otimes N}$ and $\tilde{p}_H^{\otimes N}$ is at least $1/2$. With $H^2(p, q) = \frac{1}{2}\sum_y(\sqrt{p(y)} - \sqrt{q(y)})^2$ one has $\mathrm{TV} \le \sqrt{2}H$ and $1 - H^2(p^{\otimes N}, q^{\otimes N}) = (1 - H^2(p, q))^N$, so $\frac{1}{2} \le \sqrt{2NH^2(\tilde{p}_G, \tilde{p}_H)}$ and $N \ge 1/(8H^2(\tilde{p}_G, \tilde{p}_H))$. Now $2H^2(\tilde{p}_G, \tilde{p}_H) = D + (\sqrt{1 - p_G(L_2)} - \sqrt{1 - p_H(L_2)})^2$, and the second term is at most $(p_G(L_2) - p_H(L_2))^2/(4(1-P)) \le DP/(1-P)$, because by the Cauchy–Schwarz inequality $|p_G(L_2) - p_H(L_2)| \le \sum_{L_2}|\sqrt{p_G} - \sqrt{p_H}|(\sqrt{p_G} + \sqrt{p_H}) \le \sqrt{D}\sqrt{2(p_G(L_2) + p_H(L_2))} \le 2\sqrt{DP}$. Hence $2H^2 \le D/(1-P)$ and $N \ge (1-P)/(4D)$. For the bound on D: under (14) every $p_G(x)$ and $p_H(x)$ with $x \in L_2$ is at least $\mu_2(1 - \beta F_{\max} - \frac{1}{4}\beta\Delta) \ge \frac{1}{2}\mu_2$, so $(\sqrt{p_G} - \sqrt{p_H})^2 = (p_G - p_H)^2/(\sqrt{p_G} + \sqrt{p_H})^2 \le (p_G - p_H)^2/(2\mu_2)$, while $|p_G(x) - p_H(x)| \le \mu_2(\beta|f_G(x) - f_H(x)| + \frac{1}{2}\beta\Delta)$, $f_G(x)$ denoting the class value of x in G. If G and H differ by the switch that replaces the edges ab, cd by ac, bd, the class of a pair $\{u, v\}$ changes only when $\{u, v\}$ meets $\{a, b, c, d\}$: the adjacency of u, v changes only for the four switched pairs, and the number of common neighbours of u, v changes only if an adjacency at u or at v changes, which requires u or v in $\{a, b, c, d\}$. There are $\binom{n}{2} - \binom{n-4}{2} = 4n - 10$ such pairs, with $|f_G - f_H| \le F$, and the other $\binom{n}{2} - (4n - 10)$ pairs have $f_G = f_H$. Summing the two kinds of term gives the stated bound. □

The constants at $n = 14$. For PE-X1 at the schedule and $n = 14$, $\mu_2 = 2.519 \times 10^{-4}$, the class values are -13.883, -13.699, -13.514, -13.330 for $(0, k)$ and 7.014, 7.198, 7.383 for $(1, k)$, so $\Delta = \Delta_k = 0.1846$, $F = 21.27$ and $F_{\max} = 13.88$. On the complete census the largest deviation in (14), divided by $\beta\Delta$, is 0.087, 0.17, 0.35, 0.70, 1.7 and 3.5 at $\beta = 2.5 \times 10^{-4}$, 5×10^{-4}, 10^{-3}, 2×10^{-3}, 5×10^{-3} and 10^{-2}: it is 349β throughout, the second-order term of the layer, so the window is $\beta \le 7 \times 10^{-4}$, in agreement with the exact rounding recovery at $\beta = 10^{-3}$ and its failure at 2×10^{-3} reported in Appendix C, which asks for half the gap rather than a quarter. At $\beta = 5 \times 10^{-4}$ and

Table 11: The family. Every member applies the same edge-phase operator $U_G(\gamma)$ at the same angle. The controls remove one element each.

Name	Preparation (per vertex)	Edge phase (per edge)	Mixer	Reps.	Role
PE-X1	$R_X(\eta d_v/d_{\max})$	$R_{ZZ}(\gamma)$	X	1	anchor [12]
PE-Y1	$R_X(\eta d_v/d_{\max})$	$R_{ZZ}(\gamma)$	Y	1	mixer-axis intervention
PE-X2	$R_X(\eta d_v/d_{\max})$	$R_{ZZ}(\gamma)$	X	2, tied	depth intervention
PE-QX1	H	$R_{ZZ}(\gamma)$	X	1	preparation intervention
CTRL-PHASE	$R_X(\eta d_v/d_{\max})$	$R_{ZZ}(\gamma)$	none	1	control, no mixer
CTRL-NOEDGE	$R_X(\eta d_v/d_{\max})$	none	X	1	control, no edge phase

$\delta = 10^{-2}$ part (1) gives $N = 2.35 \times 10^{14}$. For part (2) we took the first census graph G and the graph H obtained by one switch, which is not isomorphic to G and differs from it in 16 of the 46 pair classes that the switch can touch; the exact two-defect sum is $D = 2.71 \times 10^{-8}$ at $\beta = 5 \times 10^{-4}$ with layer mass $P = 0.0228$, so $N \geq 9.03 \times 10^6$, while the a priori bound on D is 6.61×10^{-7} and gives $N \geq 3.8 \times 10^5$. Over the tested angles D scales as β^2 to three digits. The squared Hellinger distance of the full distributions of G and H at the same angle, strings retained, is 1.56×10^{-8} against $D/2 = 1.35 \times 10^{-8}$ for the two-defect layer alone, so at this order 87% of the distinguishing information of the labelled distribution sits in the two-defect layer, and the corresponding bound for arbitrary procedures, $N \geq 1/(8H^2) = 8.0 \times 10^6$, is the same number to within 12%. The exponential factor $s^{-2(n-2)}$ is 1.5 at $n = 14$ and 6.9 at $n = 56$; at the orders in hand the cost is set by $\beta^{-2}c^{-4}\Delta^{-2} = 3.2 \times 10^3 \beta^{-2}\Delta^{-2}$. The run of record is `probe_shot_bound.py`.

Protocol. Exact Born probabilities are sorted in descending order. Held-out R^2 is the mean over five seed-0, fold-matched outer folds; within each training split, five unshuffled inner folds select the ridge penalty from $\{10^j : j = -14, \ldots, 2\}$; targets are standardized within training folds and features are not. Full vectors are projected to their row space by a target-free QR factorization over the population, which preserves the Gram matrix and the ridge predictions. Every representation, control and classical comparator uses the same folds. Geometry uses Spearman Mantel correlations between L_1 readout distances and absolute target differences with 10,000 permutations, seed 0 and Benjamini–Hochberg correction at $\alpha = 0.05$; conditional analyses require at least 15 graphs in fixed-C_3 strata and 12 in fixed-(C_3, C_4) strata. Sampled readouts draw independent multinomial samples for training and test graphs, sort the empirical frequencies globally, and truncate to the top 500 coordinates without renormalization, at budgets 2^{17}, 2^{20} and 2^{24} with 30 replicates; the comparator is the exact top-500 readout. The structured-readout ladder is $S \in \{2^{10}, 2^{12}, \ldots, 2^{20}\}$ with 12 paired replicates per budget, scores normalized to each readout's exact reference, monotonized over the ladder and interpolated in $\log_2 S$. The classical comparators concatenate the sorted adjacency eigenvalues with eight power traces and use ridge regression or an untuned 300-tree Extra Trees model; the folklore 2-WL comparators use cumulative and stable colour histograms from three refinement rounds with census-wide vocabulary. The continuation audit of Appendix B evaluates exact purity on the 509 cubic graphs at $n = 14$ at $\beta \in \{0.005, 0.01, 0.02, 0.05, 0.075, 0.1\}$, centres within frozen folds, rescales by $2/\beta^2$ and fits nested ordinary-least-squares bases.

Mechanism controls. On both exhaustive regular censuses CTRL-PHASE and CTRL-NOEDGE each produce one constant vector and negative held-out R^2 on every target, while every active member produces a distinct vector for every graph. Removing either the edge phase or the mixer removes all edge-sensitive probability on a regular census, as Section 2 says it must.

Table 12: Mean held-out R^2 from exact sorted readouts at the operating angle. One target-independent vector per graph is probed against every target. Girth is omitted at $n = 12$ because one test fold has zero target variance.

Census	Target	PE-X1	PE-Y1	PE-X2	PE-QX1
cubic $n = 14$	C_3	1.0000	1.0000	1.0000	0.8774
	C_4	0.9981	0.9973	0.9998	0.8688
	C_5	0.9275	0.9495	0.9604	0.7335
	C_6	0.4850	0.6980	0.4610	0.7154
	$D_\diamond$	0.9926	0.9977	0.9980	0.9940
	girth	0.9930	0.9909	0.9917	0.5775
	spectral gap	0.9443	0.9358	0.9079	0.6876
4-regular $n = 12$	C_3	1.0000	1.0000	1.0000	0.9782
	C_4	0.9993	1.0000	0.9978	0.8509
	C_5	0.7869	0.9699	0.8790	0.7775
	C_6	0.9220	0.9623	0.7991	0.8518
	$D_\diamond$	0.9987	0.9964	0.9991	0.9977
	spectral gap	0.9398	0.9525	0.9527	0.9481

Linear recovery from exact readouts. Table 12 gives held-out R^2 for one target-independent vector per graph against every target. On the cubic census the three degree-prepared members exceed 0.99 on C_3, C_4, $D_\diamond$ and girth and score 0.93 to 0.96 on C_5, in line with Table 4: the quantities determined by the first three layers are recovered linearly, and C_6, which the three-defect layer determines only through finer structure, is not (0.46 to 0.70). PE-QX1 scores lower on the cycle counts and comparably on $D_\diamond$, the one coordinate its second-order closure carries. The 4-regular census changes both degree and order; the closures do not apply there, and the recurrence of the same ranking is empirical.

Where in the ranks. Truncating the sorted readout to its first k ranks, without renormalization, and repeating the probe locates the information. For PE-X1, PE-Y1 and PE-X2, 95% of the full-vector score is reached by $k \le 106 = K_2(14)$ for C_3, C_4, $D_\diamond$ and girth and by $k = 200$ for C_5, over the grid $k \in \{25, 50, 100, 106, 200, 400, 1000, 4000, 16384\}$; at $k = 106$ these members retain at least 0.9968 of the probability mass. PE-QX1 needs $k = 4000$ for $D_\diamond$ and the full vector for the other targets, and retains 0.0137 of the mass at $k = 106$. The cutoff has a certified meaning only for PE-X1 (Proposition 6.3); for the others it is a matched rank. The C_5 onset just beyond $K_2(14)$ and the two-defect head are adjacent facts, and Table 4 says why: the pair census does not determine C_5, and the three-defect layer does.

Geometry. Spearman Mantel correlations between pairwise L_1 readout distances and absolute target differences are largest for C_3 under the degree-prepared members (0.9157 to 0.9949 on the cubic census, 0.9302 to 0.9735 on the 4-regular census) and for $D_\diamond$ under PE-QX1 (0.8583 and 0.9618), which is the second-order closure seen in the geometry of the full vector at the operating angle. After fixing C_3, the C_4 association is significant in every qualifying stratum for every member on both censuses; after fixing (C_3, C_4), the C_5 association is significant in 9 to 13 of 16 cubic strata and 13 to 24 of 37 4-regular strata. All 36 global tests remain significant after correction; no random-target control does.

Table 13: Classical placement comparators on the cubic census at $n = 14$, fold-matched to Table 12: held-out R^2 of the linear spectral readout, Extra Trees on the same features, and cumulative and stable folklore 2-WL histograms.

Target	Eigenvalues and traces	Extra Trees	2-WL cumulative	2-WL stable
C_3	1.0000	0.9999	0.9998	0.9958
C_4	1.0000	0.9973	0.9990	0.9948
C_5	1.0000	0.8959	0.9960	0.9958
C_6	0.9862	0.8530	0.9903	0.9884
$D_\diamond$	0.6358	0.8707	0.9975	0.9906
girth	0.6819	0.9817	0.8210	0.8164
spectral gap	1.0000	0.9954	0.9759	0.9726

Table 14: Finite-shot accessibility on the cubic census at $n = 14$: exact top-500 R^2 / mean R^2 at 2^{24} shots over 30 replicates. Replicate standard deviations are 0.0002 to 0.0089 for C_3, 0.0029 to 0.0089 for C_5 and 0.0095 to 0.0176 for $D_\diamond$.

Member	C_3	C_5	$D_\diamond$
PE-X1	1.000 / 0.995	0.927 / − 0.015	0.994 / 0.584
PE-Y1	1.000 / 0.905	0.951 / − 0.021	0.998 / 0.491
PE-X2	1.000 / 0.997	0.960 / − 0.012	0.998 / 0.668
PE-QX1	0.522 / 0.490	0.291 / 0.303	0.626 / 0.587

Classical placement. Fold-matched classical comparators on the cubic census (Table 13) behave as Proposition B.26 dictates: the linear spectral readout (sorted eigenvalues and eight power traces) gives $R^2 = 1.0000$ on C_3, C_4, C_5 and the spectral gap and 0.6358 on $D_\diamond$ and 0.6819 on girth; an untuned Extra Trees model on the same features gives 0.8707 and 0.9817; cumulative folklore 2-WL histograms give 0.9975 on $D_\diamond$ and 0.8210 on girth. These are target-specific accessibility results under one protocol and do not order the representations.

Irregular fixed-degree strata. Within each of four fixed-degree strata at $n = 14$ (degree sequences $[4, 4, 3^{\times 10}, 2, 2]$, $[4^{\times 7}, 2^{\times 7}]$, $[5, 5, 4, 4, 3^{\times 6}, 2^{\times 4}]$ and $[6, 4, 4, 3^{\times 8}, 2, 2, 2]$) we probe joint-degree mixing coordinates, untyped and degree-typed motif counts, and motif residuals after exact projection out of mixing; scores are macro-averaged across the equally weighted strata and no regression pools them. The degree-prepared members score 0.6002 to 0.7156 on mixing and -0.0035 to 0.0179 on the residual motifs; PE-QX1 scores 0.2161 and 0.1570. Because the comparison is within fixed degree sequences after exact projection, degree-sequence variation cannot produce the contrast.

Finite shots. Table 14 gives recovery from independently sampled training and test readouts, globally sorted, truncated to the top 500 ranks without renormalization, at 2^{24} shots over 30 replicates, against the exact top-500 reference. The degree-prepared members keep C_3 (0.905 to 0.997) and about half of $D_\diamond$ (0.491 to 0.668) and lose C_5 entirely. PE-QX1 keeps roughly what it had, and reaches $R^2 = 0.553$ on $D_\diamond$ already at 2^{17} shots although 95% of its exact score needs $k = 4000$; concentration in the early ranks does not predict finite-shot recovery. Proposition 4.6 sets the scale: the information in the two-defect block lies in within-layer differences of order $\beta\mu_2\Delta$, about 0.18β relative to probabilities of order 10^{-4}, which is what a multinomial estimate must resolve.

Structured readout. Sorting is one invariantization among many. On the four irregular strata, with Hadamard preparation and the X mixer, we compared the top 1,000 sorted probabilities with degree-sector aggregation, which pools outcomes by the number of excited vertices in each degree class and uses graph-supplied degree labels but no edge or motif information. For the joint-degree mixing basis, degree-sector aggregation preserves the exact information (R^2 from 0.9991 to 1.0000 against 0.0700 to 0.5531 for the sorted head) and reaches 50% of its own exact reference with 8.0 to 38.3 times fewer shots on the ladder $S \in \{2^{10}, \ldots, 2^{20}\}$; Hamming-weight pooling, which discards the degree classes, never reaches that threshold, and with no edge phase the sector masses are graph-independent within a stratum. The thresholds are point estimates specific to these strata, this basis, the estimator and the criterion.

G Status of the Results

Table 15 lists every result of the paper with its status and its location.

Table 15: The results and their status. "Proved" means proved for every cubic graph; "verified" means checked exhaustively on the complete connected cubic censuses at $n = 10$, 12 and 14; "certified" means established by outward-rounded interval arithmetic at the stated orders; "measured" means a computed fact about a census, with no claim beyond it; "computed" means a fact about the named graphs established by exact computation.

Result	Status	Where
Layer template for the first-order value, all four members	proved; verified	Thm. 3.2, App. A
Layer determination, X mixer (magnitude or irrationality of t^2) and Y mixer (rational independence)	proved	Thm. 3.3, App. A
Two-defect class values, injectivity, pair-census readout	proved; verified	Thm. 4.1, Cor. 4.2
Three-defect profiles; even–odd complementarity of the mixer axes	proved	Prop. A.14, Cor. 4.5
Two- and three-defect histograms at girth ≥ 5 as functions of (n, C_5, C_6); $\min(\text{girth}, 7)$ from three defects at every n	proved; checked on all 516 census graphs of girth ≥ 5 through $n = 18$	Thm. 4.3, Cor. 4.4
Shot count for the pair census from the two-defect block	proved; constants evaluated at $n = 14$	Prop. 4.6, App. F
Order-j locality and Weisfeiler–Leman placement	proved; checked at $n = 10, 12$	Thm. 5.3, App. A
Folklore 2-WL injective on the cubic censuses through $n = 20$; the CFI pair over K_4: equal first-order histograms through six defects, sorted readouts at the operating angle equal to 1.7×10^{-18} on 16,231 ranks and different at rank one	computed; the rank-one difference certified	Prop. 5.4, App. E
Second-order purity closures, PE-X1 and PE-QX1	proved; verified	Thm. 6.1, App. B
Cubic trace identities; cospectral separation at small β	proved; witnesses at $n = 18$	Prop. B.26, Cor. B.27, Cor. 6.2
Two-defect head at the operating angle, PE-X1	certified at $n = 14, 16, 18$	Prop. 6.3, App. B
C_5, C_6, girth from three defects; injectivity from four	measured on the $n = 14$ census	Sec. 7
Finite-angle map and the head-membership dip	measured on the $n = 14$ census	App. C
Linear and finite-shot recovery	measured	App. F

AI Methodology Statement

A large language model (Anthropic's Claude) was used in preparing this submission, under the authors' direction, for three purposes: drafting and typesetting derivations and prose from the authors' results and notes; generating the verification notebooks and scripts that implement the authors' circuit construction, the exact derivative, the census-level checks, the tensor contraction of Appendix E and the constants of Proposition 4.6; and assisting literature search. Every derivation was checked by the authors, and the proofs in Appendices A and B were verified against exact and

interval-arithmetic computation. Every notebook and script was run by the authors on the stated populations, and every number in this paper comes from those runs. No mathematical claim and no experimental result rests on an AI system's output without that verification.

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